THE RANKER'S MASTERBOOK

ARC.

The Inverse Circular Functions

Complete Solutions

100	**10**	**3–5**	**206**
PROBLEMS	CHAPTERS	DIFFICULTY	WORKED SOLUTIONS

arc

One hundred original problems on the inverse circular functions – where the branch you forget is the mark you lose. For the very top of JEE Advanced and the Mathematical Olympiad. Every problem solved more than one way.

Contents

Ten chapters, ten problems each.

CHAPTER I

Principal Branches & the Range Trap

sin-1

Outside the principal range, arcsin∘sin is a sawtooth.

№1 Ten Radians Out

●●●●● L3

EVALUATE

$$\sin^{-1}(\sin 10)$$

sin-inverse · reduction · quadrant · radian-measure

ANSWER

$$3\pi - 10$$

THE TRAP

The reflex answer is **10**, applying $\sin^{-1}(\sin x) = x$ blindly. But $\sin^{-1}$ only returns values in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, and $10 \notin [-\frac{\pi}{2}, \frac{\pi}{2}]$, so 10 is impossible. A subtler wrong move is $10 - 2\pi \approx 3.717$, which is correct for $\sin$ but still lies in $(\frac{\pi}{2}, \frac{3\pi}{2})$, outside the principal range.

Principal-value reduction

1. Subtract a full turn: $\sin 10 = \sin(10 - 2\pi)$ where $10 - 2\pi \approx 3.717$ lies in $(\pi, \frac{3\pi}{2})$ (third quadrant).
2. On $(\frac{\pi}{2}, \frac{3\pi}{2})$ the angle with the same sine in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is $\pi - \theta$. Take $\theta = 10 - 2\pi$: the reduced angle is $\pi - (10 - 2\pi) = 3\pi - 10$.
3. Check $3\pi - 10 \approx -0.575 \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ and $\sin(3\pi - 10) = \sin 10$. Hence $\boxed{3\pi - 10}$.

Sawtooth graph

1. The graph of $y = \sin^{-1}(\sin x)$ is a triangular wave of period 2π, equal to x on $[-\frac{\pi}{2}, \frac{\pi}{2}]$ and reflecting with slope -1 on $[\frac{\pi}{2}, \frac{3\pi}{2}]$, where it reads $\pi - x$ after shifting by the right multiple of 2π.
2. At $x = 10$: the nearest descending branch is centred near $x = \pi + 2\pi = 3\pi \approx 9.42$, on which $y = \pi + 2\pi - x = 3\pi - x$.
3. Substitute $x = 10$ to get $\boxed{3\pi - 10}$.

INSIGHT

For any x, write $x = 2k\pi + \theta$ with $\theta \in [0, 2\pi)$; then $\sin^{-1}(\sin x) = \theta$ on $[0, \frac{\pi}{2}]$, $\pi - \theta$ on $[\frac{\pi}{2}, \frac{3\pi}{2}]$, and $\theta - 2\pi$ on $[\frac{3\pi}{2}, 2\pi)$. The single safe habit is *reduce, then locate the quadrant.*

№2 The Negative Decoy

●●●●● L3

EVALUATE

$$\cos^{-1}(\cos(-5))$$

cos-inverse · even-function · reduction · quadrant

ANSWER

$$2\pi - 5$$

THE TRAP

Many write **-5 (cancellation) or 5** (using evenness $\cos(-5) = \cos 5$ then stopping). Both fail: $\cos^{-1}$ returns values in $[0, \pi]$, and neither -5 nor 5 lies there. The evenness step is legal, but $5 \approx 5$ still sits in $(\frac{3\pi}{2}, 2\pi)$, outside $[0, \pi]$.

Evenness then range

1. Use $\cos(-5) = \cos 5$ since cosine is even.
2. $5 \in (\pi, 2\pi)$; on this interval the angle in $[0, \pi]$ with equal cosine is $2\pi - \theta$. With $\theta = 5$ that is $2\pi - 5$.
3. Verify $2\pi - 5 \approx 1.283 \in [0, \pi]$ and $\cos(2\pi - 5) = \cos 5 = \cos(-5)$. Hence $\boxed{2\pi - 5}$.

Direct reduction of the negative angle

1. Add a full turn: $\cos(-5) = \cos(-5 + 2\pi) = \cos(2\pi - 5)$, and $2\pi - 5 \approx 1.283$.
2. Since $2\pi - 5 \in [0, \pi]$, it already lies in the principal range of $\cos^{-1}$.
3. Therefore $\cos^{-1}(\cos(-5)) = \boxed{2\pi - 5}$.

INSIGHT

$\cos^{-1}(\cos x) = |x|$ only when $|x| \le \pi$. Generally, fold x into $[0, 2\pi)$ then apply $\cos^{-1}(\cos\theta) = \theta$ on $[0, \pi]$ and $2\pi - \theta$ on $[\pi, 2\pi]$. Evenness disposes of the sign but never of the range obligation.

№3 Tangent's Quiet Period

L3

EVALUATE

$$\tan^{-1}(\tan 3)$$

tan-inverse · period-pi · reduction · open-interval

ANSWER

$$3 - \pi$$

THE TRAP

The seductive answer is **3**. But $\tan^{-1}$ maps into the open interval $(-\frac{\pi}{2}, \frac{\pi}{2})$, and $3 > \frac{\pi}{2} \approx 1.571$, so 3 is out of range. Because tangent has period π (not 2π), the correction is a single subtraction of π, not 2π.

Period-pi reduction

1. Tangent has period π, so $\tan 3 = \tan(3 - \pi)$ with $3 - \pi \approx -0.142$.
2. Check $3 - \pi \in (-\frac{\pi}{2}, \frac{\pi}{2})$, the principal range of $\tan^{-1}$.
3. Hence $\tan^{-1}(\tan 3) = \boxed{3 - \pi}$.

Locate the branch

1. 3 lies in $(\frac{\pi}{2}, \pi)$ (second quadrant), where tangent is negative; so the answer must be a negative number in $(-\frac{\pi}{2}, 0)$.

2. The unique such angle congruent to 3 modulo π is $3 - \pi \approx -0.142$.
3. Therefore the value is $\boxed{3 - \pi}$.

INSIGHT

For $\tan^{-1}(\tan x)$ write $x = k\pi + \theta$ with $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$; the answer is $\theta = x - k\pi$. The shift is by π because tangent's period is π – a fact that trips students drilled only on the sine sawtooth.

№4 Seven-Sixths of a Pi

●●●●● L3

EVALUATE

$$\sin^{-1}\left(\sin\frac{7\pi}{6}\right)$$

sin-inverse · exact-angle · third-quadrant · reduction

ANSWER

$$-\frac{\pi}{6}$$

THE TRAP

The instinct is **$\frac{7\pi}{6}$, or after a half-step $\frac{5\pi}{6}$** (using $\sin\frac{7\pi}{6} = -\frac{1}{2}$ and grabbing the wrong root). Neither is in $[-\frac{\pi}{2}, \frac{\pi}{2}]$. The value $\sin\frac{7\pi}{6} = -\frac{1}{2}$ is negative, so the answer must be negative — eliminating both decoys.

Quadrant reduction

1. $\frac{7\pi}{6} \in (\pi, \frac{3\pi}{2})$ (third quadrant), so $\sin^{-1}(\sin x) = \pi - x$ there: $\pi - \frac{7\pi}{6} = -\frac{\pi}{6}$.
2. Check $-\frac{\pi}{6} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ and $\sin(-\frac{\pi}{6}) = -\frac{1}{2} = \sin\frac{7\pi}{6}$.
3. Hence $\boxed{-\frac{\pi}{6}}$.

Value-first

1. Compute the inside: $\sin\frac{7\pi}{6} = -\frac{1}{2}$.
2. $\sin^{-1}(-\frac{1}{2})$ is the unique angle in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ with sine $-\frac{1}{2}$, namely $-\frac{\pi}{6}$.
3. Therefore $\boxed{-\frac{\pi}{6}}$.

INSIGHT

When the inner argument is an exact multiple of $\frac{\pi}{6}$, evaluating the inside first and then applying the inverse on $[-1, 1]$ is bulletproof – it sidesteps quadrant bookkeeping entirely, as long as you pick the root in the principal range.

№5 Cotangent's Wrong Door

●●●●● L4

EVALUATE

$$\cot^{-1}(\cot(-4))$$

cot-inverse · negative-angle · open-interval-0-pi · period-pi

ANSWER

$$2\pi - 4$$

THE TRAP

Students write **-4, or correct the period once to $-4+\pi=\pi-4\approx-0.858$**. But $\cot^{-1}$ has range $(0,\pi)$ — strictly positive — so any negative output is wrong. One π is not enough here; you must reach into $(0,\pi)$.

Reduce into (0,pi)

1. Cotangent has period π: $\cot(-4)=\cot(-4+2\pi)=\cot(2\pi-4)$.
2. $2\pi-4\approx 2.283$ lies in $(0,\pi)$, the principal range of $\cot^{-1}$.
3. Hence $\cot^{-1}(\cot(-4))=\boxed{2\pi-4}$.

Sign and quadrant

1. -4 is congruent mod π to $-4+\pi\approx-0.858$ and to $-4+2\pi\approx 2.283$; only the latter lies in $(0,\pi)$.
2. Since $\cot^{-1}$ must return a value in $(0,\pi)$, select $2\pi-4$.
3. Confirm $\cot(2\pi-4)=\cot(-4)$ by periodicity; answer $\boxed{2\pi-4}$.

INSIGHT

The convention $\cot^{-1}x=\frac{\pi}{2}-\tan^{-1}x$ forces the range $(0,\pi)$, identical to $\cos^{-1}$ in spirit but with period-π folding. Always land the reduced angle inside $(0,\pi)$ – adding π once may not suffice for very negative inputs.

№6 Eight and the Vanishing Pi

L4

EVALUATE

$$\sin^{-1}(\sin 8)+\cos^{-1}(\cos 8)$$

sin-inverse | cos-inverse | sum | complementary-cancellation

ANSWER

$$\pi$$

THE TRAP

A strong student notices $\sin^{-1}(\sin\theta)+\cos^{-1}(\cos\theta)$ *looks* like the identity $\sin^{-1}t+\cos^{-1}t=\frac{\pi}{2}$ and writes **$\frac{\pi}{2}$**. That identity needs the *same* argument t and $t\in[-1,1]$; here the arguments $\sin 8$ and $\cos 8$ differ, so it does not apply. Another trap is **16** (cancelling both inverses).

Reduce each term separately

1. $8-2\pi\approx 1.717$ lies in $(\frac{\pi}{2},\pi)$ (second quadrant).
2. $\sin^{-1}(\sin 8)=\pi-(8-2\pi)=3\pi-8$ (the $\pi-\theta$ rule for the second quadrant after de-rotating by 2π).
3. $\cos^{-1}(\cos 8)=8-2\pi$ directly, since $8-2\pi\in[0,\pi]$.
4. Add: $(3\pi-8)+(8-2\pi)=\boxed{\pi}$; the constants cancel.

Algebraic cancellation

1. Let $\theta=8-2\pi\in(\frac{\pi}{2},\pi)$, so $\sin 8=\sin\theta$, $\cos 8=\cos\theta$.
2. On $(\frac{\pi}{2},\pi)$: $\sin^{-1}(\sin\theta)=\pi-\theta$ and $\cos^{-1}(\cos\theta)=\theta$.
3. Sum $=(\pi-\theta)+\theta=\boxed{\pi}$, independent of θ in this quadrant.

INSIGHT

For any $\theta\in(\frac{\pi}{2},\pi)$, $\sin^{-1}(\sin\theta)+\cos^{-1}(\cos\theta)=\pi$ identically. The famous $\sin^{-1}t+\cos^{-1}t=\frac{\pi}{2}$ is a *different* statement about one shared input in $[-1,1]$ – confusing the two is the classic blunder.

№7 Where the Sawtooth Meets the Ray

●●●●● L4

FIND THE NUMBER OF SOLUTIONS

$$\text{the equation} \quad \cos^{-1}(\cos x) = \frac{x}{4} \quad \text{on} \quad [0, 4\pi]$$

cos-inverse · triangular-wave · equation-counting · graph-intersection

ANSWER

$$4$$

THE TRAP

Treating $\cos^{-1}(\cos x)$ as just x gives $x = \frac{x}{4} \Rightarrow x = 0$, suggesting only **1** solution. But $\cos^{-1}(\cos x)$ is the triangular wave (values capped in $[0, \pi]$), which crosses the gentle line $y = \frac{x}{4}$ several times. Forgetting the $x = 0$ boundary touch (where both sides equal 0) is the other common miscount.

Piecewise solving

1. On $[0, 4\pi]$ the wave is: x on $[0, \pi]$, $2\pi - x$ on $[\pi, 2\pi]$, $x - 2\pi$ on $[2\pi, 3\pi]$, $4\pi - x$ on $[3\pi, 4\pi]$.
2. Solve $\frac{x}{4} =$ each branch and keep roots in the branch's interval: $x = x/4 \Rightarrow x = 0$ (valid); $2\pi - x = x/4 \Rightarrow x = \frac{8\pi}{5}$ (valid); $x - 2\pi = x/4 \Rightarrow x = \frac{8\pi}{3}$ (valid); $4\pi - x = x/4 \Rightarrow x = \frac{16\pi}{5}$ (valid).
3. All four lie in their respective sub-intervals, so the count is $\boxed{4}$.

Graphical bound

1. The line $y = \frac{x}{4}$ runs from 0 to π as x goes $0 \to 4\pi$, exactly the vertical span of the wave $[0, \pi]$.
2. Each of the four monotone segments of the wave on $[0, 4\pi]$ meets this slowly rising line once (the slopes ± 1 never equal $\frac{1}{4}$, so at most one crossing per segment; checking endpoints confirms one each, including the touch at $x = 0$).
3. Total intersections $= \boxed{4}$.

INSIGHT

Replacing $\cos^{-1}(\cos x)$ (or $\sin^{-1}(\sin x)$) by its sawtooth/triangular graph turns a transcendental equation into counting line-segment intersections. Always include endpoint contacts: $x = 0$ here is a legitimate root the naive substitution would keep, but a careless sign-change count drops.

№8 Three Inverses, One Integer

●●●●● L5

EVALUATE

$$\sin^{-1}(\sin 5) + \cos^{-1}(\cos 11) + \tan^{-1}(\tan 7)$$

sin-inverse · cos-inverse · tan-inverse · mixed-sum · pi-cancellation

ANSWER

$$1$$

THE TRAP

The naive cancellation gives **$5 + 11 + 7 = 23$**. Every term must instead be folded into its principal range, and the surprise is that all the π's annihilate, leaving a bare integer. A partial-credit student who reduces only the sine term and stops will land on an irrational mess and distrust the clean answer.

Reduce, then sum

1. $5 \in (\frac{3\pi}{2}, 2\pi)$ (fourth quadrant): $\sin^{-1}(\sin 5) = 5 - 2\pi$.
2. $11 - 2\pi \approx 4.717 \in (\pi, 2\pi)$: $\cos^{-1}(\cos 11) = 2\pi - (11 - 2\pi) = 4\pi - 11$.
3. $7 - 2\pi \approx 0.717 \in (-\frac{\pi}{2}, \frac{\pi}{2})$: $\tan^{-1}(\tan 7) = 7 - 2\pi$.
4. Sum $= (5 - 2\pi) + (4\pi - 11) + (7 - 2\pi) = (5 - 11 + 7) + (-2\pi + 4\pi - 2\pi) = \boxed{1}$.

Track integer and pi parts

1. Each reduction has the form (integer combination of the argument) $+$ (integer multiple of π). Collect the two ledgers separately.
2. Integer ledger: $5 - 11 + 7 = 1$. Pi ledger: $-2\pi + 4\pi - 2\pi = 0$.
3. Since the π contributions cancel, the total is the integer ledger alone: $\boxed{1}$.

> **INSIGHT**
> Designed so the π-coefficients $(-2, +4, -2)$ sum to zero. This is the engine behind every 'evaluate to a clean number' inverse-trig sum: choose arguments whose quadrant corrections cancel the transcendental parts, leaving the rational/integer residue.

№9 Slope of the Sawtooth at Seventeen

L4

DIFFERENTIATE

$$g(x) = \cos^{-1}(\cos x); \quad \text{find } g(17) \text{ and } g'(17)$$

cos-inverse | derivative | branch-slope | piecewise

ANSWER

$$g(17) = 6\pi - 17, \quad g'(17) = -1$$

> **THE TRAP**
> Writing $g(x) = x$ gives **$g(17) = 17,\ g'(17) = 1$** — both wrong. The chain-rule slip $g'(x) = \frac{-(-\sin x)}{\sqrt{1-\cos^2 x}} = \frac{\sin x}{|\sin x|}$ is correct, but students forget the absolute value $\sqrt{1-\cos^2 x} = |\sin x|$ and report $+1$ regardless of quadrant. At $x = 17$, $\sin 17 < 0$, so the slope is -1.

Reduce then read the branch

1. $17 - 4\pi \approx 4.434 \in (\pi, 2\pi)$, so $g(17) = \cos^{-1}(\cos(17 - 4\pi)) = 2\pi - (17 - 4\pi) = 6\pi - 17 \approx 1.850$.
2. On the branch $(\pi, 2\pi)$ (after de-rotating by 4π), g equals $6\pi - x$, a line of slope -1.
3. Hence $g(17) = \boxed{6\pi - 17}$ and $g'(17) = \boxed{-1}$.

Chain rule with absolute value

1. $g'(x) = \frac{d}{dx}\cos^{-1}(\cos x) = \frac{-1}{\sqrt{1-\cos^2 x}} \cdot (-\sin x) = \frac{\sin x}{|\sin x|} = \operatorname{sgn}(\sin x)$ (where $\sin x \neq 0$).
2. At $x = 17$: $\sin 17 \approx -0.961 < 0$, so $g'(17) = -1$.
3. Combine with the value from periodic reduction $g(17) = 6\pi - 17$: answer $\boxed{g(17) = 6\pi - 17,\ g'(17) = -1}$.

> **INSIGHT**
> $\frac{d}{dx}\cos^{-1}(\cos x) = \operatorname{sgn}(\sin x)$ and $\frac{d}{dx}\sin^{-1}(\sin x) = \operatorname{sgn}(\cos x)$, each undefined at the wave's corners. The derivative is always ± 1 – the sign records which falling/rising branch you are on, and the $\sqrt{\ } = |\cdot|$ is exactly where that sign enters.

№10 Six Integers Into the Sine Mill

●●●●● L5

EVALUATE

$$\sum_{k=1}^{6} \sin^{-1}(\sin k)$$

sin-inverse · integer-arguments · telescoping-pi · quadrant-bookkeeping

ANSWER

$$3 - \pi$$

THE TRAP

The cancellation fantasy gives $1+2+3+4+5+6=$ **21**. Each k sits in a different quadrant, so each $\sin^{-1}(\sin k)$ needs its own correction; the π-terms partially survive, yielding the unexpectedly small $3-\pi$. Students who reduce the first few terms but tire and approximate often miss the exact form.

Term-by-term quadrant reduction

1. $k=1 \in (0, \frac{\pi}{2})$: value 1. $k=2,3 \in (\frac{\pi}{2}, \pi)$: values $\pi-2,\ \pi-3$. $k=4 \in (\pi, \frac{3\pi}{2})$: value $\pi-4$.
2. $k=5,6 \in (\frac{3\pi}{2}, 2\pi)$: values $5-2\pi,\ 6-2\pi$.
3. Sum the integer parts: $1-2-3-4+5+6=3$. Sum the π parts: $\pi+\pi+\pi-2\pi-2\pi=-\pi$.
4. Total $= \boxed{3-\pi}$.

Two-ledger bookkeeping

1. Note $\sin^{-1}(\sin k) = \varepsilon_k\, k + m_k\pi$ where $\varepsilon_k = \pm 1$ and $m_k \in \mathbb{Z}$ are fixed by k's quadrant: $(\varepsilon, m) = (+,0),(-,+1),(-,+1),(-,+1),(+,-2),(+,-2)$ for $k=1\ldots 6$.
2. Integer ledger $\sum \varepsilon_k k = 1-2-3-4+5+6=3$; pi ledger $\sum m_k = 0+1+1+1-2-2=-1$.
3. Therefore the sum is $3+(-1)\pi = \boxed{3-\pi}$.

INSIGHT

For $\sum_{k=1}^{n} \sin^{-1}(\sin k)$ the answer hinges on how the integers $1,\ldots,n$ distribute among the quadrants of one period. Because 2π is irrational against the integers, no k ever lands on a corner – every term reduces cleanly, and the running π-count is a genuine (non-telescoping) invariant of the partition.

CHAPTER II

Addition Formulas & ±π Corrections

Add two arctangents and the formula quietly leaks a π.

№11 The First Overflow

●●●●● L4

EVALUATE

$$\tan^{-1} 2 + \tan^{-1} 3$$

tan-inverse · addition-formula · pi-correction · principal-value

ANSWER

$$\frac{3\pi}{4}$$

THE TRAP

The reflexive move is $\tan^{-1} 2 + \tan^{-1} 3 = \tan^{-1} \frac{2+3}{1-2\cdot3} = \tan^{-1}(-1) = -\frac{\pi}{4}$. That is **wrong**: both $\tan^{-1} 2$ and $\tan^{-1} 3$ exceed $\frac{\pi}{4}$, so their sum exceeds $\frac{\pi}{2}$ and cannot be a negative principal value. The identity $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \frac{a+b}{1-ab}$ holds only when $ab < 1$; here $ab = 6 > 1$ with $a, b > 0$, so the correct relation is $\tan^{-1} a + \tan^{-1} b = \pi + \tan^{-1} \frac{a+b}{1-ab}$.

Principal-value range bound + correction

1. Since $2, 3 > 1 > 0$, we have $\tan^{-1} 2, \tan^{-1} 3 \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$, so the true sum S lies in $\left(\frac{\pi}{2}, \pi\right)$.
2. Both arguments are positive and $ab = 6 > 1$, so the corrected formula applies: $S = \pi + \tan^{-1} \dfrac{2+3}{1-6} = \pi + \tan^{-1}(-1) = \pi - \dfrac{\pi}{4}$.
3. Hence $S = \boxed{\dfrac{3\pi}{4}}$, which indeed lies in $\left(\frac{\pi}{2}, \pi\right)$, confirming the branch.

Gaussian-integer argument

1. Write $\tan^{-1} 2 = \arg(1 + 2i)$ and $\tan^{-1} 3 = \arg(1 + 3i)$, both first-quadrant arguments.
2. Multiply: $(1 + 2i)(1 + 3i) = 1 + 5i + 6i^2 = -5 + 5i$, which lies in the **second** quadrant, so $\arg(-5 + 5i) = \frac{3\pi}{4}$.
3. Since each factor has argument in $\left(0, \frac{\pi}{2}\right)$, the sum of arguments is the genuine argument of the product (no 2π wrap), giving $S = \boxed{\dfrac{3\pi}{4}}$.

INSIGHT

The Gaussian-integer view makes the $+\pi$ correction automatic: multiplying $(1 + ai)(1 + bi)$ lands you in the half-plane the true angle lives in. Whenever $ab > 1$ with $a, b > 0$, the product's real part $1 - ab$ turns negative, pushing the argument past $\frac{\pi}{2}$ – exactly the situation the naive arctan formula cannot see.

№12 Doubling Past the Line

●●●●● L4

EVALUATE

$$2\tan^{-1}\left(2+\sqrt{3}\right)$$

double-angle tan-inverse pi-correction surd

ANSWER

$$\frac{5\pi}{6}$$

THE TRAP

Applying $2\tan^{-1}x = \tan^{-1}\frac{2x}{1-x^2}$ blindly gives $\tan^{-1}\frac{2(2+\sqrt{3})}{1-(2+\sqrt{3})^2} = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$. **Wrong sign and missing π:** the double-angle identity for $\tan^{-1}$ is valid only for $|x|<1$. Here $x = 2+\sqrt{3} > 1$, so $\tan^{-1}x > \frac{\pi}{4}$ and the doubled value exceeds $\frac{\pi}{2}$, forcing $2\tan^{-1}x = \pi + \tan^{-1}\frac{2x}{1-x^2}$.

Recognise the special tangent

1. Note $\tan 75^\circ = \tan(45^\circ + 30^\circ) = \dfrac{1+\frac{1}{\sqrt{3}}}{1-\frac{1}{\sqrt{3}}} = 2+\sqrt{3}$, so $\tan^{-1}(2+\sqrt{3}) = \dfrac{5\pi}{12}$ (and $\frac{5\pi}{12} \in \left(0, \frac{\pi}{2}\right)$ is the correct principal value).
2. Therefore $2\tan^{-1}(2+\sqrt{3}) = 2\cdot\dfrac{5\pi}{12} = \boxed{\dfrac{5\pi}{6}}$.

Corrected double-angle formula

1. Since $x = 2+\sqrt{3} > 1$, the principal value $\tan^{-1}x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$, so $2\tan^{-1}x \in \left(\frac{\pi}{2}, \pi\right)$.
2. Compute $\dfrac{2x}{1-x^2}$ with $x^2 = 7+4\sqrt{3}$: $\dfrac{2(2+\sqrt{3})}{1-(7+4\sqrt{3})} = \dfrac{4+2\sqrt{3}}{-6-4\sqrt{3}} = -\dfrac{1}{\sqrt{3}}$.
3. Apply the $|x|>1$ branch: $2\tan^{-1}x = \pi + \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = \pi - \frac{\pi}{6} = \boxed{\dfrac{5\pi}{6}}$, lying in $\left(\frac{\pi}{2}, \pi\right)$ as required.

INSIGHT

The double-angle rule for $\tan^{-1}$ splits into three branches by x: $2\tan^{-1}x$ equals $\tan^{-1}\frac{2x}{1-x^2}$ for $|x|<1$, $\pi + \tan^{-1}\frac{2x}{1-x^2}$ for $x>1$, and $-\pi + \tan^{-1}\frac{2x}{1-x^2}$ for $x<-1$. The threshold $|x|=1$ is exactly where $2\tan^{-1}x$ crosses $\pm\frac{\pi}{2}$, the edge of the arctan range.

№13 Mirror of the Overflow

●●●●● L4

EVALUATE

$$\tan^{-1}(-2)+\tan^{-1}(-3)$$

tan-inverse negative-correction addition-formula quadrant

ANSWER

$$-\frac{3\pi}{4}$$

THE TRAP

Students write $\tan^{-1}\frac{-2-3}{1-(-2)(-3)} = \tan^{-1}\frac{-5}{-5} = \tan^{-1}1 = \frac{\pi}{4}$. **Wrong:** a sum of two *negative* numbers cannot be positive. With $a,b<0$ and $ab=6>1$, the correction is $-\pi$, not 0: $\tan^{-1}a+\tan^{-1}b = -\pi + \tan^{-1}\frac{a+b}{1-ab}$.

Oddness reduces to the positive case

1. Using $\tan^{-1}(-t) = -\tan^{-1} t$, the sum equals $-\left(\tan^{-1} 2 + \tan^{-1} 3\right)$.
2. From the companion result $\tan^{-1} 2 + \tan^{-1} 3 = \frac{3\pi}{4}$, the value is $\boxed{-\dfrac{3\pi}{4}}$.

Range bound + $-\pi$ correction

1. Since $-2, -3 < -1$, each arctan lies in $\left(-\frac{\pi}{2}, -\frac{\pi}{4}\right)$, so the sum $S \in \left(-\pi, -\frac{\pi}{2}\right)$.
2. Both arguments negative with $ab = 6 > 1$ forces the $-\pi$ branch: $S = -\pi + \tan^{-1} \dfrac{-5}{1-6} = -\pi + \tan^{-1} 1 = -\pi + \dfrac{\pi}{4} = \boxed{-\dfrac{3\pi}{4}}$, which lies in $\left(-\pi, -\frac{\pi}{2}\right)$.

> **INSIGHT**
> The correction's *sign* is dictated by the common sign of a, b (when $ab > 1$): $+\pi$ if both positive, $-\pi$ if both negative. The rationalized argument $\frac{a+b}{1-ab}$ is identical to the positive mirror image, which is why oddness collapses the two cases into one.

№14 Reciprocal Bookkeeping

●●●●● L4

EVALUATE

$$\tan^{-1} \tfrac{1}{3} + \tan^{-1} 3 + \tan^{-1}\left(-\tfrac{1}{5}\right) + \tan^{-1}(-5)$$

reciprocal-identity · tan-inverse · sign-of-x · telescoping

ANSWER

$$0$$

> **THE TRAP**
> A tempting shortcut is to pair $\tan^{-1} \frac{1}{3} + \tan^{-1} 3 = \frac{\pi}{2}$ and $\tan^{-1}(-\frac{1}{5}) + \tan^{-1}(-5) = \frac{\pi}{2}$ (using $\tan^{-1} x + \tan^{-1} \frac{1}{x} = \frac{\pi}{2}$ both times), giving π. **Wrong:** the reciprocal identity equals $+\frac{\pi}{2}$ only when $x > 0$; for $x < 0$ it equals $-\frac{\pi}{2}$. The second pair has $x < 0$, so it contributes $-\frac{\pi}{2}$.

Sign-aware reciprocal identity

1. Recall $\tan^{-1} x + \tan^{-1} \frac{1}{x} = \frac{\pi}{2}$ for $x > 0$ and $= -\frac{\pi}{2}$ for $x < 0$ (it is the odd extension; the two arctans cannot both be positive when $x < 0$).
2. First pair ($x = 3 > 0$): $\tan^{-1} \frac{1}{3} + \tan^{-1} 3 = +\frac{\pi}{2}$. Second pair ($x = -5 < 0$): $\tan^{-1}(-\frac{1}{5}) + \tan^{-1}(-5) = -\frac{\pi}{2}$.
3. Total $= \frac{\pi}{2} - \frac{\pi}{2} = \boxed{0}$.

Oddness symmetry

1. Group as $\left(\tan^{-1} \frac{1}{3} + \tan^{-1} 3\right) + \left(\tan^{-1}(-\frac{1}{5}) + \tan^{-1}(-5)\right)$.
2. By oddness, $\tan^{-1}(-\frac{1}{5}) + \tan^{-1}(-5) = -\left(\tan^{-1} \frac{1}{5} + \tan^{-1} 5\right) = -\frac{\pi}{2}$, while the first bracket is $+\frac{\pi}{2}$.
3. The brackets are equal in magnitude and opposite in sign, so the sum is $\boxed{0}$.

> **INSIGHT**
> $\tan^{-1} x + \tan^{-1} \frac{1}{x} = \frac{\pi}{2}\,\mathrm{sgn}(x)$ is the cleanest encoding: the function is a step, not a constant. The undefined case $x = 0$ never arises because $\frac{1}{x}$ blows up. The same sign rule governs $\cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x$ behaviour at the origin.

№15 Crossing the Sine Boundary

●●●●● L4

EVALUATE

$$\sin^{-1}\tfrac{4}{5}+\sin^{-1}\tfrac{12}{13}$$

sin-inverse · boundary-crossing · pythagorean · supplement

ANSWER

$$\pi-\sin^{-1}\frac{56}{65}$$

THE TRAP

The reflex is $\sin^{-1}\frac{4}{5}+\sin^{-1}\frac{12}{13}=\sin^{-1}\left(\frac{4}{5}\cdot\frac{5}{13}+\frac{12}{13}\cdot\frac{3}{5}\right)=\sin^{-1}\frac{56}{65}$. **Wrong:** the identity $\sin^{-1}x+\sin^{-1}y=\sin^{-1}(x\sqrt{1-y^2}+y\sqrt{1-x^2})$ requires the sum to stay in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$, i.e. $x^2+y^2\le 1$. Here $\frac{16}{25}+\frac{144}{169}\approx 1.49>1$, so the sum exceeds $\frac{\pi}{2}$ and the correct form is $\pi-\sin^{-1}\frac{56}{65}$.

Boundary test then supplement

1. Both angles exceed $\frac{\pi}{4}$? Check: $\sin^{-1}\frac{4}{5}\approx 0.927$ and $\sin^{-1}\frac{12}{13}\approx 1.176$, so the sum $S\approx 2.10\in\left(\frac{\pi}{2},\pi\right)$. Equivalently $x^2+y^2>1$ signals overshoot.
2. For $S\in\left(\frac{\pi}{2},\pi\right)$ we have $\sin S=\sin(\pi-S)$ with $\pi-S\in\left(0,\frac{\pi}{2}\right)$. Compute $\sin S=\frac{4}{5}\cdot\frac{5}{13}+\frac{12}{13}\cdot\frac{3}{5}=\frac{20+36}{65}=\frac{56}{65}$.
3. Hence $\pi-S=\sin^{-1}\frac{56}{65}$, giving $S=\boxed{\pi-\sin^{-1}\frac{56}{65}}$.

Right-triangle / cosine check

1. Let $\alpha=\sin^{-1}\frac{4}{5}$ (so $\cos\alpha=\frac{3}{5}$) and $\beta=\sin^{-1}\frac{12}{13}$ (so $\cos\beta=\frac{5}{13}$).
2. Then $\cos(\alpha+\beta)=\frac{3}{5}\cdot\frac{5}{13}-\frac{4}{5}\cdot\frac{12}{13}=\frac{15-48}{65}=-\frac{33}{65}<0$, proving $\alpha+\beta>\frac{\pi}{2}$.
3. Since $\sin(\alpha+\beta)=\frac{56}{65}>0$ and the angle is obtuse, $\alpha+\beta=\boxed{\pi-\sin^{-1}\frac{56}{65}}$ (you may also write it as $\cos^{-1}\left(-\frac{33}{65}\right)$).

INSIGHT

For $x,y>0$, the test is purely x^2+y^2 versus 1: if ≤ 1 the naive $\sin^{-1}$ formula holds; if >1 the sum is obtuse and you reflect through π. The sign of $\cos(\alpha+\beta)$ is the cleanest detector of which side of $\frac{\pi}{2}$ you are on.

№16 Two Obtuse Cosines

L4

EVALUATE

$$\cos^{-1}\left(-\tfrac{1}{2}\right)+\cos^{-1}\left(-\tfrac{1}{\sqrt{2}}\right)$$

cos-inverse · addition-formula · 2pi-correction · sum-sign

ANSWER

$$\frac{17\pi}{12}$$

THE TRAP

Plugging into $\cos^{-1}x+\cos^{-1}y=\cos^{-1}(xy-\sqrt{1-x^2}\sqrt{1-y^2})$ gives $\cos^{-1}\left(\frac{1}{2\sqrt{2}}-\frac{\sqrt{3}}{2}\cdot\frac{1}{\sqrt{2}}\right)=\cos^{-1}\left(-\frac{\sqrt{3}-1}{2\sqrt{2}}\right)=\frac{7\pi}{12}$. **Wrong:** that identity holds only when $x+y\ge 0$ (so the sum stays in $[0,\pi]$). Here $x+y=-\frac{1}{2}-\frac{1}{\sqrt{2}}<0$, so the sum lies in $(\pi,2\pi]$ and you must take 2π minus the principal value.

Known principal values (cleanest)

1. $\cos^{-1}\left(-\frac{1}{2}\right)=\frac{2\pi}{3}$ and $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)=\frac{3\pi}{4}$, both in $[0,\pi]$ as required.

2. Add: $\frac{2\pi}{3}+\frac{3\pi}{4}=\frac{8\pi+9\pi}{12}=\boxed{\frac{17\pi}{12}}$.

Corrected cosine-sum formula

1. Since $x+y<0$, the sum $S\in(\pi,2\pi]$, so $S=2\pi-\cos^{-1}\left(xy-\sqrt{1-x^2}\sqrt{1-y^2}\right)$.
2. Inner value $=\frac{1}{2\sqrt{2}}-\frac{\sqrt{3}}{2\sqrt{2}}=\frac{1-\sqrt{3}}{2\sqrt{2}}$, and $\cos^{-1}$ of this is $\frac{7\pi}{12}$ (verify: $\cos\frac{7\pi}{12}=-\sin\frac{\pi}{12}=\frac{1-\sqrt{3}}{2\sqrt{2}}$).
3. Thus $S=2\pi-\frac{7\pi}{12}=\frac{24\pi-7\pi}{12}=\boxed{\frac{17\pi}{12}}$, lying in $(\pi,2\pi]$.

INSIGHT

The $\cos^{-1}$ addition law branches on the sign of $x+y$: if $x+y\geq 0$ use $\cos^{-1}(\dots)$; if $x+y<0$ use $2\pi-\cos^{-1}(\dots)$. The correction is 2π (not π) because the $\cos^{-1}$ range $[0,\pi]$ already covers half a turn, and the cosine of the sum cannot distinguish S from $2\pi-S$
.

№17 The Möbius Argument

●●●●● L4

SIMPLIFY

$$f(x)=\tan^{-1}\left(\frac{1+x}{1-x}\right)\quad\text{for } x>1$$

tan-inverse mobius pi-correction branch-condition

ANSWER

$$f(x)=\tan^{-1}x-\frac{3\pi}{4}$$

THE TRAP

The standard simplification $\tan^{-1}\frac{1+x}{1-x}=\tan^{-1}1+\tan^{-1}x=\frac{\pi}{4}+\tan^{-1}x$ is **only valid for $x<1$**. For $x>1$ the bracket $\frac{1+x}{1-x}$ is negative and $f(x)$ is a negative principal value, so the formula overshoots by π: the correct result is $\frac{\pi}{4}+\tan^{-1}x-\pi=\tan^{-1}x-\frac{3\pi}{4}$.

Difference formula with branch correction

1. View $\frac{1+x}{1-x}=\frac{1+x}{1-x}$ as the tangent of $\frac{\pi}{4}+\theta$ where $\theta=\tan^{-1}x$, since $\tan\left(\frac{\pi}{4}+\theta\right)=\frac{1+\tan\theta}{1-\tan\theta}$.
2. For $x>1$, $\theta=\tan^{-1}x\in\left(\frac{\pi}{4},\frac{\pi}{2}\right)$, so $\frac{\pi}{4}+\theta\in\left(\frac{\pi}{2},\frac{3\pi}{4}\right)$ — **outside** the $\tan^{-1}$ range. Subtract π to return to $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$.
3. Therefore $f(x)=\left(\frac{\pi}{4}+\theta\right)-\pi=\boxed{\tan^{-1}x-\frac{3\pi}{4}}$. (Spot-check $x=2$: $\tan^{-1}(-3)=\tan^{-1}2-\frac{3\pi}{4}\approx-1.249$, correct.)

Substitution $x=\tan\theta$ and range tracking

1. Put $x=\tan\theta$ with $\theta\in\left(\frac{\pi}{4},\frac{\pi}{2}\right)$ since $x>1$. Then $\frac{1+x}{1-x}=\tan\left(\theta+\frac{\pi}{4}\right)$.
2. As $\theta+\frac{\pi}{4}$ ranges over $\left(\frac{\pi}{2},\frac{3\pi}{4}\right)$, the principal arctangent of $\tan\left(\theta+\frac{\pi}{4}\right)$ is $\theta+\frac{\pi}{4}-\pi$.
3. Re-substitute $\theta=\tan^{-1}x$: $f(x)=\boxed{\tan^{-1}x-\frac{3\pi}{4}}$.

INSIGHT

A Möbius (linear-fractional) argument inside $\tan^{-1}$ always reduces to $\tan^{-1}x$ shifted by a constant – but the constant *jumps* across the pole $x=1$ where the bracket flips sign. The jump is exactly π: $+\frac{\pi}{4}$ for $x<1$ becomes $-\frac{3\pi}{4}$ for $x>1$.

№18 The Extraneous Root

●●●●● L4

SOLVE FOR X

$$\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}$$

tan-inverse | equation | extraneous-root | pi-correction

ANSWER

$$x = \frac{1}{6}$$

THE TRAP

Rationalizing, $\frac{2x+3x}{1-6x^2} = \tan\frac{\pi}{4} = 1$ gives $6x^2 + 5x - 1 = 0$, so $x = \frac{1}{6}$ or $x = -1$. The seductive error is reporting **both** roots. But $x = -1$ makes both arctans negative (sum $\approx -2.36 < 0$), which can never equal $\frac{\pi}{4}$; it satisfies only the squared/rationalized equation, where the $-\pi$ correction was silently dropped.

Rationalize then screen by range

1. Taking tangents: $\dfrac{5x}{1-6x^2} = 1 \Rightarrow 6x^2 + 5x - 1 = 0 \Rightarrow (6x-1)(x+1) = 0$, so $x \in \left\{\frac{1}{6}, -1\right\}$.
2. The RHS $\frac{\pi}{4} > 0$ requires the LHS positive, which needs $x > 0$ (both arctans share the sign of x). This kills $x = -1$.
3. Verify $x = \frac{1}{6}$: $6x^2 = \frac{1}{6} < 1$ so no correction is needed, and $\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{2} = \frac{\pi}{4}$. Hence $\boxed{x = \dfrac{1}{6}}$ is the unique solution.

Monotonicity / counting

1. Let $g(x) = \tan^{-1}(2x) + \tan^{-1}(3x)$. Then $g'(x) = \frac{2}{1+4x^2} + \frac{3}{1+9x^2} > 0$, so g is strictly increasing on $\mathbb{R}$.
2. As $x \to +\infty$, $g \to \pi$; $g(0) = 0$. Since $\frac{\pi}{4} \in (0, \pi)$ and g is strictly increasing, there is **exactly one** solution, and it must have $x > 0$.
3. From the quadratic the only positive root is $x = \frac{1}{6}$, so $\boxed{x = \dfrac{1}{6}}$.

INSIGHT

Taking tan of both sides is a many-to-one operation that injects spurious roots differing from the truth by a multiple of π. Always re-screen candidates against the *sign and range* of the original equation – or, better, use monotonicity to bound the solution count before solving.

№19 Counting Honest Solutions

●●●●○ L4

FIND THE NUMBER OF SOLUTIONS

$$\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1}\frac{8}{31}, \quad x \in \mathbb{R}$$

tan-inverse | solution-count | extraneous-root | monotonicity

ANSWER

$$1$$

THE TRAP

Rationalizing gives $\frac{(x+1)+(x-1)}{1-(x^2-1)} = \frac{2x}{2-x^2} = \frac{8}{31}$, i.e. $4x^2 + 31x - 8 = 0$, with roots $x = \frac{1}{4}$ and $x = -8$. Counting **two** solutions is the trap: at $x = -8$ both arguments are very negative and $(x+1)(x-1) = 63 > 1$, so the true sum carries a $-\pi$ correction and equals ≈ -2.89, nowhere near the small positive target. Only $x = \frac{1}{4}$ survives.

Solve, then validate each root

1. Rationalize: $\dfrac{2x}{2-x^2}=\dfrac{8}{31}\Rightarrow 62x=16-8x^2\Rightarrow 4x^2+31x-8=0$, giving $x=\dfrac{-31\pm 33}{8}\in\{\frac14,-8\}$.
2. Test $x=\frac14$: arguments $\frac54,-\frac34$ have product <1, no correction, and the sum is a small positive angle $=\tan^{-1}\frac{8}{31}$. Valid.
3. Test $x=-8$: arguments $-7,-9$ are both <-1 with product $63>1$, so the sum lies in $\left(-\pi,-\frac{\pi}{2}\right)$ and equals $-\pi+\tan^{-1}\frac{8}{31}\neq\tan^{-1}\frac{8}{31}$. Rejected. So $\boxed{1}$ solution.

Strict monotonicity bound

1. $h(x)=\tan^{-1}(x+1)+\tan^{-1}(x-1)$ has $h'(x)=\frac{1}{1+(x+1)^2}+\frac{1}{1+(x-1)^2}>0$, so h is strictly increasing from $-\pi$ (as $x\to-\infty$) to $+\pi$.
2. The RHS $\tan^{-1}\frac{8}{31}$ is a fixed value in $(-\pi,\pi)$, and a strictly increasing function attains each value of its range exactly once.
3. Therefore the equation has exactly $\boxed{1}$ solution (namely $x=\frac14$).

> **INSIGHT**
> Whenever an inverse-trig equation reduces to a quadratic, the strictly-monotone structure of the left side caps the true solution count at one – every extra algebraic root is an artifact of the tan-induced π-periodicity. Monotonicity is a faster certificate than back-substituting both candidates.

№20 Two Overflows Cancel

L4

PROVE THAT

$$2\tan^{-1}2+\tan^{-1}\frac43=\pi$$

tan-inverse · double-angle · pi-correction · telescoping

ANSWER

$$\pi$$

> **THE TRAP**
> Writing $2\tan^{-1}2=\tan^{-1}\frac{2\cdot 2}{1-4}=\tan^{-1}\left(-\frac43\right)$ and concluding $2\tan^{-1}2+\tan^{-1}\frac43=\tan^{-1}\left(-\frac43\right)+\tan^{-1}\frac43=0$ is **wrong**. Because $2>1$, the double-angle formula needs $+\pi$: $2\tan^{-1}2=\pi+\tan^{-1}\left(-\frac43\right)$. The dropped π is exactly the missing answer.

Corrected double angle, then cancel

1. Since $x=2>1$, $2\tan^{-1}2=\pi+\tan^{-1}\dfrac{2\cdot 2}{1-2^2}=\pi+\tan^{-1}\left(-\dfrac43\right)=\pi-\tan^{-1}\dfrac43$.
2. Add $\tan^{-1}\frac43$: $2\tan^{-1}2+\tan^{-1}\frac43=\pi-\tan^{-1}\frac43+\tan^{-1}\frac43=\boxed{\pi}$.

Complementary-angle picture

1. Let $\theta=\tan^{-1}2\in\left(\frac{\pi}{4},\frac{\pi}{2}\right)$, so $2\theta\in\left(\frac{\pi}{2},\pi\right)$ is obtuse with $\tan 2\theta=-\frac43$.
2. Also $\tan^{-1}\frac43=\pi-2\theta$ because $\tan(\pi-2\theta)=-\tan 2\theta=\frac43$ and $\pi-2\theta\in\left(0,\frac{\pi}{2}\right)$ matches the principal value.
3. Hence $2\theta+\tan^{-1}\frac43=2\theta+(\pi-2\theta)=\boxed{\pi}$.

Triple-arctan identity

1. Write $2\tan^{-1}2=\tan^{-1}2+\tan^{-1}2$ and use the theorem: for positive reals with $a+b+c=abc$, $\tan^{-1}a+\tan^{-1}b+\tan^{-1}c=\pi$.
2. Take $a=b=2$, $c=\frac43$: then $a+b+c=\frac{16}{3}$ and $abc=4\cdot\frac43=\frac{16}{3}$, so the hypothesis $a+b+c=abc$ holds.
3. Therefore $\tan^{-1}2+\tan^{-1}2+\tan^{-1}\frac43=\boxed{\pi}$.

INSIGHT

The identity $a+b+c=abc \Rightarrow \sum \tan^{-1} a_i = \pi$ (for positive reals) is the angle-sum of a triangle in disguise: it says three positive angles summing to π have tangents satisfying $\tan A + \tan B + \tan C = \tan A \tan B \tan C$. The classic instance $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 = \pi$ is the same theorem with $1+2+3 = 1 \cdot 2 \cdot 3$.

CHAPTER III

Telescoping Arctangents

Each term is a difference of two angles – the sum collapses.

№21 The Gauss Cascade

●●●●● L3

EVALUATE

$$\sum_{k=1}^{\infty} \tan^{-1}\left(\frac{1}{k^2+k+1}\right)$$

telescoping infinite-series difference-formula principal-value

ANSWER

$$\frac{\pi}{4}$$

THE TRAP

The seductive wrong answer is **0**, written by the student who says "each term $\to \tan^{-1} 0 = 0$, so the sum of infinitely many vanishing terms is 0." That confuses the *terms* tending to 0 (necessary for convergence) with the *partial sums*. The terms do shrink, but they accumulate to a definite positive limit. A second trap is writing $\tan^{-1}(\infty) = \frac{\pi}{2}$ as the final answer, forgetting to subtract the residual lower-endpoint constant $\tan^{-1} 1 = \frac{\pi}{4}$.

Telescoping via the difference formula

1. Notice $\dfrac{1}{k^2+k+1} = \dfrac{(k+1)-k}{1+k(k+1)}$, which is exactly the right-hand side of $\tan^{-1} a - \tan^{-1} b = \tan^{-1} \dfrac{a-b}{1+ab}$ with $a = k+1$, $b = k$.
2. Since $a, b > 0$ we have $1+ab > 0$, so the identity holds with **no $\pm\pi$ correction**: each term equals $\tan^{-1}(k+1) - \tan^{-1} k$.
3. The partial sum collapses: $S_n = \tan^{-1}(n+1) - \tan^{-1} 1$.
4. Let $n \to \infty$: $\tan^{-1}(n+1) \to \frac{\pi}{2}$, so $S = \frac{\pi}{2} - \frac{\pi}{4} = \boxed{\dfrac{\pi}{4}}$.

Reciprocal-difference form

1. Write each term as $\tan^{-1} \dfrac{1}{k} - \tan^{-1} \dfrac{1}{k+1}$, valid since $\dfrac{\frac{1}{k} - \frac{1}{k+1}}{1 + \frac{1}{k(k+1)}} = \dfrac{1}{k^2+k+1}$ and both arguments lie in $(0, 1)$.
2. The partial sum telescopes to $\tan^{-1} 1 - \tan^{-1} \dfrac{1}{n+1}$.
3. As $n \to \infty$, $\tan^{-1} \frac{1}{n+1} \to 0$, leaving $\tan^{-1} 1 = \boxed{\dfrac{\pi}{4}}$.

INSIGHT

This is the prototype telescoper of the whole chapter. The two solutions expose the same identity read in opposite directions: $\tan^{-1}(k+1) - \tan^{-1} k$ (climbing toward $\frac{\pi}{2}$) versus $\tan^{-1}\frac{1}{k} - \tan^{-1}\frac{1}{k+1}$ (descending toward 0). Both leave exactly one surviving boundary term – the residual constant – which is the entire answer. Generalising: $\sum_{k=m}^{\infty} \tan^{-1}\frac{1}{k^2+k+1} = \frac{\pi}{2} - \tan^{-1} m$.

№22 Odd Steps, Even Squares

●●●●● L3

PROVE THAT

$$\sum_{k=1}^{n} \tan^{-1}\left(\frac{1}{2k^2}\right) = \tan^{-1}(2n+1) - \frac{\pi}{4} \qquad (n \geq 1)$$

telescoping · finite-sum · induction-free · difference-formula

ANSWER

$$\tan^{-1}(2n+1) - \frac{\pi}{4}$$

THE TRAP

Students who spot the consecutive-odd structure often write the telescoped result as $\tan^{-1}(2n+1) - \tan^{-1}(1)$ and then **silently replace $\tan^{-1} 1$ by 1** (radian/value confusion), or drop the lower term entirely and claim the sum is $\tan^{-1}(2n+1)$. The lower endpoint is the odd number $2 \cdot 1 - 1 = 1$, and $\tan^{-1} 1 = \frac{\pi}{4}$ must remain. Forgetting it overstates the sum by $\frac{\pi}{4}$ for every n.

Split the step into consecutive odds

1. Observe $2k^2 = \dfrac{(2k+1)-(2k-1)}{1} \cdot \overset{?}{-}$; precisely, $(2k-1)(2k+1) = 4k^2 - 1$, so $1 + (2k-1)(2k+1) = 4k^2$ and $(2k+1) - (2k-1) = 2$.
2. Hence $\dfrac{1}{2k^2} = \dfrac{2}{4k^2} = \dfrac{(2k+1)-(2k-1)}{1+(2k-1)(2k+1)}$, and since both $2k \pm 1 > 0$ the difference formula applies with no correction: the k-th term is $\tan^{-1}(2k+1) - \tan^{-1}(2k-1)$.
3. Summing $k = 1, \ldots, n$ telescopes (each upper odd cancels the next lower odd): $S_n = \tan^{-1}(2n+1) - \tan^{-1}(1)$.
4. Since $\tan^{-1} 1 = \frac{\pi}{4}$, we obtain $\boxed{\tan^{-1}(2n+1) - \dfrac{\pi}{4}}$. ■

Reciprocal telescoping

1. Equivalently $\dfrac{1}{2k^2} = \dfrac{\frac{1}{2k-1} - \frac{1}{2k+1}}{1 + \frac{1}{(2k-1)(2k+1)}}$, so the term is $\tan^{-1}\dfrac{1}{2k-1} - \tan^{-1}\dfrac{1}{2k+1}$ with both arguments in $(0, 1]$.
2. This telescopes to $\tan^{-1} 1 - \tan^{-1}\dfrac{1}{2n+1}$.
3. Using $\tan^{-1}\frac{1}{2n+1} = \frac{\pi}{2} - \tan^{-1}(2n+1)$ gives $\frac{\pi}{4} - \frac{\pi}{2} + \tan^{-1}(2n+1) = \boxed{\tan^{-1}(2n+1) - \dfrac{\pi}{4}}$. ■

Numerical sanity check

1. For $n = 1$: LHS $= \tan^{-1}\frac{1}{2} = 0.46365$; RHS $= \tan^{-1} 3 - \frac{\pi}{4} = 1.24905 - 0.78540 = 0.46365$. Match.
2. For $n = 2$: LHS $= \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{8} = 0.46365 + 0.12435 = 0.58800$; RHS $= \tan^{-1} 5 - \frac{\pi}{4} = 1.37340 - 0.78540 = 0.58800$. Match, confirming the closed form $\boxed{\tan^{-1}(2n+1) - \dfrac{\pi}{4}}$.

INSIGHT

The step size here is 2 between *consecutive odd integers*, so the residual is $\tan^{-1} 1$, not $\tan^{-1} 0$. Letting $n \to \infty$ gives the clean corollary $\sum_{k=1}^{\infty} \tan^{-1}\frac{1}{2k^2} = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$. The general principle: a unit numerator over $2k^2$ secretly encodes a gap of two units between odd ladder-rungs.

№23 The Skip-a-Step Sum

●●●●● L4

EVALUATE

$$\sum_{k=1}^{\infty} \tan^{-1}\left(\frac{2}{k^2}\right)$$

telescoping · two-step · infinite-series · boundary-terms

ANSWER

$$\frac{3\pi}{4}$$

THE TRAP

The classic wrong answer is **$\frac{\pi}{4}$**. A student decomposes the term as $\tan^{-1}(k+1) - \tan^{-1}(k-1)$ but then telescopes it as if only *one* boundary term survives at each end, writing $S = \frac{\pi}{2} - \tan^{-1} 0 = \frac{\pi}{2}$, or worse mimics Problem 1's $\frac{\pi}{4}$. Because the index jumps by **two**, *two* terms survive at the top (both $\to \frac{\pi}{2}$) and *two* at the bottom ($\tan^{-1} 1$ and $\tan^{-1} 0$). Missing the second surviving top term loses a whole $\frac{\pi}{4}$.

Two-step telescope

1. Write $\frac{2}{k^2} = \frac{(k+1)-(k-1)}{1+(k-1)(k+1)}$, since $(k-1)(k+1) = k^2 - 1$ so $1 + (k-1)(k+1) = k^2$.
2. For $k \geq 2$ both $k \pm 1 \geq 1 > 0$, and for $k = 1$ the lower argument is 0; in all cases $1 + (k-1)(k+1) = k^2 > 0$, so no $\pm\pi$ correction: the term is $\tan^{-1}(k+1) - \tan^{-1}(k-1)$.
3. This is a **gap-2** telescope. The partial sum keeps the top two and bottom two: $S_n = \left[\tan^{-1}(n+1) + \tan^{-1}(n)\right] - \left[\tan^{-1}(1) + \tan^{-1}(0)\right]$.
4. As $n \to \infty$: $\tan^{-1}(n+1) + \tan^{-1}(n) \to \frac{\pi}{2} + \frac{\pi}{2} = \pi$, and the bottom is $\frac{\pi}{4} + 0 = \frac{\pi}{4}$. Thus $S = \pi - \frac{\pi}{4} = \boxed{\frac{3\pi}{4}}$.

Split into two single-step series

1. Group by parity is unnecessary; instead split the difference: $S_n = \sum_{k=1}^{n} \tan^{-1}(k+1) - \sum_{k=1}^{n} \tan^{-1}(k-1)$.
2. Re-index the second sum: $\sum_{k=1}^{n} \tan^{-1}(k-1) = \sum_{j=0}^{n-1} \tan^{-1}(j)$, and the first $= \sum_{j=2}^{n+1} \tan^{-1}(j)$.
3. Their difference leaves $\tan^{-1}(n+1) + \tan^{-1}(n) - \tan^{-1}(1) - \tan^{-1}(0)$, exactly as above.
4. Taking the limit gives $\pi - \frac{\pi}{4} = \boxed{\frac{3\pi}{4}}$.

INSIGHT

A numerator of 2 over k^2 encodes a *two-unit* jump in the arctan ladder, so the telescope has a width-2 overlap and leaves **two** surviving constants at each frontier. The signature $\frac{3\pi}{4} = \pi - \frac{\pi}{4}$ is the fingerprint of a gap-2 collapse. Generally a wider numerator that opens a gap of j in the arctan ladder produces $j \cdot \frac{\pi}{2}$ minus the lower surviving constants.

№24 A Parameter That Forgets Its Sign

●●●●● L4

EVALUATE

$$S(x) = \sum_{k=0}^{\infty} \tan^{-1}\left(\frac{x}{1+k(k+1)x^2}\right), \qquad x \in \mathbb{R}$$

telescoping · parametric · sign-trap · infinite-series

ANSWER

$$S(x)=\begin{cases}\dfrac{\pi}{2}, & x>0\\ 0, & x=0\\ -\dfrac{\pi}{2}, & x<0\end{cases}$$

THE TRAP

The dominant wrong answer is the flat **$S(x)=\dfrac{\pi}{2}$ for all $x\neq 0$**. Writing the term as $\tan^{-1}((k+1)x)-\tan^{-1}(kx)$ and telescoping gives $\lim_{n\to\infty}\tan^{-1}((n+1)x)-\tan^{-1}0$. The student plugs in $\frac{\pi}{2}$ without noticing that $\tan^{-1}((n+1)x)\to+\frac{\pi}{2}$ only when $x>0$; for $x<0$ the argument $\to-\infty$ and the limit is $-\frac{\pi}{2}$. The function S is **odd and discontinuous at 0**, not constant.

Telescoping with sign bookkeeping

1. Set $a_k=kx$. Then $a_{k+1}-a_k=x$ and $1+a_ka_{k+1}=1+k(k+1)x^2$, so $\dfrac{x}{1+k(k+1)x^2}=\dfrac{a_{k+1}-a_k}{1+a_ka_{k+1}}$.
2. Crucially $1+k(k+1)x^2\geq 1>0$ for every $k\geq 0$, so the difference formula needs **no $\pm\pi$ correction** regardless of x: the term equals $\tan^{-1}((k+1)x)-\tan^{-1}(kx)$.
3. Telescoping from $k=0$: $S_n=\tan^{-1}((n+1)x)-\tan^{-1}(0)=\tan^{-1}((n+1)x)$.
4. Now take the limit honoring the sign: if $x>0$, $\to\frac{\pi}{2}$; if $x<0$, $\to-\frac{\pi}{2}$; if $x=0$ every term is 0. Hence $\boxed{S(x)=\frac{\pi}{2}\operatorname{sgn}(x)}$.

Oddness shortcut

1. Each term is an odd function of x (numerator x, denominator even in x), so $S(-x)=-S(x)$; thus it suffices to find S for $x>0$ and reflect.
2. For $x>0$, all arguments are positive and the telescope gives $S_n=\tan^{-1}((n+1)x)\to\frac{\pi}{2}$.
3. By oddness $S(x)=-\frac{\pi}{2}$ for $x<0$, and $S(0)=0$, giving the three-case answer $\boxed{\frac{\pi}{2}\operatorname{sgn}(x)}$.

INSIGHT

The telescope itself is sign-blind because the denominator $1+k(k+1)x^2$ stays positive, so the per-term identity holds verbatim for all x. The subtlety lives entirely in the **single surviving boundary limit** $\tan^{-1}((n+1)x)$, whose endpoint depends on $\operatorname{sgn} x$. This is the cleanest illustration that a uniform algebraic collapse can still hide a discontinuity in the closed form.

№25 Squares Within Squares

●●●●● L4

EVALUATE

$$\sum_{k=1}^{\infty}\tan^{-1}\left(\frac{2k+1}{1+k^2(k+1)^2}\right)$$

telescoping · perfect-squares · infinite-series · difference-formula

ANSWER

$$\frac{\pi}{4}$$

THE TRAP

The tempting wrong answer is **$\dfrac{\pi}{2}$**. A student correctly finds the term equals $\tan^{-1}((k+1)^2)-\tan^{-1}(k^2)$ and sees the top boundary $\tan^{-1}((n+1)^2)\to\frac{\pi}{2}$, then reports $\frac{\pi}{2}$ — forgetting the lower endpoint is $\tan^{-1}(1^2)=\tan^{-1}1=\frac{\pi}{4}$, **not** $\tan^{-1}0$. The series starts at $k=1$, so the first surviving constant is $\tan^{-1}(1)$, costing $\frac{\pi}{4}$.

Recognising consecutive squares

1. The numerator factors a difference of squares: $(k+1)^2-k^2=2k+1$, and the denominator is $1+k^2(k+1)^2=1+(k^2)\big((k+1)^2\big)$.
2. Therefore $\dfrac{2k+1}{1+k^2(k+1)^2}=\dfrac{(k+1)^2-k^2}{1+k^2(k+1)^2}$, the difference-formula RHS with $a=(k+1)^2$, $b=k^2$. Both positive $\Rightarrow$ no correction.
3. So the term is $\tan^{-1}((k+1)^2)-\tan^{-1}(k^2)$ and the sum telescopes to $\tan^{-1}((n+1)^2)-\tan^{-1}(1)$.
4. Letting $n\to\infty$: $\frac{\pi}{2}-\frac{\pi}{4}=\boxed{\dfrac{\pi}{4}}$.

Substitution $u=k^2$

1. Define $f(u)=\tan^{-1}u$. The term is $f((k+1)^2)-f(k^2)$, a telescoping difference along the subsequence of perfect squares $1,4,9,16,\dots$
2. Partial sum $=f((n+1)^2)-f(1)$ because every interior square cancels.
3. Limit: $\tan^{-1}(\infty)-\tan^{-1}1=\frac{\pi}{2}-\frac{\pi}{4}=\boxed{\dfrac{\pi}{4}}$.

INSIGHT

Any strictly increasing positive sequence $a_k\to\infty$ generates a telescoper $\tan^{-1}\frac{a_{k+1}-a_k}{1+a_ka_{k+1}}$ summing to $\frac{\pi}{2}-\tan^{-1}a_1$. Here $a_k=k^2$ gives lower constant $\tan^{-1}1=\frac{\pi}{4}$. Choosing $a_k=k^3$, 2^k, $k!$ etc. all give $\frac{\pi}{2}-\tan^{-1}a_1$ – the device is universal; only the first term's value changes the answer.

№26 Start at Two, Land on a Third

L4

EVALUATE

$$\sum_{k=2}^{\infty}\tan^{-1}\left(\frac{2k}{k^4+k^2+2}\right)$$

telescoping · shifted-index · biquadratic · factorisation

ANSWER

$$\tan^{-1}\left(\frac{1}{3}\right)$$

THE TRAP

The seductive errors are twofold. First, a student factors $k^4+k^2+1=(k^2-k+1)(k^2+k+1)$ on autopilot and mis-handles the stray $+1$, getting the wrong term. Second — even after the correct decomposition $\tan^{-1}\frac{1}{k^2-k+1}-\tan^{-1}\frac{1}{k^2+k+1}$ — they reflexively start the telescope at $k=1$ and report **$\dfrac{\pi}{4}$** (the $k\geq 1$ value). But the sum here **starts at $k=2$**, so the surviving lower constant is $\tan^{-1}\frac{1}{2^2-2+1}=\tan^{-1}\frac{1}{3}$, not $\tan^{-1}\frac{1}{1}=\frac{\pi}{4}$.

Reciprocal telescoping with a careful denominator

1. Let $b_k=k^2+k+1$. Then $b_{k-1}=(k-1)^2+(k-1)+1=k^2-k+1$, so $\tan^{-1}\dfrac{1}{b_{k-1}}-\tan^{-1}\dfrac{1}{b_k}=\tan^{-1}\dfrac{b_k-b_{k-1}}{1+b_kb_{k-1}}$.
2. Compute $b_k-b_{k-1}=2k$ and $b_kb_{k-1}=(k^2+1)^2-k^2=k^4+k^2+1$, so $1+b_kb_{k-1}=k^4+k^2+2$. Both reciprocals lie in $(0,1)$, no correction.
3. Hence the term equals $\tan^{-1}\dfrac{1}{k^2-k+1}-\tan^{-1}\dfrac{1}{k^2+k+1}$, a clean telescope.
4. Summing $k=2$ to n: $S_n=\tan^{-1}\dfrac{1}{b_1}-\tan^{-1}\dfrac{1}{b_n}=\tan^{-1}\dfrac{1}{3}-\tan^{-1}\dfrac{1}{n^2+n+1}$ (note $b_1=k^2-k+1$ at $k=2$ equals 3).

5. Let $n \to \infty$: the tail vanishes, giving $\boxed{\tan^{-1}\left(\frac{1}{3}\right)}$.

Relate to the full series

1. The same term for $k \geq 1$ sums to $\tan^{-1} \frac{1}{1^2-1+1} - 0 = \tan^{-1} 1 = \frac{\pi}{4}$ (the standard $k^4 + k^2 + 2$ telescope).
2. Our sum omits the $k = 1$ term, which is $\tan^{-1} \frac{1}{1} - \tan^{-1} \frac{1}{3} = \frac{\pi}{4} - \tan^{-1} \frac{1}{3}$.
3. Therefore the answer is $\frac{\pi}{4} - (\frac{\pi}{4} - \tan^{-1} \frac{1}{3}) = \boxed{\tan^{-1}\left(\frac{1}{3}\right)}$.

INSIGHT

The factorisation $k^4 + k^2 + 1 = (k^2 - k + 1)(k^2 + k + 1)$ tempts you, but the actual denominator is $k^4 + k^2 + \mathbf{2}$, i.e. $1 + (k^2 - k + 1)(k^2 + k + 1)$ – the $\boxed{+1}$ from the difference formula is built in. The lesson: when a series starts at $k = m$, the surviving constant is $\tan^{-1} \frac{1}{(m-1)^2+(m-1)+1} = \tan^{-1} \frac{1}{m^2-m+1}$, *not* automatically $\frac{\pi}{4}$.

№27 The Ladder to a Right Angle

●●●●● L4

EVALUATE

$$\sum_{k=1}^{\infty} \tan^{-1}\left(\frac{1}{k^2 - k + 1}\right)$$

telescoping · infinite-series · half-pi · difference-formula

ANSWER

$$\frac{\pi}{2}$$

THE TRAP

The tempting wrong answer is $**\frac{\pi}{4}**$, by false analogy with $\sum \tan^{-1} \frac{1}{k^2+k+1} = \frac{\pi}{4}$ (Problem 1). The signs differ: here the term is $\tan^{-1} k - \tan^{-1}(k-1)$, so the lower surviving constant is $\tan^{-1}(0) = 0$, **not** $\tan^{-1} 1$. The whole climb from $\tan^{-1} 0$ up to $\frac{\pi}{2}$ is retained, giving $\frac{\pi}{2}$. Mistaking the lower endpoint as $\frac{\pi}{4}$ halves... actually subtracts $\frac{\pi}{4}$ erroneously.

Telescoping anchored at zero

1. Write $\dfrac{1}{k^2 - k + 1} = \dfrac{k - (k-1)}{1 + k(k-1)}$, the difference-formula RHS with $a = k$, $b = k - 1$.
2. For $k \geq 1$ we have $a \geq 1 > 0$ and $b \geq 0$, so $1 + ab \geq 1 > 0$: no correction. The term equals $\tan^{-1} k - \tan^{-1}(k-1)$.
3. Telescoping: $S_n = \tan^{-1} n - \tan^{-1} 0 = \tan^{-1} n$.
4. As $n \to \infty$, $\tan^{-1} n \to \boxed{\frac{\pi}{2}}$.

Pairing with the companion series

1. Let $A = \sum_{k\geq1} \tan^{-1} \frac{1}{k^2-k+1} = \tan^{-1} n \to \frac{\pi}{2}$ and recall $B = \sum_{k\geq1} \tan^{-1} \frac{1}{k^2+k+1} = \frac{\pi}{4}$.
2. Note $k^2 - k + 1 = (k-1)^2 + (k-1) + 1$, so A is just B re-indexed to include the $k = 0$ block: $A = \tan^{-1} \frac{1}{0+0+1} + B = \frac{\pi}{4} + \frac{\pi}{4}$.
3. Both routes confirm $A = \boxed{\frac{\pi}{2}}$.

INSIGHT

The pair $k^2 \pm k + 1$ are the two faces of the same telescoper shifted by one index. The $+$ version lands at $\frac{\pi}{4}$ (lower constant $\tan^{-1} 1$); the $-$ version lands at $\frac{\pi}{2}$ (lower constant $\tan^{-1} 0$). Memorising both as a couplet inoculates against the single most common arctan-series slip: grabbing the wrong boundary term.

№28 Halving the Odd Reciprocals

●●●●● L4

EVALUATE

$$\sum_{k=1}^{\infty} \tan^{-1}\left(\frac{4}{4k^2+3}\right)$$

telescoping · infinite-series · odd-denominators · closed-form

ANSWER

$$\tan^{-1}(2)$$

THE TRAP

The natural false move is to force this into a $k^2 + k + 1$ shape and declare the answer $\frac{\pi}{4}$ or $\frac{\pi}{2}$. The correct decomposition is $\tan^{-1}\frac{2}{2k-1} - \tan^{-1}\frac{2}{2k+1}$, telescoping to $\tan^{-1}\frac{2}{1} - 0 = \tan^{-1} 2$. A student who instead writes the top argument as $\frac{2}{2n+1}$ but then **rounds the surviving lower term $\tan^{-1}\frac{2}{1}$ to $\frac{\pi}{2}$** (treating $\tan^{-1} 2 \approx \tan^{-1} \infty$) reports $\frac{\pi}{2}$, overshooting by $\frac{\pi}{2} - \tan^{-1} 2 = \tan^{-1}\frac{1}{2}$.

Reciprocal odd telescoping

1. Set $c_k = \dfrac{2}{2k-1}$. Then $c_k - c_{k+1} = \dfrac{2}{2k-1} - \dfrac{2}{2k+1} = \dfrac{2 \cdot 2}{(2k-1)(2k+1)} = \dfrac{4}{4k^2-1}$.
2. Also $1 + c_k c_{k+1} = 1 + \dfrac{4}{4k^2-1} = \dfrac{4k^2+3}{4k^2-1}$, so $\dfrac{c_k - c_{k+1}}{1 + c_k c_{k+1}} = \dfrac{4}{4k^2+3}$.
3. Since $c_k, c_{k+1} > 0$, the term is $\tan^{-1} c_k - \tan^{-1} c_{k+1} = \tan^{-1}\dfrac{2}{2k-1} - \tan^{-1}\dfrac{2}{2k+1}$, no correction.
4. Telescoping: $S_n = \tan^{-1}\dfrac{2}{1} - \tan^{-1}\dfrac{2}{2n+1} \to \tan^{-1} 2 - 0 = \boxed{\tan^{-1} 2}$.

Direct verification of the first surviving term

1. The telescope leaves only the very first $c_1 = \frac{2}{2 \cdot 1 - 1} = 2$ and a vanishing tail $\frac{2}{2n+1} \to 0$.
2. Hence $S = \tan^{-1} 2$. Numerically $\tan^{-1} 2 = 1.10715$; partial sum to $k = 4$ is $\tan^{-1}\frac{4}{7} + \tan^{-1}\frac{4}{19} + \tan^{-1}\frac{4}{39} + \tan^{-1}\frac{4}{67} = 0.51915 + 0.20750 + 0.10221 + 0.05963 = 0.88848$, and since $S_4 = \tan^{-1} 2 - \tan^{-1}\frac{2}{9}$, adding the surviving tail $\tan^{-1}\frac{2}{9} = 0.21867$ gives exactly 1.10715. Confirms $\boxed{\tan^{-1} 2}$.

INSIGHT

Numerator 4 over $4k^2 + 3$ is the disguise of a gap-1 telescope among the *scaled odd reciprocals* $\frac{2}{2k-1}$. The answer $\tan^{-1} 2$ is irrational-looking but exact and clean. General template: $\sum_{k \geq 1} \tan^{-1}\frac{2c}{c^2 + (2k-1)(2k+1)} = \tan^{-1}\frac{c}{1}$ with $c = 2$ here. Don't force every arctan series into a π-multiple – many collapse to a single $\tan^{-1}$ of a rational.

№29 An Irrational Step Size

●●●●● L5

EVALUATE

$$\sum_{k=1}^{\infty} \tan^{-1}\left(\frac{\sqrt{2}}{2k^2+2k+1}\right)$$

telescoping · surd · infinite-series · co-function · principal-value

ANSWER

$$\tan^{-1}\left(\frac{1}{\sqrt{2}}\right) = \cot^{-1}\left(\sqrt{2}\right)$$

THE TRAP

The slick wrong answer is **$\frac{\pi}{2} - \tan^{-1}\sqrt{2}$ reported as 0**, by a student who writes the term as $\tan^{-1}((k+1)\sqrt{2}) - \tan^{-1}(k\sqrt{2})$, sees the top $\to \frac{\pi}{2}$, but then incorrectly subtracts the *top* limit again as the lower term, getting $\frac{\pi}{2} - \frac{\pi}{2} = 0$. The genuine lower constant is $\tan^{-1}(1 \cdot \sqrt{2}) = \tan^{-1}\sqrt{2}$, so $S = \frac{\pi}{2} - \tan^{-1}\sqrt{2}$. The further trap is leaving the answer in the form $\frac{\pi}{2} - \tan^{-1}\sqrt{2}$ and not simplifying it to the clean co-function $\cot^{-1}\sqrt{2} = \tan^{-1}\frac{1}{\sqrt{2}}$.

Irrational-step telescope

1. Let $a_k = k\sqrt{2}$. Then $a_{k+1} - a_k = \sqrt{2}$ and $1 + a_k a_{k+1} = 1 + 2k(k+1) = 2k^2 + 2k + 1$.
2. So $\dfrac{\sqrt{2}}{2k^2+2k+1} = \dfrac{a_{k+1} - a_k}{1 + a_k a_{k+1}}$; with $a_k > 0$ the formula needs no correction, and the term is $\tan^{-1}((k+1)\sqrt{2}) - \tan^{-1}(k\sqrt{2})$.
3. Telescoping: $S_n = \tan^{-1}((n+1)\sqrt{2}) - \tan^{-1}(\sqrt{2}) \to \frac{\pi}{2} - \tan^{-1}\sqrt{2}$.
4. Convert via the co-function identity $\frac{\pi}{2} - \tan^{-1} t = \cot^{-1} t = \tan^{-1}\frac{1}{t}$ (valid for $t > 0$): $S = \cot^{-1}\sqrt{2} = \boxed{\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)}$.

Right-triangle reading of the limit

1. After telescoping, $S = \frac{\pi}{2} - \tan^{-1}\sqrt{2}$. Interpret $\tan^{-1}\sqrt{2}$ as the angle in a right triangle with opposite $\sqrt{2}$, adjacent 1.
2. Its complement has opposite 1, adjacent $\sqrt{2}$, i.e. angle $\tan^{-1}\frac{1}{\sqrt{2}}$. Since both angles are acute and sum to $\frac{\pi}{2}$, $S = \tan^{-1}\frac{1}{\sqrt{2}}$.
3. Numerical check: $\frac{\pi}{2} - \tan^{-1}\sqrt{2} = 1.5708 - 0.9553 = 0.6155$ and $\tan^{-1}\frac{1}{\sqrt{2}} = \tan^{-1} 0.7071 = 0.6155$. Hence $\boxed{\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)}$.

INSIGHT

An irrational step $a_k = k\sqrt{2}$ telescopes just as cleanly as an integer one – the algebra only needs $1 + a_k a_{k+1} > 0$. The lesson at difficulty 5 is the *finishing move*: a sum of $\tan^{-1}$ of rationals/surds can converge to $\frac{\pi}{2} - \tan^{-1}\sqrt{2}$, which is **not** in lowest terms until you apply $\cot^{-1} t = \tan^{-1}\frac{1}{t}$ (legitimate only for $t > 0$ on the principal branch). General: $a_k = ck$ gives $\sum = \frac{\pi}{2} - \tan^{-1} c = \tan^{-1}\frac{1}{c}$.

№30 Two Constants Escape

●●●●● L5

EVALUATE

$$\sum_{k=1}^{\infty} \tan^{-1}\left(\frac{4}{4k^2 - 3}\right)$$

telescoping · two-step · boundary-terms · infinite-series · branch-care

ANSWER

$$\pi - \tan^{-1}(2)$$

THE TRAP

Two distinct wrong answers compete. (1) **$\frac{\pi}{2}$**: the student finds the term $= \tan^{-1}(2k+2) - \tan^{-1}(2k-2)$, sees one top boundary $\to \frac{\pi}{2}$, and forgets this is a **gap-2** telescope where *two* top constants survive. (2) **$\frac{\pi}{2}$ again from a sign slip**: at $k=1$, $4k^2-3=1>0$, but a careless student who writes the term as $\tan^{-1}\frac{(2k+2)-(2k-2)}{1+(2k+2)(2k-2)}$ may panic when $(2k+2)(2k-2)=4k^2-4$ makes $1+(\cdot)=4k^2-3$ small for $k=1$; it is still positive ($=1$), so no $\pm\pi$ correction is needed, yet some add a spurious π. The truth: two top terms ($\to \pi$) minus the two bottom terms $\tan^{-1}2+\tan^{-1}0$.

Gap-2 telescope with two surviving constants

1. Set $a_k = 2k$. Then $a_{k+1} - a_{k-1} = (2k+2)-(2k-2) = 4$ and $1 + a_{k+1}a_{k-1} = 1 + (2k+2)(2k-2) = 1 + 4k^2 - 4 = 4k^2 - 3$.
2. For all $k \geq 1$, $4k^2 - 3 \geq 1 > 0$, so the difference formula holds with **no correction**: the term is $\tan^{-1}(2k+2) - \tan^{-1}(2k-2)$.
3. This is a width-2 telescope. The partial sum keeps the **top two** and **bottom two** terms: $S_n = \left[\tan^{-1}(2n+2) + \tan^{-1}(2n)\right] - \left[\tan^{-1}(2) + \tan^{-1}(0)\right]$.
4. As $n \to \infty$, the two top terms each $\to \frac{\pi}{2}$, summing to π; the bottom is $\tan^{-1}2 + 0$. Therefore $S = \pi - \tan^{-1}2 = \boxed{\pi - \tan^{-1}(2)}$.

Split into two single-step series

1. $S_n = \sum_{k=1}^{n}\tan^{-1}(2k+2) - \sum_{k=1}^{n}\tan^{-1}(2k-2)$. Re-index: first $= \sum_{j=2}^{n+1}\tan^{-1}(2j)$, second $= \sum_{j=0}^{n-1}\tan^{-1}(2j)$.
2. Subtracting, the overlap $j = 2, \ldots, n-1$ cancels, leaving $\tan^{-1}(2(n+1)) + \tan^{-1}(2n) - \tan^{-1}(2\cdot 1) - \tan^{-1}(2\cdot 0)$.
3. Limit $= \frac{\pi}{2} + \frac{\pi}{2} - \tan^{-1}2 - 0 = \boxed{\pi - \tan^{-1}(2)}$. Numerically $\pi - 1.10715 = 2.03444$, matching the partial sums.

INSIGHT

This is the chapter's capstone: a gap-2 telescope whose answer **exceeds $\frac{\pi}{2}$** precisely because *two* upper constants escape to $\frac{\pi}{2}$ each. The signature $\pi - \tan^{-1}2$ packages both subtleties – counting the right number of surviving boundary terms (two, not one) and refusing to round $\tan^{-1}2$ to $\frac{\pi}{2}$. A width-d telescope of $a_k \to \infty$ always leaves d top terms (total $d \cdot \frac{\pi}{2}$) minus the d initial constants.

CHAPTER IV

Identities & Collapse

Most nested inverse-trig expressions are secretly a straight line.

№31 The Sign Trap at Three-Quarters

●●●●● L3

EVALUATE

$$\text{Evaluate } \sin^{-1}\left(2x\sqrt{1-x^2}\right) \text{ at } x = \sin\frac{5\pi}{12}.$$

sine-double-angle · branch · principal-value · collapse

ANSWER

$$\frac{\pi}{6}$$

THE TRAP

The reflex is $\sin^{-1}(2x\sqrt{1-x^2}) = 2\sin^{-1}x = 2\cdot\frac{5\pi}{12} = \frac{5\pi}{6}$. But $\frac{5\pi}{6} > \frac{\pi}{2}$ lies **outside** the range of $\sin^{-1}$, so it cannot be the value. Here $x = \sin 75^\circ \approx 0.966 > \frac{1}{\sqrt{2}}$, so the correct branch is $\pi - 2\sin^{-1}x$.

Branch reduction by quadrant

1. Write $x = \sin\theta$ with $\theta = \sin^{-1}x = \frac{5\pi}{12}$ (since $\frac{5\pi}{12} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$). Then $2x\sqrt{1-x^2} = 2\sin\theta\cos\theta = \sin 2\theta = \sin\frac{5\pi}{6}$.
2. The double angle $2\theta = \frac{5\pi}{6} \notin [-\frac{\pi}{2}, \frac{\pi}{2}]$, so $\sin^{-1}(\sin 2\theta) = \pi - 2\theta$.
3. Hence the value is $\pi - \frac{5\pi}{6} = \boxed{\frac{\pi}{6}}$.

Direct angle computation

1. $2x\sqrt{1-x^2} = \sin 150^\circ = \frac{1}{2}$, so we need $\sin^{-1}\frac{1}{2}$.
2. Since $x > \frac{1}{\sqrt{2}}$ the value is in the first quadrant and small: $\sin^{-1}\frac{1}{2} = \frac{\pi}{6}$.
3. Thus the answer is $\boxed{\frac{\pi}{6}}$, not $\frac{5\pi}{6}$.

INSIGHT

The full identity is $\sin^{-1}(2x\sqrt{1-x^2}) = 2\sin^{-1}x$ only on $|x| \le \frac{1}{\sqrt{2}}$; for $\frac{1}{\sqrt{2}} < x \le 1$ it is $\pi - 2\sin^{-1}x$, and for $-1 \le x < -\frac{1}{\sqrt{2}}$ it is $-\pi - 2\sin^{-1}x$. The collapse point $|x| = \frac{1}{\sqrt{2}}$ is exactly where $2\sin^{-1}x$ exits $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

№32 Piecewise or Bust

●●●●● L4

SIMPLIFY

$$\text{Simplify } f(x) = \sin^{-1}\left(2x\sqrt{1-x^2}\right) \text{ on } [-1,1], \text{ giving the full piecewise form with domains.}$$

sine-double-angle · piecewise · domain · collapse

ANSWER

$$f(x)=\begin{cases}-\pi-2\sin^{-1}x, & -1\le x<-\frac{1}{\sqrt2}\\ 2\sin^{-1}x, & -\frac{1}{\sqrt2}\le x\le\frac{1}{\sqrt2}\\ \pi-2\sin^{-1}x, & \frac{1}{\sqrt2}<x\le 1\end{cases}$$

THE TRAP

Writing $f(x)=2\sin^{-1}x$ for all $x\in[-1,1]$. This forces f to range over $[-\pi,\pi]$, but $\sin^{-1}$ of anything lives in $[-\frac{\pi}{2},\frac{\pi}{2}]$. The single line is correct only where $2\sin^{-1}x$ already lies in $[-\frac{\pi}{2},\frac{\pi}{2}]$, i.e. $|x|\le\frac{1}{\sqrt2}$.

Substitution and arcsin-of-sine reduction

1. Put $\theta=\sin^{-1}x\in[-\frac{\pi}{2},\frac{\pi}{2}]$, so $2x\sqrt{1-x^2}=\sin 2\theta$ and $f=\sin^{-1}(\sin 2\theta)$ with $2\theta\in[-\pi,\pi]$.
2. Use $\sin^{-1}(\sin t)=t$ on $[-\frac{\pi}{2},\frac{\pi}{2}]$, $=\pi-t$ on $[\frac{\pi}{2},\frac{3\pi}{2}]$ (here $t\in(\frac{\pi}{2},\pi]$ gives $\pi-t$), and $=-\pi-t$ on $[-\frac{3\pi}{2},-\frac{\pi}{2}]$ (here $t\in[-\pi,-\frac{\pi}{2})$).
3. Translate 2θ ranges back via $\theta=\sin^{-1}x$: $2\theta\in[-\frac{\pi}{2},\frac{\pi}{2}]\iff|x|\le\frac{1}{\sqrt2}$, giving the three pieces of the $\square$ form above.

Continuity / endpoint check

1. The two collapse points are $x=\pm\frac{1}{\sqrt2}$ where $\sin^{-1}x=\pm\frac{\pi}{4}$, so $2\sin^{-1}x=\pm\frac{\pi}{2}$ touches the range boundary.
2. On $(\frac{1}{\sqrt2},1]$, f decreases from $\frac{\pi}{2}$ to 0 (at $x=1$, $2x\sqrt{1-x^2}=0$), forcing the reflected branch $\pi-2\sin^{-1}x$; by oddness of the argument f is odd, giving $-\pi-2\sin^{-1}x$ on $[-1,-\frac{1}{\sqrt2})$.
3. Matching values at $x=\pm\frac{1}{\sqrt2}$ confirms continuity, yielding the boxed piecewise answer.

INSIGHT

This is the canonical 'collapse' object: a degree-2 nested arcsine that is *globally* a sawtooth, not a line. The graph is a tent peaking at $\frac{\pi}{2}$ over $x=\frac{1}{\sqrt2}$ and dipping to $-\frac{\pi}{2}$ over $x=-\frac{1}{\sqrt2}$.

№33 Triple Angle Overshoot

L3

EVALUATE

$$\text{Evaluate }\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)-3\tan^{-1}x\text{ at }x=2.$$

tangent-triple-angle · branch · collapse · principal-value

ANSWER

$$-\pi$$

THE TRAP

Many declare the difference is 0, reasoning $\tan^{-1}\frac{3x-x^3}{1-3x^2}=3\tan^{-1}x$. But $3\tan^{-1}2\approx 3.32$ exceeds $\frac{\pi}{2}$, so it cannot equal a principal $\tan^{-1}$ value. For $x>\frac{1}{\sqrt3}$ the identity carries a $+\pi$ correction: $\tan^{-1}(\cdot)=3\tan^{-1}x-\pi$.

Branch-corrected triple angle

1. Let $\theta=\tan^{-1}x$; then $\frac{3x-x^3}{1-3x^2}=\tan 3\theta$, so the expression is $\tan^{-1}(\tan 3\theta)-3\theta$.
2. For $x=2$, $\theta=\tan^{-1}2\approx 1.107$, so $3\theta\approx 3.32\in(\frac{\pi}{2},\frac{3\pi}{2})$, where $\tan^{-1}(\tan 3\theta)=3\theta-\pi$.
3. The difference is $(3\theta-\pi)-3\theta=\boxed{-\pi}$.

Direct numerical collapse

1. Compute the argument: $\frac{3\cdot 2-8}{1-12} = \frac{-2}{-11} = \frac{2}{11}$, so the first term is $\tan^{-1}\frac{2}{11} \approx 0.1799$.
2. $3\tan^{-1} 2 \approx 3.3214$; their difference is $0.1799 - 3.3214 = -3.1416 \approx -\pi$.
3. Hence the exact value is $\boxed{-\pi}$.

INSIGHT

$\tan^{-1}\frac{3x-x^3}{1-3x^2} = 3\tan^{-1} x$ holds only for $|x| < \frac{1}{\sqrt{3}}$; it equals $3\tan^{-1} x - \pi$ for $x > \frac{1}{\sqrt{3}}$ and $3\tan^{-1} x + \pi$ for $x < -\frac{1}{\sqrt{3}}$. The jumps occur at $x = \pm\frac{1}{\sqrt{3}}$ where $\tan 3\theta$ blows up ($3\theta = \pm\frac{\pi}{2}$).

№34 Where Three Times Holds

●●●●● L4

SOLVE FOR X

$$\text{Find all } x \text{ for which } \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right) = 3\tan^{-1} x.$$

tangent-triple-angle | equation | domain | collapse

ANSWER

$$-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$$

THE TRAP

Concluding 'all real x except $x = \pm\frac{1}{\sqrt{3}}$' (where the argument is undefined). The equation is an *identity only on the central interval*; outside it the two sides differ by exactly $\pm\pi$, so they are never equal there — excluding the poles is far too generous.

Range argument

1. The left side is a principal $\tan^{-1}$, hence lies in $(-\frac{\pi}{2}, \frac{\pi}{2})$.
2. The right side $3\tan^{-1} x$ lies in $(-\frac{\pi}{2}, \frac{\pi}{2})$ exactly when $\tan^{-1} x \in (-\frac{\pi}{6}, \frac{\pi}{6})$, i.e. $|x| < \tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$.
3. On that interval $3\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ so $\tan^{-1}(\tan 3\theta) = 3\theta$ and equality holds; outside, the sides differ by π. Answer: $\boxed{-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}}$.

Branch-difference equation

1. Set $D(x) = \tan^{-1}\frac{3x-x^3}{1-3x^2} - 3\tan^{-1} x$. By the triple-angle reduction, $D = 0,\ -\pi,\ +\pi$ on the three regions $|x| < \frac{1}{\sqrt{3}}$, $x > \frac{1}{\sqrt{3}}$, $x < -\frac{1}{\sqrt{3}}$ respectively.
2. $D = 0$ only on the middle region; the constants $\pm\pi$ are never 0.
3. Therefore the solution set is the open interval $\boxed{\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)}$.

INSIGHT

A clean illustration that an 'identity' is really a conditional equation. The endpoints $\pm\frac{1}{\sqrt{3}}$ are excluded both because the argument is undefined *and* because the limit jumps by π there.

№35 Cosine Knows No Sign

●●●●● L3

SIMPLIFY

$$\text{Simplify } g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) \ \text{ for all real } x, \text{ giving the piecewise form.}$$

cosine-double-angle | arctangent | absolute-value | collapse

ANSWER

$$g(x) = \begin{cases} 2\tan^{-1} x, & x \ge 0 \\ -2\tan^{-1} x, & x < 0 \end{cases} = 2\tan^{-1}|x|$$

THE TRAP

Writing $g(x) = 2\tan^{-1} x$ for all x. Since $\cos^{-1}$ outputs values in $[0, \pi]$, $g \ge 0$ always; but $2\tan^{-1} x < 0$ for $x < 0$. The clean single line holds only for $x \ge 0$.

Substitution

1. Let $x = \tan\theta$ with $\theta = \tan^{-1} x \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Then $\frac{1-x^2}{1+x^2} = \cos 2\theta$, so $g = \cos^{-1}(\cos 2\theta)$ with $2\theta \in (-\pi, \pi)$.
2. Use $\cos^{-1}(\cos t) = |t|$ on $(-\pi, \pi)$: thus $g = |2\theta| = 2|\tan^{-1} x|$.
3. Since $\tan^{-1} x$ has the sign of x, $g = 2\tan^{-1} x$ for $x \ge 0$ and $-2\tan^{-1} x$ for $x < 0$, i.e. $\boxed{2\tan^{-1}|x|}$.

Evenness observation

1. The argument $\frac{1-x^2}{1+x^2}$ is an even function of x, so g is even: $g(-x) = g(x)$.
2. For $x \ge 0$ the standard reduction gives $g(x) = 2\tan^{-1} x$ (both sides increase from 0 at $x = 0$ to π as $x \to \infty$).
3. Evenness then forces $g(x) = 2\tan^{-1}|x|$ on all of $\mathbb{R}$, matching the boxed form.

INSIGHT

Because cos is even, the nested $\cos^{-1}$ collapses to $2\tan^{-1}|x|$ – a V-shape, not a line. Contrast with the $\sin^{-1}/(2x/(1+x^2))$ object, which is odd. The parity of the outer inverse function dictates the global shape.

№36 Two Faces of 2arctan

●●●●● L4

PROVE THAT

$$\text{Prove that for } x > 1, \quad 2\tan^{-1} x = \pi - \sin^{-1}\left(\frac{2x}{1+x^2}\right).$$

arctangent | sine-form | branch | collapse

ANSWER

$$\text{Proved: for } x > 1,\ 2\tan^{-1} x = \pi - \sin^{-1}\left(\frac{2x}{1+x^2}\right).$$

THE TRAP

Stating $2\tan^{-1} x = \sin^{-1}\frac{2x}{1+x^2}$ without the $\pi - (\cdot)$ correction. That clean form holds only for $|x| \le 1$; for $x > 1$, $2\tan^{-1} x > \frac{\pi}{2}$ leaves the range of $\sin^{-1}$, so a reflection through π is mandatory.

Quadrant of the double angle

1. Let $\theta = \tan^{-1} x \in (\frac{\pi}{4}, \frac{\pi}{2})$ since $x > 1$. Then $\frac{2x}{1+x^2} = \sin 2\theta$, and $2\theta \in (\frac{\pi}{2}, \pi)$.
2. On $(\frac{\pi}{2}, \pi)$ we have $\sin^{-1}(\sin 2\theta) = \pi - 2\theta$.
3. Therefore $\sin^{-1}\frac{2x}{1+x^2} = \pi - 2\tan^{-1} x$, i.e. $2\tan^{-1} x = \pi - \sin^{-1}\frac{2x}{1+x^2}$. ■

Differentiate and pin a constant

1. Let $L(x) = 2\tan^{-1} x + \sin^{-1} \frac{2x}{1+x^2}$. For $x > 1$, $\frac{d}{dx} \sin^{-1} \frac{2x}{1+x^2} = -\frac{2}{1+x^2}$ (the negative branch, since $\frac{2x}{1+x^2}$ is decreasing there), which cancels $\frac{d}{dx} 2\tan^{-1} x = \frac{2}{1+x^2}$.
2. So $L'(x) = 0$ on $(1, \infty)$, i.e. L is constant; evaluate the limit as $x \to \infty$: $2\tan^{-1} x \to \pi$ and $\sin^{-1} \frac{2x}{1+x^2} \to 0$, giving $L = \pi$.
3. Thus $2\tan^{-1} x = \pi - \sin^{-1} \frac{2x}{1+x^2}$ for all $x > 1$. ■

INSIGHT

The three-piece law is $2\tan^{-1} x = \sin^{-1} \frac{2x}{1+x^2}$ for $|x| \le 1$, $= \pi - \sin^{-1} \frac{2x}{1+x^2}$ for $x > 1$, and $= -\pi - \sin^{-1} \frac{2x}{1+x^2}$ for $x < -1$. The constant-derivative trick is the cleanest way to nail which constant appears on each piece.

№37 The Slope That Flips

●●●●● L4

DIFFERENTIATE

$$\text{For } f(x) = \sin^{-1}\left(2x\sqrt{1-x^2}\right), \text{ find } f'(x) \text{ at } x = \frac{\sqrt{3}}{2}.$$

sine-double-angle · derivative · branch · collapse

ANSWER

$$-4$$

THE TRAP

Differentiating the line $2\sin^{-1} x$ to get $f' = \frac{2}{\sqrt{1-x^2}} = \frac{2}{1/2} = +4$. At $x = \frac{\sqrt{3}}{2} > \frac{1}{\sqrt{2}}$ the active branch is $\pi - 2\sin^{-1} x$, whose derivative is $-\frac{2}{\sqrt{1-x^2}} = -4$. The sign is the whole point.

Differentiate the correct piece

1. Since $\frac{1}{\sqrt{2}} < \frac{\sqrt{3}}{2} \le 1$, the relevant branch is $f(x) = \pi - 2\sin^{-1} x$.
2. Then $f'(x) = -\dfrac{2}{\sqrt{1-x^2}}$; at $x = \frac{\sqrt{3}}{2}$, $\sqrt{1-x^2} = \sqrt{1 - \frac{3}{4}} = \frac{1}{2}$.
3. So $f' = -\frac{2}{1/2} = \boxed{-4}$.

Chain rule with signed simplification

1. Directly, $f'(x) = \dfrac{1}{\sqrt{1-(2x\sqrt{1-x^2})^2}} \cdot \dfrac{d}{dx}\left(2x\sqrt{1-x^2}\right) = \dfrac{2(1-2x^2)}{|1-2x^2|\sqrt{1-x^2}}$.
2. For $x = \frac{\sqrt{3}}{2}$, $1 - 2x^2 = 1 - \frac{3}{2} = -\frac{1}{2} < 0$, so $\frac{1-2x^2}{|1-2x^2|} = -1$.
3. Hence $f' = \dfrac{-2}{\sqrt{1-x^2}} = \dfrac{-2}{1/2} = \boxed{-4}$.

INSIGHT

The factor $\frac{1-2x^2}{|1-2x^2|} = \operatorname{sgn}(1-2x^2)$ encodes the branch switch at $|x| = \frac{1}{\sqrt{2}}$. The derivative is $+\frac{2}{\sqrt{1-x^2}}$ on the central interval and $-\frac{2}{\sqrt{1-x^2}}$ on the two outer pieces – the tent's two slopes.

№38 Counting the Honest x

●●●●● L3

FIND THE NUMBER OF SOLUTIONS

$$\text{Find the number of integers } x \in [-5, 5] \text{ satisfying } \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = 2\tan^{-1} x.$$

cosine-double-angle · arctangent · integer-count · collapse

ANSWER

$$6$$

THE TRAP

Answering 11 (all integers in $[-5,5]$), believing the identity is universal. It fails for every $x<0$ because the left side is ≥ 0 while $2\tan^{-1}x<0$. Equality holds only for $x \geq 0$.

Sign/range filter

1. The left side $\cos^{-1}(\cdot) \in [0,\pi]$ is always ≥ 0; the right side $2\tan^{-1}x$ has the sign of x.
2. Equality requires $2\tan^{-1}x \geq 0$, i.e. $x \geq 0$; and on $x \geq 0$ the identity $\cos^{-1}\frac{1-x^2}{1+x^2} = 2\tan^{-1}x$ does hold.
3. Integers with $0 \leq x \leq 5$ are $0,1,2,3,4,5$: $\boxed{6}$ solutions.

Use the collapsed form

1. From the simplification $\cos^{-1}\frac{1-x^2}{1+x^2} = 2\tan^{-1}|x|$, the equation becomes $2\tan^{-1}|x| = 2\tan^{-1}x$.
2. This holds iff $|x| = x$, i.e. $x \geq 0$.
3. Counting integers in $[0,5]$ gives $\boxed{6}$.

INSIGHT

Reducing an equation to its collapsed identity ($2\tan^{-1}|x| = 2\tan^{-1}x \iff x \geq 0$) makes the count instant. The seductive '11' is the classic domain-blindness error.

№39 A Telescope in Disguise

●●●●● L4

EVALUATE

$$\text{Evaluate } \sum_{k=1}^{4} \tan^{-1}\left(\frac{1}{1+k+k^2}\right).$$

arctangent · difference-formula · telescoping · collapse

ANSWER

$$\tan^{-1}5 - \frac{\pi}{4}$$

THE TRAP

Trying to collapse each term as $\tan^{-1}\frac{1}{1+k+k^2} = \tan^{-1}(k+1) - \tan^{-1}k$ and then *summing the arctans of the arguments* via the addition law in one shot — but the addition law $\tan^{-1}a + \tan^{-1}b = \tan^{-1}\frac{a+b}{1-ab}$ silently adds $\pm\pi$ once $ab>1$. The safe route is the telescoping difference, where all terms lie in $(-\frac{\pi}{2}, \frac{\pi}{2})$ and the partial differences never overflow.

Telescoping difference

1. Note $\frac{1}{1+k+k^2} = \frac{(k+1)-k}{1+(k+1)k}$, so $\tan^{-1}\frac{1}{1+k+k^2} = \tan^{-1}(k+1) - \tan^{-1}k$ (valid since $\tan^{-1}(k+1), \tan^{-1}k \in (0, \frac{\pi}{2})$ and their difference is in $(0, \frac{\pi}{2})$, never overflowing).
2. Summing $k=1$ to 4 telescopes to $\tan^{-1}5 - \tan^{-1}1$.
3. Since $\tan^{-1}1 = \frac{\pi}{4}$, the sum is $\boxed{\tan^{-1}5 - \frac{\pi}{4}}$.

Numerical confirmation

1. Compute terms: $\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{13} + \tan^{-1}\frac{1}{21} \approx 0.3218 + 0.1419 + 0.0768 + 0.0476 = 0.5880$.

2. Compare $\tan^{-1} 5 - \frac{\pi}{4} \approx 1.3734 - 0.7854 = 0.5880$. They match.

3. Hence the closed form is $\boxed{\tan^{-1} 5 - \frac{\pi}{4}}$.

INSIGHT
The hidden structure is $1 + k + k^2 = 1 + k(k+1)$, the denominator of the $\tan^{-1}$ subtraction formula. Telescoping is branch-safe here because each partial difference stays inside $(-\frac{\pi}{2}, \frac{\pi}{2})$ – the very reason to prefer differences over naive addition when arguments exceed 1.

№40 The Tent's Full Reach

●●●●● L3

FIND THE RANGE

Find the range of $f(x) = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$ over its natural domain $[-1, 1]$.

sine-double-angle | range | piecewise | collapse

ANSWER

$$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

THE TRAP
Reasoning '$f = 2\sin^{-1} x$, and $\sin^{-1} x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, so $f \in [-\pi, \pi]$.' The output of $\sin^{-1}$ of *any* real argument cannot exceed $[-\frac{\pi}{2}, \frac{\pi}{2}]$, so $[-\pi, \pi]$ is impossible — the line form is exactly what misleads here.

Range of the inner expression

1. Let $u = 2x\sqrt{1-x^2}$. For $x \in [-1, 1]$ write $x = \sin t$, $t \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, so $u = \sin 2t$ with $2t \in [-\pi, \pi]$; thus $u \in [-1, 1]$ and attains every value there.
2. $f = \sin^{-1} u$ with u ranging over the full $[-1, 1]$, so f ranges over the full $[-\frac{\pi}{2}, \frac{\pi}{2}]$.
3. Range $= \boxed{[-\frac{\pi}{2}, \frac{\pi}{2}]}$.

Read off the piecewise tent

1. From the piecewise form, f rises $0 \to \frac{\pi}{2}$ on $[0, \frac{1}{\sqrt{2}}]$, falls $\frac{\pi}{2} \to 0$ on $[\frac{1}{\sqrt{2}}, 1]$, and is odd on the negative side reaching $-\frac{\pi}{2}$ at $x = -\frac{1}{\sqrt{2}}$.
2. The maximum $\frac{\pi}{2}$ and minimum $-\frac{\pi}{2}$ are attained, with all intermediate values covered by continuity.
3. Hence the range is $\boxed{[-\frac{\pi}{2}, \frac{\pi}{2}]}$.

INSIGHT
Even though f is *not* the line $2\sin^{-1} x$, its range coincidentally fills the same $[-\frac{\pi}{2}, \frac{\pi}{2}]$ as a single $\sin^{-1}$ – because the tent peaks exactly at the boundary values $\pm\frac{\pi}{2}$. Same range, completely different graph: a reminder that range alone never certifies an identity.

CHAPTER V

Equations & Root-Counting

Every inverse-trig equation is a domain problem in disguise.

№41 A Tilted Pair of Sines

●●●●● L4

SOLVE FOR X

$$\sin^{-1} x + \sin^{-1}(2x) = \frac{\pi}{3}$$

arcsin · squaring · extraneous-root · principal-range

ANSWER

$$x = \frac{\sqrt{21}}{14}$$

THE TRAP

After squaring, students reach $28x^2 = 3$ and write **both** $x = \pm\frac{\sqrt{21}}{14}$. But the negative root is extraneous: if $x < 0$ then both $\sin^{-1} x$ and $\sin^{-1}(2x)$ are negative, so the left side is negative and can never equal the positive $\frac{\pi}{3}$. Squaring forgot the sign.

Isolate-and-square

1. Write the equation as $\sin^{-1}(2x) = \frac{\pi}{3} - \sin^{-1} x$. For this to make sense the right side must lie in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, and since the left side equals a positive angle target we will need $x \geq 0$.
2. Take sine of both sides. With $s = \sin^{-1} x$ (so $\sin s = x,\ \cos s = \sqrt{1-x^2}$):
$$2x = \sin\left(\frac{\pi}{3} - s\right) = \frac{\sqrt{3}}{2}\sqrt{1-x^2} - \frac{1}{2}x.$$
3. Collect: $\frac{5x}{2} = \frac{\sqrt{3}}{2}\sqrt{1-x^2} \Rightarrow 5x = \sqrt{3}\,\sqrt{1-x^2}$. This already **forces $x \geq 0$**.
4. Square: $25x^2 = 3(1-x^2) \Rightarrow 28x^2 = 3 \Rightarrow x = \frac{\sqrt{21}}{14}$ (positive root only). Domain check: $2x = \frac{\sqrt{21}}{7} < 1$, fine.

$$\boxed{x = \frac{\sqrt{21}}{14}}$$

Monotonicity (uniqueness)

1. The function $f(x) = \sin^{-1} x + \sin^{-1}(2x)$ is **strictly increasing** on its domain $|x| \leq \frac{1}{2}$, since both summands increase.
2. Hence $f(x) = \frac{\pi}{3}$ has **at most one** solution. As $f(0) = 0 < \frac{\pi}{3}$ and $f(\frac{1}{2}) = \sin^{-1}\frac{1}{2} + \frac{\pi}{2} > \frac{\pi}{3}$, exactly one root lies in $(0, \frac{1}{2})$.
3. Solving the sine equation as above pins it to $\boxed{x = \frac{\sqrt{21}}{14}}$, and monotonicity guarantees no second root exists — confirming the negative candidate is spurious.

INSIGHT

Whenever you square an inverse-sine equation, the equation $5x = \sqrt{3}\sqrt{1-x^2}$ **before** squaring already encodes the sign: a non-negative left side forces $x \geq 0$. Read the sign off the pre-squared form rather than re-deriving it.

№42 Two Arctangents, One Target

●●●●● L4

SOLVE FOR X

$$\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1}\left(\frac{8}{31}\right)$$

arctan · addition-formula · ab-less-than-one · extraneous-root

ANSWER

$$x = \frac{1}{4}$$

THE TRAP

The quadratic $4x^2 + 31x - 8 = 0$ gives $x = \frac{1}{4}$ and $x = -8$, and the careless student boxes both. But at $x = -8$ the two summands are $\tan^{-1}(-7) + \tan^{-1}(-9)$, **both negative**, summing to about -2.89, which can never equal the positive $\tan^{-1}\frac{8}{31}$. The addition formula $\tan^{-1} a + \tan^{-1} b = \tan^{-1}\frac{a+b}{1-ab}$ requires $ab < 1$; at $x = -8$, $ab = x^2 - 1 = 63 > 1$, so the formula has silently shifted the result by $-\pi$.

Addition formula with the $ab<1$ guard

1. Set $a = x+1,\ b = x-1$, so $a + b = 2x$ and $ab = x^2 - 1$. The identity $\tan^{-1} a + \tan^{-1} b = \tan^{-1}\frac{a+b}{1-ab}$ is valid **only when $ab < 1$**, i.e. $x^2 < 2$.
2. Within that range: $\dfrac{2x}{1-(x^2-1)} = \dfrac{2x}{2-x^2} = \dfrac{8}{31}$.
3. Cross-multiply: $62x = 8(2-x^2) \Rightarrow 8x^2 + 62x - 16 = 0 \Rightarrow 4x^2 + 31x - 8 = 0$, giving $x = \frac{1}{4}$ or $x = -8$.
4. Apply the guard $x^2 < 2$: only $x = \frac{1}{4}$ survives ($x = -8$ violates $ab < 1$ and is extraneous).

$$\boxed{x = \tfrac{1}{4}}$$

Direct angle check

1. Compute each candidate's left side directly. At $x = \frac{1}{4}$: $\tan^{-1}\frac{5}{4} + \tan^{-1}(-\frac{3}{4}) \approx 0.896 - 0.644 = 0.2526$ rad, and $\tan^{-1}\frac{8}{31} \approx 0.2526$. Match.
2. At $x = -8$: $\tan^{-1}(-7) + \tan^{-1}(-9) \approx -1.429 - 1.460 = -2.889$ rad $\neq 0.2526$.
3. So the genuine solution is $\boxed{x = \frac{1}{4}}$; the second root is an artifact of tangent's π-periodicity.

INSIGHT

The tangent addition identity is a statement modulo π. The quadratic you get is correct for tan of both sides, but you must independently verify which roots keep the **actual angles** (not just their tangents) equal. The single inequality $ab < 1$ is the cleanest filter.

№43 Where Arccos Meets Arctan

●●●●● L4

FIND THE NUMBER OF SOLUTIONS

Find the number of real solutions of $\cos^{-1} x = \tan^{-1} x$.

arccos · arctan · range-intersection · monotonicity

ANSWER

$$1$$

THE TRAP

A common move is to take the tangent of both sides. Done correctly that gives $\dfrac{\sqrt{1-x^2}}{x} = x$, i.e. $x^4 + x^2 - 1 = 0$, whose only root in $[-1, 1]$ is $x = \sqrt{\frac{\sqrt{5}-1}{2}} \approx 0.786$ — so even the algebra yields just **one** value. The deeper trap is forgetting that $\tan(\cos^{-1} x) = \sqrt{1-x^2}/x$, not x; a careless $\sqrt{1-x^2} = x^2$-style slip produces $x = \frac{1}{\sqrt{2}}$, which is **not** a solution ($\cos^{-1} \frac{1}{\sqrt{2}} = \frac{\pi}{4} \neq \tan^{-1} \frac{1}{\sqrt{2}} \approx 0.615$). The clean way avoids algebra entirely: the **range constraint** plus monotonicity pins the count at one.

Strict monotonic crossing

1. The domain is $[-1, 1]$ (intersection of the two domains). Define $g(x) = \cos^{-1} x - \tan^{-1} x$. Since $\cos^{-1} x$ is **strictly decreasing** and $\tan^{-1} x$ is **strictly increasing**, g is strictly decreasing on $[-1, 1]$.
2. Endpoints: $g(0) = \frac{\pi}{2} - 0 > 0$ and $g(1) = 0 - \frac{\pi}{4} < 0$. By the Intermediate Value Theorem g has a zero in $(0, 1)$.
3. A strictly decreasing continuous function changes sign at most once, so the zero is unique. For $x < 0$, $g(x) > g(0) > 0$, ruling out any negative root.

$$\boxed{\text{exactly 1 solution}}$$

Range pinching

1. For equality we need $\cos^{-1} x = \tan^{-1} x$. On $[-1, 1]$ we have $\tan^{-1} x \in [-\frac{\pi}{4}, \frac{\pi}{4}]$, while $\cos^{-1} x \in [0, \pi]$, so a common value must lie in $[0, \frac{\pi}{4}]$.
2. From $\cos^{-1} x \leq \frac{\pi}{4}$ (decreasing) we get $x \geq \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$, and from $\tan^{-1} x \geq 0$ we get $x \geq 0$. Hence any solution lies in $\left[\frac{1}{\sqrt{2}}, 1\right]$.
3. On this short interval $\cos^{-1} x$ **decreases** from $\frac{\pi}{4}$ (at $x = \frac{1}{\sqrt{2}}$) down to 0 (at $x = 1$), while $\tan^{-1} x$ **increases** from $\tan^{-1} \frac{1}{\sqrt{2}} \approx 0.615$ up to $\frac{\pi}{4}$. At the left end $\cos^{-1} > \tan^{-1}$ and at the right end $\cos^{-1} < \tan^{-1}$, so the strictly decreasing gap crosses zero exactly once $\Rightarrow \boxed{1}$ solution (numerically $x \approx 0.786$).

INSIGHT

Counting problems for transcendental inverse-trig equations are almost always won by **monotonicity + endpoint signs**, not by solving. Compare a strictly decreasing function with a strictly increasing one: at most one crossing, ever. (Here the lone root happens to be algebraic, $x = \sqrt{(\sqrt{5}-1)/2}$, but you never needed to find it.)

№44 The Half-Angle Disguise

●●●●● L4

SOLVE FOR X

$$\tan^{-1}\left(\frac{1-x}{1+x}\right) = \frac{1}{2} \tan^{-1} x, \qquad x > 0$$

arctan · subtraction-formula · half-angle · domain-validity

ANSWER

$$x = \frac{1}{\sqrt{3}}$$

THE TRAP

Students often rewrite $\tan^{-1} \frac{1-x}{1+x} = \frac{\pi}{4} - \tan^{-1} x$ **without** the restriction. That identity holds only for $x > -1$; if one applies it for $x < -1$ a stray π appears. Inside the given $x > 0$ the identity is safe, but the unwary student who later "checks" by squaring may pick up $x = -\frac{1}{\sqrt{3}}$, which violates $x > 0$ and corresponds to a different branch.

Subtraction identity

1. Write $\dfrac{1-x}{1+x} = \dfrac{1-x}{1+1\cdot x}$, which is exactly $\tan(A-B) = \dfrac{\tan A - \tan B}{1+\tan A \tan B}$ with $\tan A = 1$ and $\tan B = x$. Since the product $\tan A \tan B = x > -1$ keeps the difference inside the principal range, we get the exact identity $\tan^{-1}\dfrac{1-x}{1+x} = \tan^{-1} 1 - \tan^{-1} x = \dfrac{\pi}{4} - \tan^{-1} x$ for all $x > -1$.
2. Substitute: $\dfrac{\pi}{4} - \tan^{-1} x = \dfrac{1}{2}\tan^{-1} x \Rightarrow \dfrac{\pi}{4} = \dfrac{3}{2}\tan^{-1} x$.
3. So $\tan^{-1} x = \dfrac{\pi}{6} \Rightarrow x = \tan\dfrac{\pi}{6} = \dfrac{1}{\sqrt{3}}$, which indeed satisfies $x > 0$.

$$\boxed{x = \tfrac{1}{\sqrt{3}}}$$

Substitution $x=\tan\theta$

1. Let $x = \tan\theta$ with $\theta \in (0, \frac{\pi}{2})$ since $x > 0$, so $\tan^{-1} x = \theta$.
2. Then $\dfrac{1-x}{1+x} = \dfrac{1-\tan\theta}{1+\tan\theta} = \tan\left(\dfrac{\pi}{4} - \theta\right)$, and for $\theta \in (0, \frac{\pi}{2})$ the angle $\frac{\pi}{4} - \theta \in (-\frac{\pi}{4}, \frac{\pi}{4})$ lies in the principal range, so the left side is $\frac{\pi}{4} - \theta$.
3. Equation becomes $\frac{\pi}{4} - \theta = \frac{\theta}{2} \Rightarrow \theta = \frac{\pi}{6} \Rightarrow x = \tan\frac{\pi}{6} = \boxed{\dfrac{1}{\sqrt{3}}}$.

INSIGHT

The pattern $\frac{1-x}{1+x}$ is the tangent of a 45° difference, valid as a clean subtraction only when $x > -1$ keeps the result inside $(-\frac{\pi}{2}, \frac{\pi}{2})$. Always state the interval on which the collapse $\tan^{-1}\frac{1-x}{1+x} = \frac{\pi}{4} - \tan^{-1} x$ is legal.

№45 A Sine Splits in Two

●●●●● L4

SOLVE FOR X

$$\sin^{-1} x + \sin^{-1}(1-x) = \cos^{-1} x$$

arcsin · arccos · double-angle · complementary-identity

ANSWER

$$x \in \{0,\ \tfrac{1}{2}\}$$

THE TRAP

Using $\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$ and then taking sine gives $1 - x = 1 - 2x^2$, i.e. $x(2x-1) = 0$, so $x = 0, \frac{1}{2}$. The trap is to **discard $x = 0$** as "trivial" or to doubt it — but $x = 0$ is genuine: $\sin^{-1} 0 + \sin^{-1} 1 = \frac{\pi}{2} = \cos^{-1} 0$. Both roots are valid; rejecting either loses half the answer.

Complementary reduction

1. Replace $\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$. The equation becomes $\sin^{-1}(1-x) = \frac{\pi}{2} - 2\sin^{-1} x$.
2. The right side must lie in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (the range of $\sin^{-1}$): $-\frac{\pi}{2} \le \frac{\pi}{2} - 2\sin^{-1} x \le \frac{\pi}{2} \Rightarrow \sin^{-1} x \in [0, \frac{\pi}{2}] \Rightarrow x \in [0, 1]$.
3. Take sine: $1 - x = \sin(\frac{\pi}{2} - 2\sin^{-1} x) = \cos(2\sin^{-1} x) = 1 - 2x^2$. Hence $2x^2 - x = 0 \Rightarrow x = 0$ or $x = \frac{1}{2}$.
4. Both lie in $[0, 1]$ and check directly.

$$\boxed{x = 0 \text{ or } x = \tfrac{1}{2}}$$

Direct verification of candidates

1. From the algebra $x(2x-1) = 0$, test each. $x = 0$: LHS $= 0 + \frac{\pi}{2} = \frac{\pi}{2}$, RHS $= \cos^{-1} 0 = \frac{\pi}{2}$. Valid.
2. $x = \frac{1}{2}$: LHS $= \sin^{-1}\frac{1}{2} + \sin^{-1}\frac{1}{2} = \frac{\pi}{6} + \frac{\pi}{6} = \frac{\pi}{3}$, RHS $= \cos^{-1}\frac{1}{2} = \frac{\pi}{3}$. Valid.

3. Both survive, so the solution set is $\boxed{\{0, \frac{1}{2}\}}$.

INSIGHT

Replacing $\cos^{-1} x$ by $\frac{\pi}{2} - \sin^{-1} x$ converts the equation into one inverse function, and the range box on $\frac{\pi}{2} - 2\sin^{-1} x$ pre-restricts x to $[0, 1]$. Here the restriction admits both algebraic roots – so this is a rare case where nothing is extraneous, and the lesson is *not* to over-prune.

№46 Cotangent Minus Tangent

●●●●● L4

SOLVE FOR X

$$\cot^{-1} x - \tan^{-1} x = \frac{\pi}{6}$$

arccot · arctan · complementary-identity · principal-branch

ANSWER

$$x = \frac{1}{\sqrt{3}}$$

THE TRAP

A student who (incorrectly) treats $\cot^{-1} x$ as $\tan^{-1}(1/x)$ will get $\tan^{-1} \frac{1}{x} - \tan^{-1} x = \frac{\pi}{6}$ and, for **negative** x, an extra sign error. The JEE convention is $\cot^{-1} x \in (0, \pi)$ with the identity $\cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x$ holding for **all real x** — whereas $\tan^{-1}(1/x)$ equals $\cot^{-1} x$ only when $x > 0$. Mixing them up admits a phantom negative root $x = -\sqrt{3}$ (where the true left side equals $\frac{7\pi}{6}$, not $\frac{\pi}{6}$).

Complementary identity

1. Use $\cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x$, valid for **every** real x under the principal convention $\cot^{-1} : \mathbb{R} \to (0, \pi)$.
2. Substitute: $\left(\frac{\pi}{2} - \tan^{-1} x\right) - \tan^{-1} x = \frac{\pi}{6} \Rightarrow \frac{\pi}{2} - 2\tan^{-1} x = \frac{\pi}{6}$.
3. So $2\tan^{-1} x = \frac{\pi}{3} \Rightarrow \tan^{-1} x = \frac{\pi}{6} \Rightarrow x = \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$.

$$\boxed{x = \frac{1}{\sqrt{3}}}$$

Single-variable angle

1. Let $\theta = \tan^{-1} x \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Then $\cot^{-1} x = \frac{\pi}{2} - \theta$ regardless of the sign of x.
2. The equation reads $(\frac{\pi}{2} - \theta) - \theta = \frac{\pi}{6} \Rightarrow \theta = \frac{\pi}{6}$.
3. Therefore $x = \tan \frac{\pi}{6} = \boxed{\frac{1}{\sqrt{3}}}$, the unique solution since θ is determined uniquely.

INSIGHT

The reliable JEE identity is $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ for all real x. The shortcut $\cot^{-1} x = \tan^{-1}(1/x)$ is true only for $x > 0$ and is off by π for $x < 0$ – never use it without a sign caveat.

№47 The Vanishing Radical

●●●●● L4

SOLVE FOR X

$$\tan^{-1}\sqrt{x^2 + x} \;+\; \sin^{-1}\sqrt{x^2 + x + 1} = \frac{\pi}{2}$$

arctan · arcsin · domain-pinching · radical

ANSWER

$$x \in \{0,\ -1\}$$

THE TRAP

Seeing the structure $\tan^{-1}\sqrt{t} + \sin^{-1}\sqrt{t+1}$, a student may apply $\tan^{-1} a + \sin^{-1} b = \frac{\pi}{2}$ and write $\sqrt{x^2+x} = \cot(\sin^{-1}\sqrt{x^2+x+1})$, then grind algebra producing spurious roots. The real key is the **domain**: $\sin^{-1}$ needs $x^2+x+1 \le 1$ i.e. $x^2+x \le 0$, while the radical $\sqrt{x^2+x}$ needs $x^2+x \ge 0$. Together $x^2+x=0$, instantly forcing $x \in \{0,-1\}$ with no algebra at all.

Domain squeeze

1. Inside $\sin^{-1}$: need $0 \le x^2+x+1 \le 1 \Rightarrow x^2+x \le 0 \Rightarrow x \in [-1,0]$.
2. Inside the radical of $\tan^{-1}$: need $x^2+x \ge 0 \Rightarrow x \le -1$ or $x \ge 0$.
3. Intersecting the two constraints leaves only $x^2+x=0$, i.e. $x=0$ or $x=-1$.
4. Verify: at both, $\tan^{-1}\sqrt{0} + \sin^{-1}\sqrt{1} = 0 + \frac{\pi}{2} = \frac{\pi}{2}$.

$$\boxed{x = 0 \text{ or } x = -1}$$

Let $t=x^2+x$

1. Set $t = x^2+x \ge 0$ (forced by the radical). The equation is $\tan^{-1}\sqrt{t} + \sin^{-1}\sqrt{t+1} = \frac{\pi}{2}$.
2. $\sin^{-1}\sqrt{t+1}$ requires $\sqrt{t+1} \le 1 \Rightarrow t \le 0$. Combined with $t \ge 0$ gives $t=0$.
3. At $t=0$ the left side is $0 + \sin^{-1} 1 = \frac{\pi}{2}$ — an identity. Solving $x^2+x=0$ gives $\boxed{x \in \{0,-1\}}$.

INSIGHT

Many "hard" radical inverse-trig equations are really domain puzzles: opposite inequalities from a square root and from an $\sin^{-1}$ argument collapse the feasible set to a single equation. Check domains *before* manipulating – it often solves the problem outright.

№48 Reciprocal Twins

●●●●● L4

FIND THE NUMBER OF SOLUTIONS

Find the number of real solutions of $\tan^{-1}\left(\dfrac{1}{2x-1}\right) + \tan^{-1}\left(\dfrac{1}{2x+1}\right) = \tan^{-1}\left(\dfrac{4}{7}\right)$.

arctan · addition-formula · ab-less-than-one · root-counting

ANSWER

$$2$$

THE TRAP

After applying the addition formula one lands on $\frac{2x}{2x^2-1} = \frac{4}{7}$, i.e. $4x^2-7x-2=0$, with roots $x = 2,\ -\frac{1}{4}$. The tempting shortcut is to declare the negative root extraneous by analogy with the usual "π-shift" lore. But here check the product: at $x=-\frac{1}{4}$, $a=-\frac{2}{3}$, $b=2$, $ab=-\frac{4}{3}<1$, so the formula stays in-branch and the root is **genuine**. Discarding it gives the wrong count of 1.

Addition formula then branch audit

1. With $a = \frac{1}{2x-1}$, $b = \frac{1}{2x+1}$: $a+b = \dfrac{4x}{4x^2-1}$ and $1-ab = \dfrac{4x^2-2}{4x^2-1}$, so $\dfrac{a+b}{1-ab} = \dfrac{4x}{4x^2-2} = \dfrac{2x}{2x^2-1}$ (valid when $ab<1$).
2. Set equal to $\frac{4}{7}$: $\dfrac{2x}{2x^2-1} = \dfrac{4}{7} \Rightarrow 14x = 4(2x^2-1) \Rightarrow 4x^2-7x-2=0$, giving $x=2,\ x=-\frac{1}{4}$.

3. Audit the branch via $ab=\frac{1}{(2x-1)(2x+1)}=\frac{1}{4x^2-1}$. At $x=2$: $ab=\frac{1}{15}<1$. At $x=-\frac{1}{4}$: $ab=\frac{1}{4\cdot\frac{1}{16}-1}=\frac{1}{-\frac{3}{4}}=-\frac{4}{3}<1$. **Both** satisfy $ab<1$, so both are in-branch.
4. Direct check: $x=2\Rightarrow\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{5}=\tan^{-1}\frac{4}{7}$ ✓; $x=-\frac{1}{4}\Rightarrow\tan^{-1}(-\frac{2}{3})+\tan^{-1}2=\tan^{-1}\frac{4}{7}$ ✓.

$$\boxed{2\text{ solutions}}$$

Numerical sign-scan

1. Define $F(x)=\tan^{-1}\frac{1}{2x-1}+\tan^{-1}\frac{1}{2x+1}-\tan^{-1}\frac{4}{7}$, treating the poles at $x=\pm\frac{1}{2}$ as breaks.
2. Scanning x over $(-\infty,\infty)$, F changes sign exactly at $x=-\frac{1}{4}$ and $x=2$ (the apparent jumps at $x=\pm\frac{1}{2}$ are branch discontinuities of individual $\tan^{-1}$ terms, not roots of F).
3. Hence the equation has $\boxed{2}$ real solutions, $x=2$ and $x=-\frac{1}{4}$.

INSIGHT

The product test $ab<1$ is not a blanket "reject the negative root" rule. Compute ab for **each** candidate; a negative x can keep $ab<1$ and stay valid. The branch condition, not the sign of x, is what counts.

№49 The Triple-Angle Plateau

●●●●● L5

FIND THE NUMBER OF SOLUTIONS

Find the number of real solutions of $3\cos^{-1}x-\cos^{-1}(4x^3-3x)=\pi$.

arccos · triple-angle · branch-piecewise · root-counting

ANSWER

$$1$$

THE TRAP

The identity $\cos^{-1}(4x^3-3x)=3\cos^{-1}x$ is widely "known", and a student who applies it everywhere gets $3\cos^{-1}x-3\cos^{-1}x=0\neq\pi$, concluding **no** solutions. But that identity holds only on $x\in[\frac{1}{2},1]$; off that interval $\cos^{-1}(\cos 3t)$ folds, making the left side a continuous piecewise-linear function of $t=\cos^{-1}x$ that actually passes through π once.

Folding via $t=\cos^{-1}x$

1. Let $t=\cos^{-1}x\in[0,\pi]$, so $x=\cos t$ and $4x^3-3x=\cos 3t$. The equation is $3t-\cos^{-1}(\cos 3t)=\pi$.
2. As t ranges over $[0,\pi]$, $3t$ ranges over $[0,3\pi]$ and $\cos^{-1}(\cos 3t)$ folds: it equals $3t$ on $[0,\frac{\pi}{3}]$, equals $2\pi-3t$ on $[\frac{\pi}{3},\frac{2\pi}{3}]$, and $3t-2\pi$ on $[\frac{2\pi}{3},\pi]$.
3. On the middle piece $t\in[\frac{\pi}{3},\frac{2\pi}{3}]$ (i.e. $x\in[-\frac{1}{2},\frac{1}{2}]$): LHS $=3t-(2\pi-3t)=6t-2\pi$. Set $=\pi$: $6t-2\pi=\pi\Rightarrow t=\frac{\pi}{2}\Rightarrow x=\cos\frac{\pi}{2}=0$.
4. On the outer pieces LHS is 2π (for $x\in[-1,-\frac{1}{2}]$) and 0 (for $x\in[\frac{1}{2},1]$), neither equal to π. So there is exactly **one** solution.

$$\boxed{1\text{ solution, at }x=0}$$

Monotone middle branch

1. The full left side $g(x)=3\cos^{-1}x-\cos^{-1}(4x^3-3x)$ is continuous and equals the constant 2π on $[-1,-\frac{1}{2}]$ and the constant 0 on $[\frac{1}{2},1]$.
2. On the bridge $(-\frac{1}{2},\frac{1}{2})$ it is **strictly decreasing** from 2π to 0 (it equals $6\cos^{-1}x-2\pi$ there), so it attains each value in $(0,2\pi)$ exactly once.
3. Since $\pi\in(0,2\pi)$, $g(x)=\pi$ has exactly one root, found by $6\cos^{-1}x-2\pi=\pi\Rightarrow\cos^{-1}x=\frac{\pi}{2}\Rightarrow\boxed{x=0}$, total count $\boxed{1}$.

INSIGHT

$\cos^{-1}(4x^3-3x)=3\cos^{-1}x$ only where $3\cos^{-1}x\in[0,\pi]$, i.e. $x\in[\frac{1}{2},1]$. Elsewhere $\cos^{-1}(\cos 3t)$ reflects off 0 and π, turning the expression into a tent of slopes ± 6 – and the equation's solution lives precisely on the folded middle, where the naive identity fails.

№50 The Sum of Squared Angles

●●●●● L5

SOLVE FOR X

$$(\sin^{-1}x)^2+(\cos^{-1}x)^2=\frac{5\pi^2}{8}$$

arcsin arccos sum-pi-over-2 range-rejection

ANSWER

$$x=-\frac{1}{\sqrt{2}}$$

THE TRAP

Writing $s=\sin^{-1}x$ and using $\cos^{-1}x=\frac{\pi}{2}-s$ gives $16s^2-8\pi s-3\pi^2=0$, so $s=\frac{3\pi}{4}$ or $s=-\frac{\pi}{4}$. The seductive error is taking $s=\frac{3\pi}{4}$ and writing $x=\sin\frac{3\pi}{4}=\frac{1}{\sqrt{2}}$. But $\frac{3\pi}{4}$ is **outside** the range $[-\frac{\pi}{2},\frac{\pi}{2}]$ of $\sin^{-1}$, so it is not a legitimate value of s and $x=\frac{1}{\sqrt{2}}$ is extraneous.

Reduce to one angle, then range-filter

1. Let $s=\sin^{-1}x\in[-\frac{\pi}{2},\frac{\pi}{2}]$. Then $\cos^{-1}x=\frac{\pi}{2}-s$, and the equation becomes $s^2+\left(\frac{\pi}{2}-s\right)^2=\frac{5\pi^2}{8}$.
2. Expand: $2s^2-\pi s+\frac{\pi^2}{4}=\frac{5\pi^2}{8}\Rightarrow 2s^2-\pi s-\frac{3\pi^2}{8}=0\Rightarrow 16s^2-8\pi s-3\pi^2=0$.
3. Solve: $s=\dfrac{8\pi\pm\sqrt{64\pi^2+192\pi^2}}{32}=\dfrac{8\pi\pm 16\pi}{32}=\frac{3\pi}{4}\text{ or }-\frac{\pi}{4}$.
4. Range filter: $s=\frac{3\pi}{4}\notin[-\frac{\pi}{2},\frac{\pi}{2}]$ is rejected; only $s=-\frac{\pi}{4}$ survives, giving $x=\sin(-\frac{\pi}{4})=-\frac{1}{\sqrt{2}}$.

$$\boxed{x=-\tfrac{1}{\sqrt{2}}}$$

Quadratic in the gap $d=s-c$

1. With $s+c=\frac{\pi}{2}$ (always), use $s^2+c^2=\frac{(s+c)^2+(s-c)^2}{2}=\frac{\frac{\pi^2}{4}+(s-c)^2}{2}=\frac{5\pi^2}{8}$.
2. Thus $(s-c)^2=\frac{5\pi^2}{4}-\frac{\pi^2}{4}=\pi^2\Rightarrow s-c=\pm\pi$. Combined with $s+c=\frac{\pi}{2}$: $(s,c)=(\frac{3\pi}{4},-\frac{\pi}{4})$ or $(-\frac{\pi}{4},\frac{3\pi}{4})$.
3. Only the second pair is admissible ($s\in[-\frac{\pi}{2},\frac{\pi}{2}]$ and $c\in[0,\pi]$): $s=-\frac{\pi}{4}$, $c=\frac{3\pi}{4}$, so $x=\sin(-\frac{\pi}{4})=\boxed{-\dfrac{1}{\sqrt{2}}}$. The first pair has $c=-\frac{\pi}{4}<0$, impossible for $\cos^{-1}$.

INSIGHT

Because $\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$ is rigid, any symmetric combination reduces to one variable – but the quadratic in s will routinely throw a root outside $[-\frac{\pi}{2},\frac{\pi}{2}]$. The range of $\sin^{-1}$ (and the non-negativity of $\cos^{-1}$) is the filter that keeps exactly the legitimate root.

CHAPTER VI

Trig of Inverse Trig

The trig of a sum of inverse functions – drawn on a triangle.

№51 The Three-Step Collapse

L3

EVALUATE

$$\tan\left(2\tan^{-1}\tfrac{1}{3}+\tan^{-1}\tfrac{1}{7}\right)$$

tangent · double-angle · addition-formula · right-triangle

ANSWER

$$1$$

THE TRAP

The seductive error is to write $2\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{7}=\frac{\pi}{4}$ and stop — students often *assume* the answer is an angle and report $\frac{\pi}{4}$ as the **answer** to the evaluation, when the question asks for the **tangent**, which is 1. A subtler trap: applying the triple-sum arctan formula and blindly adding π. Here $\tan^{-1}\frac{1}{3}\in(0,\frac{\pi}{4})$ so $2\tan^{-1}\frac{1}{3}\in(0,\frac{\pi}{2})$ and adding $\tan^{-1}\frac{1}{7}$ keeps the sum strictly inside $(0,\frac{\pi}{2})$ — **no π correction is needed** and $\tan$ of the sum is genuinely $+1$.

Iterated tangent addition

1. Double angle first: $\tan\left(2\tan^{-1}\frac{1}{3}\right)=\dfrac{2\cdot\frac{1}{3}}{1-\frac{1}{9}}=\dfrac{\frac{2}{3}}{\frac{8}{9}}=\dfrac{3}{4}$.
2. Now add: with $\tan A=\frac{3}{4}$ and $\tan B=\frac{1}{7}$, $\tan(A+B)=\dfrac{\frac{3}{4}+\frac{1}{7}}{1-\frac{3}{4}\cdot\frac{1}{7}}=\dfrac{\frac{25}{28}}{\frac{25}{28}}=1$.
3. Since the sum lies in $(0,\frac{\pi}{2})$, the value $\boxed{1}$ is exact.

Gaussian-integer phases

1. Encode each arctangent as the argument of a complex number: $2\tan^{-1}\frac{1}{3}=\arg(3+i)^2=\arg(8+6i)$ and $\tan^{-1}\frac{1}{7}=\arg(7+i)$.
2. Multiply: $(8+6i)(7+i)=56+8i+42i+6i^2=50+50i$. The total angle is $\arg(50+50i)$.
3. $\arg(50+50i)=\tan^{-1}\frac{50}{50}=\frac{\pi}{4}$, so the tangent equals $\boxed{1}$, and the positive real part confirms the sum is in $(0,\frac{\pi}{2})$ — no quadrant shift.

INSIGHT

Multiplying $(3+i)^2(7+i)$ and reading arg is the fastest sanity check on any chain of $\tan^{-1}\frac{1}{k}$ values; the sign of the **real part** tells you instantly whether you have stayed inside $(-\frac{\pi}{2},\frac{\pi}{2})$, killing the $\pm\pi$ ambiguity for free.

№52 When the Obtuse Angle Cancels

L3

EVALUATE

$$\sin\big(\cos^{-1}(-\tfrac{4}{5}) + \tan^{-1}\tfrac{3}{4}\big)$$

sine | addition-formula | obtuse-quadrant | sign-of-root

ANSWER

$$0$$

THE TRAP

The fatal slip is reading $\cos^{-1}(-\frac{4}{5})$ as a *first-quadrant* triangle and taking $\cos = -\frac{4}{5}$, $\sin = -\frac{3}{5}$ — copying the negative sign onto the sine. But $\cos^{-1}$ outputs an angle in $[0, \pi]$, where **sine is non-negative**, so $\sin\big(\cos^{-1}(-\frac{4}{5})\big) = +\frac{3}{5}$. With the wrong $-\frac{3}{5}$ a student gets $\sin(a+b) = -\frac{24}{25}$; the correct sign gives the clean cancellation to 0.

Right-triangle with quadrant check

1. Let $a = \cos^{-1}(-\frac{4}{5})$. Then $a \in (\frac{\pi}{2}, \pi)$, so $\cos a = -\frac{4}{5}$ but $\sin a = +\frac{3}{5}$ (sine positive on $[0, \pi]$).
2. Let $b = \tan^{-1}\frac{3}{4} \in (0, \frac{\pi}{2})$, so $\sin b = \frac{3}{5}$, $\cos b = \frac{4}{5}$.
3. $\sin(a+b) = \sin a \cos b + \cos a \sin b = \frac{3}{5} \cdot \frac{4}{5} + \big(-\frac{4}{5}\big) \cdot \frac{3}{5} = \frac{12}{25} - \frac{12}{25} = \boxed{0}$.

Angle identification

1. Note $\cos^{-1}(-\frac{4}{5}) = \pi - \cos^{-1}\frac{4}{5} = \pi - \tan^{-1}\frac{3}{4}$ (since a 3-4-5 triangle gives $\cos^{-1}\frac{4}{5} = \tan^{-1}\frac{3}{4}$).
2. Then the argument is $\big(\pi - \tan^{-1}\frac{3}{4}\big) + \tan^{-1}\frac{3}{4} = \pi$.
3. $\sin\pi = \boxed{0}$, and the cancellation is structural — the two 3-4-5 angles are exactly supplementary minus equal.

INSIGHT

Whenever $\cos^{-1}$ of a negative number appears, write it as $\pi - \cos^{-1}|x|$ before doing anything else: this makes the supplementary structure visible and stops the reflex of dragging the minus sign onto the sine, which is always ≥ 0 on the $\cos^{-1}$ range.

№53 Half of an Obtuse Angle

●●●●○ L4

EVALUATE

$$\tan\big(\tfrac{1}{2}\cos^{-1}(-\tfrac{7}{25})\big)$$

half-angle | tangent | obtuse-quadrant | sign-of-root

ANSWER

$$\frac{4}{3}$$

THE TRAP

Students reach for $\tan\frac{t}{2} = \pm\sqrt{\frac{1-\cos t}{1+\cos t}} = \pm\sqrt{\frac{1+7/25}{1-7/25}} = \pm\sqrt{\frac{32}{18}} = \pm\frac{4}{3}$ and then guess the sign — many pick $-\frac{4}{3}$ because the input is negative. The branch is forced: $t = \cos^{-1}(-\frac{7}{25}) \in (\frac{\pi}{2}, \pi)$, so $\frac{t}{2} \in (\frac{\pi}{4}, \frac{\pi}{2})$, where tangent is **positive**. The answer is $+\frac{4}{3}$.

Half-angle via sine (sign-free)

1. Set $t = \cos^{-1}(-\frac{7}{25}) \in (\frac{\pi}{2}, \pi)$, so $\cos t = -\frac{7}{25}$ and $\sin t = +\frac{24}{25}$ (positive on $[0, \pi]$).
2. Use the sign-free identity $\tan\frac{t}{2} = \dfrac{1-\cos t}{\sin t} = \dfrac{1+\frac{7}{25}}{\frac{24}{25}} = \dfrac{32/25}{24/25} = \dfrac{32}{24}$.
3. Simplify: $\boxed{\dfrac{4}{3}}$. Because $\sin t > 0$ the formula automatically delivers the correct positive value.

Surd form with explicit branch

1. $\tan\frac{t}{2} = \pm\sqrt{\dfrac{1-\cos t}{1+\cos t}} = \pm\sqrt{\dfrac{1+7/25}{1-7/25}} = \pm\sqrt{\dfrac{32/25}{18/25}} = \pm\sqrt{\dfrac{16}{9}} = \pm\dfrac{4}{3}$.
2. Locate the half-angle: $t \in (\frac{\pi}{2}, \pi) \Rightarrow \frac{t}{2} \in (\frac{\pi}{4}, \frac{\pi}{2})$, first quadrant, so $\tan\frac{t}{2} > 0$.
3. Therefore $\tan\frac{t}{2} = \boxed{\dfrac{4}{3}}$.

INSIGHT
The identity $\tan\frac{t}{2} = \frac{1-\cos t}{\sin t}$ is the professional's choice: since $\cos^{-1}$ outputs $t \in [0, \pi]$ where $\sin t \geq 0$, this form carries the correct sign on its own and removes the $\pm$ that the surd version forces you to resolve by hand.

№54 A Difference of Two Signs

●●●●● L4

EVALUATE

$$\sin\left(\cos^{-1}\left(-\tfrac{5}{13}\right) - \sin^{-1}\left(-\tfrac{3}{5}\right)\right)$$

sine · difference-formula · two-quadrants · sign-of-root

ANSWER

$$\frac{33}{65}$$

THE TRAP
Two independent sign traps stack here. First, $\cos^{-1}(-\frac{5}{13}) \in (\frac{\pi}{2}, \pi)$ gives $\sin = +\frac{12}{13}$, NOT $-\frac{12}{13}$. Second, $\sin^{-1}(-\frac{3}{5}) \in (-\frac{\pi}{2}, 0)$ gives $\cos = +\frac{4}{5}$ (cosine positive on $[-\frac{\pi}{2}, \frac{\pi}{2}]$), and crucially $\sin = -\frac{3}{5}$ stays negative. Mishandling either flip produces $\frac{63}{65}$, $-\frac{33}{65}$ or $-\frac{63}{65}$ instead of the correct $\frac{33}{65}$.

Difference formula with quadrant bookkeeping

1. $a = \cos^{-1}(-\frac{5}{13}) \in (\frac{\pi}{2}, \pi)$: $\cos a = -\frac{5}{13}$, $\sin a = +\frac{12}{13}$.
2. $b = \sin^{-1}(-\frac{3}{5}) \in (-\frac{\pi}{2}, 0)$: $\sin b = -\frac{3}{5}$, $\cos b = +\frac{4}{5}$.
3. $\sin(a-b) = \sin a\cos b - \cos a \sin b = \frac{12}{13}\cdot\frac{4}{5} - \left(-\frac{5}{13}\right)\left(-\frac{3}{5}\right) = \frac{48}{65} - \frac{15}{65} = \boxed{\dfrac{33}{65}}$.

Rewrite to principal-positive angles

1. Use $\sin^{-1}(-\frac{3}{5}) = -\sin^{-1}\frac{3}{5}$, so the argument becomes $\cos^{-1}(-\frac{5}{13}) + \sin^{-1}\frac{3}{5}$.
2. Let $a = \cos^{-1}(-\frac{5}{13})$ ($\sin a = \frac{12}{13}, \cos a = -\frac{5}{13}$) and $b = \sin^{-1}\frac{3}{5}$ ($\sin b = \frac{3}{5}, \cos b = \frac{4}{5}$).
3. $\sin(a+b) = \frac{12}{13}\cdot\frac{4}{5} + \left(-\frac{5}{13}\right)\frac{3}{5} = \frac{48}{65} - \frac{15}{65} = \boxed{\dfrac{33}{65}}$.

INSIGHT
Pulling the odd-function fact $\sin^{-1}(-u) = -\sin^{-1}u$ out front converts a difference of awkwardly-signed angles into a sum of standard ones, after which only the single $\cos^{-1}(\text{negative}) \Rightarrow \sin > 0$ flip remains to track.

№55 Doubling a Sharp Angle

●●●●● L3

EVALUATE

$$\cos\left(2\cos^{-1}\tfrac{1}{4}\right)$$

cosine · double-angle · right-triangle · rational-value

ANSWER

$$-\frac{7}{8}$$

THE TRAP

The tempting shortcut is to think "$\frac{1}{4}$ is small, so $\cos^{-1}\frac{1}{4}$ is near $\frac{\pi}{2}$, doubling lands near π, answer near -1" and then sloppily compute $\cos\left(2\cos^{-1}\frac{1}{4}\right)$ via $1-2\sin^2$ using the WRONG $\sin=\frac{1}{4}$. Worse, some use $2\cos^2\theta-1$ but plug $\cos\theta=\frac{1}{4}$ as $\frac{1}{2}$ from a careless half-step. The clean identity $\cos 2\theta = 2\cos^2\theta - 1$ with $\cos\theta=\frac{1}{4}$ gives exactly $-\frac{7}{8}$; mistaking $\cos\theta$ for $\sin\theta$ yields $+\frac{7}{8}$.

Direct double-angle in cosine

1. Let $\theta=\cos^{-1}\frac{1}{4}$, so $\cos\theta=\frac{1}{4}$ directly (no root needed).
2. $\cos 2\theta = 2\cos^2\theta - 1 = 2\cdot\frac{1}{16}-1=\frac{1}{8}-1$.
3. $= \boxed{-\frac{7}{8}}$.

Via the sine of the angle

1. With $\theta=\cos^{-1}\frac{1}{4}\in[0,\pi]$, $\sin\theta=+\sqrt{1-\frac{1}{16}}=\frac{\sqrt{15}}{4}$ (positive on $[0,\pi]$).
2. $\cos 2\theta = 1-2\sin^2\theta = 1-2\cdot\frac{15}{16}=1-\frac{30}{16}=1-\frac{15}{8}$.
3. $= \boxed{-\frac{7}{8}}$, matching the cosine route.

INSIGHT

For $\cos\left(2\cos^{-1}x\right)$ always prefer $2x^2-1$ – it uses the given value verbatim with no square root and no sign decision. The form $1-2\sin^2\theta$ is correct too but needlessly drags in $\sqrt{1-x^2}$ whose sign you would then have to justify.

№56 The Surd in the Denominator

●●●●● L4

EVALUATE

$$\tan\left(2\cos^{-1}\left(-\frac{3}{\sqrt{13}}\right)\right)$$

tangent · double-angle · obtuse-quadrant · surd-input · sign-of-root

ANSWER

$$-\frac{12}{5}$$

THE TRAP

The naive student builds a right triangle with legs 3 and 2 (hypotenuse $\sqrt{13}$), reads $\tan\theta=\frac{2}{3}$, and gets $\tan 2\theta=\frac{2(2/3)}{1-4/9}=+\frac{12}{5}$. But $\theta=\cos^{-1}(-\frac{3}{\sqrt{13}})\in(\frac{\pi}{2},\pi)$ where cosine is negative and sine positive, so $\tan\theta=\frac{+2/\sqrt{13}}{-3/\sqrt{13}}=-\frac{2}{3}$. The negative tangent flips the double-angle result to $-\frac{12}{5}$. Forgetting the obtuse-quadrant sign of $\tan\theta$ is the whole trap.

Tangent of the angle, then double

1. $\theta=\cos^{-1}(-\frac{3}{\sqrt{13}})\in(\frac{\pi}{2},\pi)$: $\cos\theta=-\frac{3}{\sqrt{13}}$, $\sin\theta=+\frac{2}{\sqrt{13}}$, hence $\tan\theta=-\frac{2}{3}$.
2. $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta} = \dfrac{2(-\frac{2}{3})}{1-\frac{4}{9}} = \dfrac{-\frac{4}{3}}{\frac{5}{9}} = -\frac{4}{3}\cdot\frac{9}{5}$.
3. $= \boxed{-\frac{12}{5}}$.

Build $\sin2\theta$ and $\cos2\theta$ separately

1. With $\cos\theta = -\frac{3}{\sqrt{13}}$, $\sin\theta = \frac{2}{\sqrt{13}}$: $\cos 2\theta = 2\cos^2\theta - 1 = 2\cdot\frac{9}{13} - 1 = \frac{5}{13}$.
2. $\sin 2\theta = 2\sin\theta\cos\theta = 2\cdot\frac{2}{\sqrt{13}}\cdot\left(-\frac{3}{\sqrt{13}}\right) = -\frac{12}{13}$.
3. $\tan 2\theta = \dfrac{\sin 2\theta}{\cos 2\theta} = \dfrac{-12/13}{5/13} = \boxed{-\dfrac{12}{5}}$.

INSIGHT

A surd input like $-\frac{3}{\sqrt{13}}$ hides the quadrant in plain sight: the minus sign means $\theta \in (\frac{\pi}{2}, \pi)$, so $\tan\theta$ inherits the sign of $\frac{(+)}{(-)}$. Drawing the reference triangle is fine, but the leg signs must be read off the quadrant, not off the bare magnitudes.

№57 Secant Meets Cosecant

L4

EVALUATE

$$\sin\left(\sec^{-1}\left(-\tfrac{13}{5}\right) + \csc^{-1}\tfrac{5}{3}\right)$$

sine · addition-formula · secant-inverse · cosecant-inverse · sign-of-root

ANSWER

$$\frac{33}{65}$$

THE TRAP

Two reciprocal-inverse pitfalls. For $\sec^{-1}(-\frac{13}{5})$: the range is $[0,\pi]\setminus\{\frac{\pi}{2}\}$, and since the input is negative the angle is in $(\frac{\pi}{2},\pi]$, so $\cos = -\frac{5}{13}$ but $\sin = +\frac{12}{13}$ — a student who treats it like a 5-12-13 first-quadrant triangle writes $\cos = +\frac{5}{13}$ and gets $\frac{63}{65}$. For $\csc^{-1}\frac{5}{3} \in (0, \frac{\pi}{2}]$: $\sin = \frac{3}{5}, \cos = +\frac{4}{5}$. The correct combination is $\frac{33}{65}$, not $\frac{63}{65}$.

Convert to sine/cosine, track ranges

1. $a = \sec^{-1}(-\frac{13}{5}) \in (\frac{\pi}{2}, \pi]$: $\cos a = \frac{1}{\sec a} = -\frac{5}{13}$, and $\sin a = +\frac{12}{13}$ (sine ≥ 0 on $[0,\pi]$).
2. $b = \csc^{-1}\frac{5}{3} \in (0, \frac{\pi}{2}]$: $\sin b = \frac{1}{\csc b} = \frac{3}{5}$, $\cos b = +\frac{4}{5}$.
3. $\sin(a+b) = \sin a\cos b + \cos a \sin b = \frac{12}{13}\cdot\frac{4}{5} + \left(-\frac{5}{13}\right)\frac{3}{5} = \frac{48-15}{65} = \boxed{\dfrac{33}{65}}$.

Rewrite via $\cos^{-1}$ and $\sin^{-1}$

1. $\sec^{-1}(-\frac{13}{5}) = \cos^{-1}(-\frac{5}{13})$ and $\csc^{-1}\frac{5}{3} = \sin^{-1}\frac{3}{5}$, both standard.
2. $\cos^{-1}(-\frac{5}{13}) = \pi - \cos^{-1}\frac{5}{13}$, an obtuse angle with $\sin = \frac{12}{13}, \cos = -\frac{5}{13}$; and $\sin^{-1}\frac{3}{5}$ has $\sin = \frac{3}{5}, \cos = \frac{4}{5}$.
3. $\sin\left[(\pi - \cos^{-1}\frac{5}{13}) + \sin^{-1}\frac{3}{5}\right] = \frac{12}{13}\cdot\frac{4}{5} - \frac{5}{13}\cdot\frac{3}{5} = \boxed{\dfrac{33}{65}}$.

INSIGHT

The first move with $\sec^{-1}$ and $\csc^{-1}$ should always be to flip to $\cos^{-1}\frac{1}{x}$ and $\sin^{-1}\frac{1}{x}$ – their ranges then match the familiar $[0,\pi]$ and $[-\frac{\pi}{2}, \frac{\pi}{2}]$, and the sign of the cosine for a negative-input $\sec^{-1}$ becomes obvious.

№58 The Symbolic Mirror

L3

SIMPLIFY

$$\cos\left(\sin^{-1}x - \cos^{-1}x\right) \quad \text{for } x \in [-1,1]$$

cosine · symbolic · complementary-identity · double-angle

ANSWER

$$2x\sqrt{1-x^2}$$

THE TRAP

A common wrong move is to expand $\cos(\sin^{-1}x - \cos^{-1}x) = \cos(\sin^{-1}x)\cos(\cos^{-1}x) + \sin(\sin^{-1}x)\sin(\cos^{-1}x) = \sqrt{1-x^2}\cdot x + x\cdot\sqrt{1-x^2} = 2x\sqrt{1-x^2}$ — which is actually correct, but students often mis-sign $\sin(\cos^{-1}x)$ as $\pm$ depending on x. Since $\cos^{-1}x \in [0,\pi]$, $\sin(\cos^{-1}x) = +\sqrt{1-x^2}$ for **all** $x \in [-1,1]$ (never negative), and $\cos(\sin^{-1}x) = +\sqrt{1-x^2}$ likewise. Both roots are non-negative across the whole domain; the result $2x\sqrt{1-x^2}$ holds with no case split.

Reduce via the complementary identity

1. Use $\cos^{-1}x = \frac{\pi}{2} - \sin^{-1}x$ (valid for all $x \in [-1,1]$). Then $\sin^{-1}x - \cos^{-1}x = 2\sin^{-1}x - \frac{\pi}{2}$.
2. $\cos\left(2\sin^{-1}x - \frac{\pi}{2}\right) = \cos(2\theta - \frac{\pi}{2}) = \sin 2\theta$ where $\theta = \sin^{-1}x$.
3. $\sin 2\theta = 2\sin\theta\cos\theta = 2x\sqrt{1-x^2}$, since $\cos\theta = +\sqrt{1-x^2}$ on $[-\frac{\pi}{2}, \frac{\pi}{2}]$. Answer: $\boxed{2x\sqrt{1-x^2}}$.

Direct cosine-difference expansion

1. $\cos(A-B) = \cos A\cos B + \sin A\sin B$ with $A = \sin^{-1}x,\ B = \cos^{-1}x$.
2. $\cos A = \sqrt{1-x^2},\ \sin A = x,\ \cos B = x,\ \sin B = \sqrt{1-x^2}$ — all roots positive because $A \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ and $B \in [0,\pi]$.
3. $= \sqrt{1-x^2}\cdot x + x\cdot\sqrt{1-x^2} = \boxed{2x\sqrt{1-x^2}}$.

INSIGHT

The complementary identity $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$ collapses any combination of these two into a single $\sin^{-1}x$, turning the problem into a one-variable double-angle. And the key sign fact: on their respective principal ranges, both $\cos(\sin^{-1}x)$ and $\sin(\cos^{-1}x)$ equal $+\sqrt{1-x^2}$ everywhere – no $\pm$.

№59 Halving a Near-Straight Angle

●●●●● L4

EVALUATE

$$\sin\left(\tfrac{1}{2}\cos^{-1}\left(-\tfrac{119}{169}\right)\right)$$

half-angle · sine · obtuse-quadrant · sign-of-root · pythagorean-triple

ANSWER

$$\frac{12}{13}$$

THE TRAP

The half-angle formula gives $\sin\frac{t}{2} = \pm\sqrt{\frac{1-\cos t}{2}} = \pm\sqrt{\frac{1+119/169}{2}} = \pm\sqrt{\frac{288/169}{2}} = \pm\frac{12}{13}$. Students who anchor on the negative input often pick $-\frac{12}{13}$. But $t = \cos^{-1}(-\frac{119}{169}) \in (\frac{\pi}{2}, \pi)$, so $\frac{t}{2} \in (\frac{\pi}{4}, \frac{\pi}{2})$ — first quadrant — where sine is **positive**. The answer is $+\frac{12}{13}$. (The numbers come from the 119-120-169 Pythagorean triple.)

Half-angle surd with branch resolution

1. $\sin\frac{t}{2} = \pm\sqrt{\dfrac{1-\cos t}{2}} = \pm\sqrt{\dfrac{1+\frac{119}{169}}{2}} = \pm\sqrt{\dfrac{288}{338}} = \pm\sqrt{\dfrac{144}{169}} = \pm\dfrac{12}{13}$.
2. Branch: $t \in (\frac{\pi}{2}, \pi) \Rightarrow \frac{t}{2} \in (\frac{\pi}{4}, \frac{\pi}{2})$, where $\sin > 0$.
3. Hence $\sin\frac{t}{2} = \boxed{\dfrac{12}{13}}$.

Cross-check through the full triple

1. $\cos t = -\frac{119}{169} \Rightarrow \sin t = +\frac{120}{169}$ (from $169^2 - 119^2 = 120^2$, sine positive on $[0, \pi]$).
2. Then $\cos\frac{t}{2} = +\sqrt{\frac{1+\cos t}{2}} = \sqrt{\frac{50/169}{1} \cdot \frac{1}{2}} = \sqrt{\frac{25}{169}} = \frac{5}{13}$ (also positive in QI).
3. Verify $2\sin\frac{t}{2}\cos\frac{t}{2} = 2 \cdot \frac{12}{13} \cdot \frac{5}{13} = \frac{120}{169} = \sin t$. Consistent, so $\sin\frac{t}{2} = \boxed{\frac{12}{13}}$.

INSIGHT

When you halve an angle from $\cos^{-1}$, the output $t \in [0, \pi]$ forces $\frac{t}{2} \in [0, \frac{\pi}{2}]$ – the **first quadrant** – so both $\sin\frac{t}{2}$ and $\cos\frac{t}{2}$ come out positive every time. The $\pm$ in the half-angle surd is never genuinely ambiguous for a $\cos^{-1}$ input.

№60 The Sum That Overshoots

●●●●● L5

EVALUATE

$$\sin\left(\tan^{-1} 2 + \tan^{-1} 3\right)$$

sine | tangent-addition | branch-overflow | principal-value

ANSWER

$$\frac{1}{\sqrt{2}}$$

THE TRAP

The textbook reflex: $\tan(\tan^{-1} 2 + \tan^{-1} 3) = \dfrac{2+3}{1-2\cdot 3} = \dfrac{5}{-5} = -1$, conclude the sum is $-\frac{\pi}{4}$, hence $\sin = -\frac{1}{\sqrt{2}}$. This is WRONG. Both $\tan^{-1} 2$ and $\tan^{-1} 3$ exceed $\frac{\pi}{4}$, so their sum exceeds $\frac{\pi}{2}$ — it actually equals $\frac{3\pi}{4}$, which lies **outside** $(-\frac{\pi}{2}, \frac{\pi}{2})$. The arctangent-addition formula needs a $+\pi$ correction because $\tan^{-1} 2$ and $\tan^{-1} 3$ have product of arguments > 1. The true value is $\sin\frac{3\pi}{4} = +\frac{1}{\sqrt{2}}$.

Pin the angle by the addition rule with π-shift

1. $\tan$ of the sum is $\dfrac{2+3}{1-6} = -1$, so the sum is $\equiv -\frac{\pi}{4} \pmod{\pi}$. Candidates: $-\frac{\pi}{4}$ or $\frac{3\pi}{4}$.
2. Bound the sum: $\tan^{-1} 2 > \frac{\pi}{4}$ and $\tan^{-1} 3 > \frac{\pi}{4}$, so the sum $> \frac{\pi}{2}$; also each is $< \frac{\pi}{2}$, so the sum $< \pi$. Thus the sum is in $(\frac{\pi}{2}, \pi)$, forcing $\frac{3\pi}{4}$.
3. $\sin\frac{3\pi}{4} = \boxed{\frac{1}{\sqrt{2}}}$.

Right triangles plus quadrant of the resultant

1. Let $a = \tan^{-1} 2$ ($\sin a = \frac{2}{\sqrt{5}}, \cos a = \frac{1}{\sqrt{5}}$), $b = \tan^{-1} 3$ ($\sin b = \frac{3}{\sqrt{10}}, \cos b = \frac{1}{\sqrt{10}}$), both in QI.
2. $\cos(a+b) = \cos a\cos b - \sin a \sin b = \frac{1}{\sqrt{5}}\frac{1}{\sqrt{10}} - \frac{2}{\sqrt{5}}\frac{3}{\sqrt{10}} = \frac{1-6}{\sqrt{50}} = -\frac{5}{5\sqrt{2}} = -\frac{1}{\sqrt{2}}$.
3. $\sin(a+b) = \sin a\cos b + \cos a \sin b = \frac{2+3}{\sqrt{50}} = \frac{5}{5\sqrt{2}} = \boxed{\frac{1}{\sqrt{2}}}$ (positive, consistent with the QII sum $\frac{3\pi}{4}$).

Gaussian-integer argument

1. $\tan^{-1} 2 + \tan^{-1} 3 = \arg(1+2i) + \arg(1+3i) = \arg\left[(1+2i)(1+3i)\right]$.
2. $(1+2i)(1+3i) = 1 + 3i + 2i + 6i^2 = -5 + 5i$, which lies in the **second quadrant**, so $\arg = \frac{3\pi}{4}$.
3. $\sin\frac{3\pi}{4} = \boxed{\frac{1}{\sqrt{2}}}$ — the negative real part is exactly the signal that a naive $\tan$-formula would have missed.

INSIGHT

The arctangent addition formula $\tan^{-1} u + \tan^{-1} v = \tan^{-1} \frac{u+v}{1-uv} + k\pi$ needs $k = +1$ when $u, v > 0$ and $uv > 1$ (the sum spills past $\frac{\pi}{2}$). The Gaussian-integer trick exposes this instantly: the sign of the real part of the product fixes the quadrant, so you never have to memorize the k-rule.

CHAPTER VII

Domain, Range & Graphs

Where is the composite defined, and how often does its graph cross?

№61 A Logarithm Inside the Cosine

L3

FIND THE DOMAIN

$$f(x) = \cos^{-1}(\log_2 x)$$

domain composite logarithm cosine-inverse

ANSWER

$$\left[\frac{1}{2}, 2\right]$$

THE TRAP

The seductive wrong answer is $(0, \infty)$, written by the student who only enforces the domain of $\log_2 x$ (namely $x > 0$) and forgets that the **output** of the logarithm must itself land in $[-1, 1]$ to be a legal input of $\cos^{-1}$. The composite is only defined where *both* the inner function is defined **and** its value lies in the outer function's domain.

Chain the two domain conditions

1. $\cos^{-1} u$ requires $-1 \leq u \leq 1$. With $u = \log_2 x$ this forces $-1 \leq \log_2 x \leq 1$.
2. Exponentiate base 2 (an increasing function, so inequalities are preserved): $2^{-1} \leq x \leq 2^1$, i.e. $\frac{1}{2} \leq x \leq 2$.
3. This automatically satisfies $x > 0$, so the domain is $\boxed{[\frac{1}{2}, 2]}$.

Solve the boundary equations

1. The endpoints of the inner range occur when $\log_2 x = -1$ and $\log_2 x = 1$.
2. These give $x = 2^{-1} = \frac{1}{2}$ and $x = 2^1 = 2$; since $\log_2$ is continuous and increasing, every x between them keeps $\log_2 x \in [-1, 1]$.
3. Hence $\boxed{[\frac{1}{2}, 2]}$.

INSIGHT

For any composite $g^{-1}(h(x))$ where g^{-1} is an inverse trig function, the domain is the **pre-image** under h of g^{-1}'s natural domain, intersected with the domain of h. Here the $\cos^{-1}$ constraint $[-1, 1]$ pulls back through $\log_2$ to the symmetric-in-log interval $[\frac{1}{2}, 2]$.

№62 Sine-Inverse Meets a Logarithmic Tail

L3

FIND THE DOMAIN

$$f(x) = \sin^{-1}\left(\frac{x-3}{2}\right) + \ln(4 - x)$$

domain · sine-inverse · logarithm · intersection

ANSWER

$$[1,\ 4)$$

THE TRAP

The tempting wrong answer is $[1,5]$, from solving only $-1 \le \frac{x-3}{2} \le 1$. That ignores the $\ln(4-x)$ term, which demands the **strict** inequality $4-x>0$. The right endpoint is therefore open at 4, not closed at 5 — the logarithm both shrinks the interval and removes its endpoint.

Intersect the two requirements

1. $\sin^{-1}$ needs $-1 \le \frac{x-3}{2} \le 1 \Rightarrow -2 \le x-3 \le 2 \Rightarrow 1 \le x \le 5$.
2. $\ln(4-x)$ needs $4-x>0 \Rightarrow x<4$.
3. Intersecting $[1,5]\cap(-\infty,4)$ gives $\boxed{[1,4)}$.

Number-line overlay

1. Shade $[1,5]$ for the arcsine piece and the ray $x<4$ for the log piece on one number line.
2. The overlap starts at the closed left end $x=1$ and stops just before $x=4$, where the log blows up.
3. Read off $\boxed{[1,4)}$.

INSIGHT

When a sum of functions is involved, the domain is the **intersection** of the individual domains. A logarithm contributes a strict inequality, so it can convert a closed endpoint of another piece into an open one – always track open vs. closed separately.

№63 Two Increasing Arcs Stacked

●●●●● L4

FIND THE RANGE

$$f(x) = \sin^{-1} x + \tan^{-1} x$$

range · monotonic · sine-inverse · tangent-inverse

ANSWER

$$\left[-\frac{3\pi}{4}, \frac{3\pi}{4}\right]$$

THE TRAP

A common wrong answer is $\left(-\frac{3\pi}{4}, \frac{3\pi}{4}\right)$ with **open** ends, copied reflexively from $\tan^{-1}$'s open range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. But the *domain* of f is forced down to $[-1,1]$ by the $\sin^{-1}$ term, and at the attainable endpoint $x=1$ we get $\frac{\pi}{2}+\frac{\pi}{4}=\frac{3\pi}{4}$ exactly. The extreme values **are** reached, so the range is closed.

Monotonicity on the shared domain

1. $\sin^{-1} x$ is defined only on $[-1,1]$, so f has domain $[-1,1]$; on it $\tan^{-1} x$ is also defined.
2. Both $\sin^{-1} x$ and $\tan^{-1} x$ are strictly increasing, so their sum f is strictly increasing on $[-1,1]$.
3. Therefore the range is $[\,f(-1),\ f(1)\,] = \left[-\frac{\pi}{2}-\frac{\pi}{4},\ \frac{\pi}{2}+\frac{\pi}{4}\right] = \boxed{\left[-\frac{3\pi}{4}, \frac{3\pi}{4}\right]}$.

Odd-function symmetry

1. Both summands are odd, so $f(-x)=-f(x)$; the range is symmetric about 0.

2. Hence it suffices to find the maximum, which (by monotonicity) is at $x = 1$: $f(1) = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$.

3. By symmetry the minimum is $-\frac{3\pi}{4}$, giving $\boxed{\left[-\frac{3\pi}{4}, \frac{3\pi}{4}\right]}$.

INSIGHT

The domain of a *sum* is the intersection of domains, so the most restrictive term ($\sin^{-1}$, domain $[-1,1]$) caps everything. Whenever the function is monotone on a **closed** domain, endpoints are attained and the range is closed – the openness of $\tan^{-1}$'s range is irrelevant once x is confined to $[-1,1]$.

№64 A Weighted Difference of Arcs

●●●●● L4

FIND THE RANGE

$$f(x) = 2\cos^{-1}x - \sin^{-1}x$$

range · monotonic · cosine-inverse · sine-inverse · weighted

ANSWER

$$\left[-\frac{\pi}{2}, \frac{5\pi}{2}\right]$$

THE TRAP

The classic blunder is to write $\cos^{-1}x = \frac{\pi}{2} - \sin^{-1}x$, substitute, and get $f = \pi - 3\sin^{-1}x$ with range $[\pi - \frac{3\pi}{2}, \pi + \frac{3\pi}{2}] = [-\frac{\pi}{2}, \frac{5\pi}{2}]$ — which is in fact correct here, but students who instead mis-pair the endpoints (taking $\cos^{-1}$ and $\sin^{-1}$ at the *same* sign of extreme) report $[0, 2\pi]$ or $[\frac{\pi}{2}, \frac{3\pi}{2}]$. The pitfall is forgetting that $\cos^{-1}$ is **decreasing** while $-\sin^{-1}$ is also decreasing, so both extremes line up at the **same** endpoint, not opposite ones.

Reduce to a single arc

1. On $[-1,1]$ use $\cos^{-1}x = \frac{\pi}{2} - \sin^{-1}x$, so $f = 2\left(\frac{\pi}{2} - \sin^{-1}x\right) - \sin^{-1}x = \pi - 3\sin^{-1}x$.
2. As $\sin^{-1}x$ ranges over $[-\frac{\pi}{2}, \frac{\pi}{2}]$, $-3\sin^{-1}x$ ranges over $[-\frac{3\pi}{2}, \frac{3\pi}{2}]$.
3. Adding π: $f \in \left[\pi - \frac{3\pi}{2},\ \pi + \frac{3\pi}{2}\right] = \boxed{\left[-\frac{\pi}{2}, \frac{5\pi}{2}\right]}$.

Direct monotonicity at endpoints

1. $\cos^{-1}x$ and $-\sin^{-1}x$ are both decreasing, so $f = 2\cos^{-1}x - \sin^{-1}x$ is strictly decreasing on $[-1,1]$.
2. Maximum at $x = -1$: $2\pi - (-\frac{\pi}{2}) = \frac{5\pi}{2}$; minimum at $x = 1$: $2 \cdot 0 - \frac{\pi}{2} = -\frac{\pi}{2}$.
3. Range $= \boxed{\left[-\frac{\pi}{2}, \frac{5\pi}{2}\right]}$.

INSIGHT

The identity $\cos^{-1}x + \sin^{-1}x = \frac{\pi}{2}$ (valid for all $x \in [-1,1]$) collapses any linear combination $a\cos^{-1}x + b\sin^{-1}x$ into a single affine function of $\sin^{-1}x$, after which the range is read off the endpoints $\pm\frac{\pi}{2}$ – provided you respect that both pieces are decreasing.

№65 Counting Lattice Points in a Split Domain

●●●●● L4

FIND THE NUMBER OF INTEGERS IN THE DOMAIN

$$f(x) = \sin^{-1}\left(\log_3 \frac{x^2}{3}\right)$$

domain · integer-count · logarithm · disconnected · sine-inverse

ANSWER

$$6$$

THE TRAP

A strong student often lands on 7 by treating the domain as the full interval $[-3,3]$ and counting $-3,\ldots,3$. Two slips drive this: (i) the domain is the *disconnected* set $[-3,-1]\cup[1,3]$, not one interval, because the lower bound $|x|\ge 1$ carves out the whole middle; and (ii) $x=0$ is excluded outright since $\log_3(0/3)$ is undefined. Dropping the $|x|\ge 1$ condition wrongly keeps $x=0$ (and the rest of $(-1,1)$ in x), giving 7 instead of the correct 6.

Pull the arcsine constraint through the log

1. Require $-1\le \log_3\frac{x^2}{3}\le 1$, i.e. $3^{-1}\le \frac{x^2}{3}\le 3^1$.
2. Multiply by 3: $1\le x^2\le 9$, so $1\le |x|\le 3$, giving the domain $[-3,-1]\cup[1,3]$ (note $x=0$ already excluded since $x^2>0$ is needed).
3. Integers in this set: $\{-3,-2,-1\}\cup\{1,2,3\}$, a total of $\boxed{6}$.

Substitute $t=x^2$

1. Let $t=x^2\ge 0$; the condition becomes $1\le t\le 9$.
2. For each such t the solutions are $x=\pm\sqrt{t}$, so admissible $|x|\in[1,3]$.
3. The only integers with $|x|\in[1,3]$ are $\pm1,\pm2,\pm3$ — exactly $\boxed{6}$.

INSIGHT

Even powers (x^2) turn a single interval condition on the inner expression into a **symmetric pair** of intervals in x, automatically excluding any neighbourhood of 0 that would violate the lower bound. Always solve $|x|\in[a,b]$ explicitly and count both sign branches; never assume the domain is one connected piece.

№66 A Circle Caps the Arcsine

●●●●● L4

FIND THE NUMBER OF INTEGERS IN THE DOMAIN

$$f(x)=\sin^{-1}\left(\frac{|x|-2}{3}\right)+\sqrt{16-x^2}$$

domain · integer-count · square-root · absolute-value · intersection

ANSWER

$$9$$

THE TRAP

The seductive wrong answer is 11, from solving only the arcsine condition $-1\le\frac{|x|-2}{3}\le 1\Rightarrow |x|\le 5\Rightarrow[-5,5]$ and counting $-5,\ldots,5$. That ignores the square root, which clamps x to $[-4,4]$. The $\sqrt{16-x^2}$ factor is the binding constraint, not the arcsine — the more restrictive interval wins.

Intersect interval conditions

1. Arcsine: $-1\le\frac{|x|-2}{3}\le 1\Rightarrow -3\le |x|-2\le 3\Rightarrow |x|\le 5$, i.e. $x\in[-5,5]$.
2. Square root: $16-x^2\ge 0\Rightarrow x\in[-4,4]$.
3. Intersection is $[-4,4]$; integers $-4,-3,\ldots,4$ number $\boxed{9}$.

Test the binding constraint directly

1. Since $[-4, 4] \subset [-5, 5]$, the square root is the tighter requirement, so the domain equals $[-4, 4]$.
2. Count integers from -4 to 4 inclusive: $4 - (-4) + 1 = 9$.
3. Hence $\boxed{9}$ integers.

INSIGHT
With a sum, intersect the domains and let the **smaller** interval govern. A handy check: list the candidate intervals, find which is a subset of the others, and that subset is the answer – no need to re-test every point.

№67 Sum of Squared Arcs

●●●●● L4

FIND THE RANGE

$$f(x) = \left(\sin^{-1} x\right)^2 + \left(\cos^{-1} x\right)^2$$

range · quadratic · sine-inverse · cosine-inverse · optimization

ANSWER

$$\left[\frac{\pi^2}{8}, \frac{5\pi^2}{4}\right]$$

THE TRAP
The classic trap gives $\left[0, \frac{\pi^2}{2}\right]$ or even $\left[\frac{\pi^2}{16}, \frac{\pi^2}{4}\right]$ by minimizing each square independently. But the two squares are **coupled** through $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$; you cannot make both small at once. The true minimum is at the *interior* point where $\sin^{-1} x = \cos^{-1} x = \frac{\pi}{4}$, not at an endpoint.

Single-variable quadratic

1. Let $a = \sin^{-1} x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$; then $\cos^{-1} x = \frac{\pi}{2} - a$ and $f = a^2 + \left(\frac{\pi}{2} - a\right)^2 = 2a^2 - \pi a + \frac{\pi^2}{4}$.
2. This upward parabola in a has vertex at $a = \frac{\pi}{4}$ (inside the interval), giving the minimum $f = \frac{\pi^2}{8}$.
3. The maximum is at the endpoint $a = -\frac{\pi}{2}$: $f = 2 \cdot \frac{\pi^2}{4} + \frac{\pi^2}{2} + \frac{\pi^2}{4} = \frac{5\pi^2}{4}$. Range $= \boxed{\left[\frac{\pi^2}{8}, \frac{5\pi^2}{4}\right]}$.

Sum-and-product identity

1. With $s = a + b = \frac{\pi}{2}$ fixed and $p = ab$, write $f = a^2 + b^2 = s^2 - 2p = \frac{\pi^2}{4} - 2p$.
2. As a runs over $[-\frac{\pi}{2}, \frac{\pi}{2}]$, the product $p = a\left(\frac{\pi}{2} - a\right)$ is maximized at $a = \frac{\pi}{4}$ ($p = \frac{\pi^2}{16}$, giving $f_{\min} = \frac{\pi^2}{8}$) and minimized at $a = -\frac{\pi}{2}$ ($p = -\frac{\pi^2}{2}$, giving $f_{\max} = \frac{5\pi^2}{4}$).
3. Therefore the range is $\boxed{\left[\frac{\pi^2}{8}, \frac{5\pi^2}{4}\right]}$.

INSIGHT
Whenever two inverse-trig values are linked by $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, substitute to one variable. A sum of squares of two quantities with fixed sum is minimized when they are **equal** and maximized at the **extreme** of the allowed interval – the AM–QM intuition made rigorous by the parabola.

№68 The Branch-Folded Constant

●●●●● L4

FIND THE RANGE

$$f(x) = 2\tan^{-1} x + \sin^{-1}\left(\frac{2x}{1 + x^2}\right)$$

range · branch · double-angle · tangent-inverse · piecewise

ANSWER

$$[-\pi,\ \pi]$$

THE TRAP

The fatal shortcut is to declare $\sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2\tan^{-1}x$ for **all** x, get $f = 4\tan^{-1}x$, and report the *open* range $(-2\pi, 2\pi)$. That identity holds only on $|x| \le 1$. For $x > 1$ it becomes $\pi - 2\tan^{-1}x$ and for $x < -1$ it is $-\pi - 2\tan^{-1}x$, so f is the **constant** π (resp. $-\pi$) on those tails — capping the range at $\pm\pi$, both attained.

Piecewise branch reduction

1. Use the branched identity: $\sin^{-1}\frac{2x}{1+x^2} = 2\tan^{-1}x$ for $|x| \le 1$, $= \pi - 2\tan^{-1}x$ for $x > 1$, $= -\pi - 2\tan^{-1}x$ for $x < -1$.
2. Then $f = 4\tan^{-1}x$ on $[-1, 1]$ (range $[-\pi, \pi]$ since $\tan^{-1}(\pm 1) = \pm\frac{\pi}{4}$); $f = \pi$ for $x > 1$; $f = -\pi$ for $x < -1$.
3. Combining, every value in $[-\pi, \pi]$ is hit on $[-1, 1]$ and the tails only repeat $\pm\pi$. Range $= \boxed{[-\pi, \pi]}$.

Continuity plus boundedness

1. f is continuous and odd; on $[-1, 1]$ it equals $4\tan^{-1}x$, monotonically sweeping $[-\pi, \pi]$.
2. For $|x| > 1$ the two terms' $\tan^{-1}x$ parts cancel, freezing f at $\pm\pi$ — so f never exceeds π in absolute value.
3. Thus the supremum π and infimum $-\pi$ are attained (at $x = 1$ and $x = -1$), and the range is $\boxed{[-\pi, \pi]}$.

INSIGHT

The identity $\sin^{-1}\frac{2x}{1+x^2} = 2\tan^{-1}x$ is a **branch trap**: it is valid only on $[-1, 1]$ because the left side is bounded by $\frac{\pi}{2}$ while $2\tan^{-1}x$ escapes that band for $|x| > 1$. Outside, the principal-value fold turns the function constant. Always state the validity interval of any inverse-trig identity before using it in a range computation.

№69 When the Secant Argument Never Fails

●●●●● L5

FIND THE DOMAIN

$$f(x) = \sec^{-1}\left(\frac{x^2+3}{2x}\right)$$

domain · secant-inverse · rational · am-gm · absolute-value

ANSWER

$$\mathbb{R} \setminus \{0\}$$

THE TRAP

The plausible wrong answer is $[\sqrt{3}, \infty) \cup (-\infty, -\sqrt{3}]$, obtained by a student who solves $\frac{x^2+3}{2x} \ge 1$ as if only the *positive* branch mattered, forgetting that $\sec^{-1}$ admits arguments with $|\cdot| \ge 1$ — including all the large-negative values produced when $x < 0$. The correct condition is $\left|\frac{x^2+3}{2x}\right| \ge 1$, which by AM–GM holds for **every** nonzero x.

AM–GM on the absolute value

1. $\sec^{-1}u$ requires $|u| \ge 1$. Here $|u| = \dfrac{x^2+3}{2|x|}$ (numerator positive).
2. Set $t = |x| > 0$. By AM–GM, $\dfrac{t^2+3}{2t} = \dfrac{t}{2} + \dfrac{3}{2t} \ge 2\sqrt{\dfrac{t}{2}\cdot\dfrac{3}{2t}} = 2\sqrt{\dfrac{3}{4}} = \sqrt{3}$, with equality at $t = \sqrt{3}$.
3. Since $\sqrt{3} > 1$, we get $|u| \ge \sqrt{3} > 1$ for all $x \ne 0$ (and $x = 0$ is excluded by the denominator). Domain $= \boxed{\mathbb{R} \setminus \{0\}}$.

Quadratic positivity

1. The condition $\left|\frac{x^2+3}{2x}\right| \geq 1$ is equivalent to $x^2+3 \geq 2|x|$, i.e. $|x|^2 - 2|x| + 3 \geq 0$.
2. The quadratic $u^2 - 2u + 3 = (u-1)^2 + 2$ in $u = |x|$ is always $\geq 2 > 0$, so the inequality holds for every real x.
3. Only $x = 0$ (where the expression is undefined) is barred, so the domain is $\boxed{\mathbb{R} \setminus \{0\}}$.

INSIGHT

For $\sec^{-1}$ and $\csc^{-1}$ the gate is $|u| \geq 1$, an **absolute-value** condition – never just $u \geq 1$. Rational arguments of the form $\frac{x^2+c}{2x}$ recur because $\frac{x^2+c}{2|x|} = \frac{|x|}{2} + \frac{c}{2|x|}$ has a minimum $\sqrt{c}$ by AM–GM; here $\sqrt{3} > 1$ guarantees the whole punctured line qualifies.

№70 A Gentle Line Through the Sawtooth

●●●●● L3

FIND THE NUMBER OF SOLUTIONS

$$\sin^{-1}(\sin x) = \frac{x}{10}$$

graph · sawtooth · intersection · sine-inverse · periodic

ANSWER

$$11$$

THE TRAP

The lazy wrong answer is 1 (just $x = 0$), from a student who never sketches the left side and assumes $\sin^{-1}(\sin x) = x$. In fact $y = \sin^{-1}(\sin x)$ is the **triangular wave** of amplitude $\frac{\pi}{2}$ and period 2π, and the shallow line $y = \frac{x}{10}$ slices through many of its rising/falling arms before escaping the band $|y| \leq \frac{\pi}{2}$ at $|x| = 5\pi$.

Graph the wave against the line

1. $y = \sin^{-1}(\sin x)$ is a triangular wave oscillating between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ with period 2π; the line $y = \frac{x}{10}$ stays in $|y| \leq \frac{\pi}{2}$ only for $|x| \leq 5\pi \approx 15.7$.
2. Within $[-5\pi, 5\pi]$ the line crosses the wave once near the origin and twice per relevant peak/valley on each side; careful counting of the upward and downward arms gives 5 intersections for $x > 0$, 5 for $x < 0$, plus the crossing at $x = 0$.
3. Total $= 5 + 5 + 1 = \boxed{11}$.

Symmetry and arm-counting

1. Both sides are odd, so non-zero roots pair up $\pm x$ and $x = 0$ is automatically a solution; the count is therefore odd.
2. On $x > 0$ the line $\frac{x}{10}$ rises slowly enough to meet every wave arm whose value-band it still overlaps, up to the last arm before $x = 5\pi$ — five intersections.
3. Doubling for symmetry and adding the origin: $2 \cdot 5 + 1 = \boxed{11}$.

INSIGHT

Equations $\sin^{-1}(\sin x) =$ line are pure **graph-reading** problems: replace the left side by its triangular-wave picture (slopes ± 1, amplitude $\frac{\pi}{2}$, period 2π) and count crossings of the line within the strip $|y| \leq \frac{\pi}{2}$. A shallower slope $\frac{1}{k}$ reaches the strip edge at $|x| = k\frac{\pi}{2}$, admitting more arms and hence more solutions.

CHAPTER VIII

The Calculus of Inverse Trig

d/dx

Differentiate across a branch point and the chain rule lies by a sign.

№71 The Phantom Jump

●●●●● L4

DIFFERENTIATE

Let $f(x) = \tan^{-1}\left(\dfrac{2x}{1-x^2}\right)$. Find $f'(x)$ for all $x \neq \pm 1$, and decide whether f is differentiable at $x = 2$.

derivative · branch · tangent-double-angle · discontinuity

ANSWER

$$f'(x) = \frac{2}{1+x^2} \text{ for all } x \neq \pm 1; \quad f \text{ is differentiable at } x = 2 \text{ with } f'(2) = \tfrac{2}{5}.$$

THE TRAP

The seductive move is to write $f(x) = 2\tan^{-1} x$ everywhere (the double-angle identity), then claim a clean global derivative — or, worse, to declare f non-differentiable at $x = 2$ because 'the formula breaks at ± 1.' The identity $\tan^{-1}\frac{2x}{1-x^2} = 2\tan^{-1} x$ holds ONLY on $|x| < 1$. For $x > 1$ the true value is $2\tan^{-1} x - \pi$. The constant $-\pi$ has zero derivative, so the *derivative* $\frac{2}{1+x^2}$ is identical on every interval — the jumps live in f itself, at $x = \pm 1$, not at $x = 2$. So $x = 2$ is perfectly smooth.

Piecewise identity reduction

1. With $x = \tan\theta$, $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$, the argument is $\tan 2\theta$, so $f(x) = \tan^{-1}(\tan 2\theta)$ which equals 2θ only when $2\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$, i.e. $|x| < 1$.
2. On $|x| < 1$: $f(x) = 2\tan^{-1} x$. On $x > 1$: $2\theta \in (\frac{\pi}{2}, \pi)$, so $f(x) = 2\tan^{-1} x - \pi$. On $x < -1$: $f(x) = 2\tan^{-1} x + \pi$.
3. Differentiating each piece, the additive constants $\pm\pi$ vanish: $f'(x) = \dfrac{2}{1+x^2}$ on every interval, $x \neq \pm 1$.
4. Since $2 \in (1, \infty)$ lies in the interior of a smooth piece, $f'(2) = \dfrac{2}{1+4} = \boxed{\dfrac{2}{5}}$ and f is differentiable there.

Direct chain rule

1. Let $u = \dfrac{2x}{1-x^2}$. Then $\dfrac{du}{dx} = \dfrac{2(1-x^2) - 2x(-2x)}{(1-x^2)^2} = \dfrac{2(1+x^2)}{(1-x^2)^2}$.
2. $f' = \dfrac{1}{1+u^2} \cdot \dfrac{du}{dx}$. Compute $1 + u^2 = \dfrac{(1-x^2)^2 + 4x^2}{(1-x^2)^2} = \dfrac{(1+x^2)^2}{(1-x^2)^2}$.
3. Hence $f' = \dfrac{(1-x^2)^2}{(1+x^2)^2} \cdot \dfrac{2(1+x^2)}{(1-x^2)^2} = \dfrac{2}{1+x^2}$ for all $x \neq \pm 1$ (the $(1-x^2)^2$ cancels cleanly).
4. This is finite and continuous at $x = 2$, giving $f'(2) = \boxed{\dfrac{2}{5}}$; the only failures of differentiability are the jump points $x = \pm 1$ where f is undefined.

INSIGHT
The single-line formula $\frac{2}{1+x^2}$ is *globally* correct for the derivative precisely because the branch corrections are **constants**. The function jumps by π at $x = \pm 1$ (e.g. $f(0.99) \approx 1.56$ but $f(1.01) \approx -1.56$), yet its slope is seamless everywhere it is defined. Moral: a discontinuity in f and a discontinuity in f' are different beasts – locating one tells you nothing about the other.

№72 Sawtooth Slopes

●●●●● L4

EVALUATE

$$\text{Let } g(x) = \sin^{-1}(\sin x). \text{ Evaluate } g'(2) + g'(5) + g'(8).$$

sawtooth · sin-inverse · piecewise · derivative-at-point

ANSWER

$$g'(2) + g'(5) + g'(8) = -1 + 1 - 1 = -1.$$

THE TRAP
The reflex is to apply the chain rule blindly: $g'(x) = \dfrac{\cos x}{\sqrt{1-\sin^2 x}} = \dfrac{\cos x}{|\cos x|}$, then carelessly cancel to 1, giving $g'(2) + g'(5) + g'(8) = 3$. The cancellation is illegal whenever $\cos x < 0$: $\sqrt{\cos^2 x} = |\cos x|$, NOT $\cos x$. So $g'(x) = \operatorname{sgn}(\cos x) = \pm 1$. At $x = 2$ and $x = 8$, $\cos$ is negative, so each contributes -1.

Sign of cosine

1. For x not an odd multiple of $\frac{\pi}{2}$, $g'(x) = \dfrac{\cos x}{\sqrt{1-\sin^2 x}} = \dfrac{\cos x}{|\cos x|} = \operatorname{sgn}(\cos x)$.
2. $x = 2$ rad lies in $(\frac{\pi}{2}, \pi) \approx (1.57, 3.14)$, so $\cos 2 < 0$ and $g'(2) = -1$.
3. $x = 5$ rad lies in $(\frac{3\pi}{2}, 2\pi) \approx (4.71, 6.28)$, so $\cos 5 > 0$ and $g'(5) = +1$.
4. $x = 8$ rad: subtract $2\pi \approx 6.28$ to get $1.72 \in (\frac{\pi}{2}, \pi)$, so $\cos 8 < 0$ and $g'(8) = -1$. Sum $= -1 + 1 - 1 = \boxed{-1}$.

Graph / triangle-wave picture

1. $g(x) = \sin^{-1}(\sin x)$ is the triangle wave: it equals $x - 2\pi k$ on rising stretches (slope $+1$) and $\pi - (x - 2\pi k)$ or similar on falling stretches (slope -1), with corners at odd multiples of $\frac{\pi}{2}$.
2. The slope is $+1$ exactly where the original $\sin$ curve is increasing (i.e. $\cos x > 0$) and -1 where it is decreasing.
3. $\sin$ decreases through $x = 2$, increases through $x = 5$, decreases through $x = 8$ (since $8 - 2\pi \approx 1.72$ is on a decreasing stretch).
4. Slopes $-1, +1, -1$ sum to $\boxed{-1}$.

INSIGHT
$\frac{d}{dx} \sin^{-1}(\sin x) = \operatorname{sgn}(\cos x)$ is the cleanest example of branch sign in calculus: the derivative is a square wave ± 1 with no in-between. The same logic gives $\frac{d}{dx} \cos^{-1}(\cos x) = \operatorname{sgn}(\sin x)$ and $\frac{d}{dx} \tan^{-1}(\tan x) = 1$ wherever defined. The error of dropping the absolute value is the single most common branch mistake in the entire subject.

№73 Two Faces of a Double Angle

●●●●● L4

DIFFERENTIATE

$$\text{Let } h(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right). \text{ Find } h'(x) \text{ for } |x| < 1 \text{ and for } |x| > 1, \text{ and state } h'\left(\tfrac{1}{\sqrt{3}}\right) \text{ and } h'(\sqrt{3}).$$

sin-inverse · double-angle · branch-sign · piecewise

ANSWER

$$h'(x) = \frac{2}{1+x^2}\ (|x|<1), \quad h'(x) = -\frac{2}{1+x^2}\ (|x|>1); \quad h'\Big(\frac{1}{\sqrt{3}}\Big) = \frac{3}{2}, \quad h'(\sqrt{3}) = -\frac{1}{2}.$$

THE TRAP

The trap is to substitute $x = \tan\theta$, recognise $\frac{2x}{1+x^2} = \sin 2\theta$, conclude $h(x) = 2\tan^{-1} x$ globally, and report $h'(x) = \frac{2}{1+x^2}$ everywhere — so $h'(\sqrt{3}) = \frac{2}{4} = \frac{1}{2}$. Wrong sign. $\sin^{-1}(\sin 2\theta) = 2\theta$ only when $2\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, i.e. $|x| \le 1$. For $|x| > 1$ ($|2\theta| > \frac{\pi}{2}$), $\sin^{-1}(\sin 2\theta) = \pi - 2\theta$ (or $-\pi - 2\theta$), whose derivative is $-\frac{2}{1+x^2}$. The correct value is $h'(\sqrt{3}) = -\frac{1}{2}$.

Tangent substitution with range tracking

1. Put $x = \tan\theta$, $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$; then the argument is $\sin 2\theta$, and $h = \sin^{-1}(\sin 2\theta)$.
2. If $|x| < 1$ then $|\theta| < \frac{\pi}{4}$, so $2\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ and $h = 2\theta = 2\tan^{-1} x \Rightarrow h' = \dfrac{2}{1+x^2}$.
3. If $x > 1$ then $\theta \in (\frac{\pi}{4}, \frac{\pi}{2})$, so $2\theta \in (\frac{\pi}{2}, \pi)$ and $\sin^{-1}(\sin 2\theta) = \pi - 2\theta \Rightarrow h' = -\dfrac{2}{1+x^2}$ (same for $x < -1$ by oddness).
4. Thus $h'(\frac{1}{\sqrt{3}}) = \dfrac{2}{1+1/3} = \boxed{\dfrac{3}{2}}$ and $h'(\sqrt{3}) = -\dfrac{2}{1+3} = \boxed{-\dfrac{1}{2}}$.

Brute chain rule exposing the modulus

1. $\dfrac{d}{dx}\sin^{-1}(u) = \dfrac{u'}{\sqrt{1-u^2}}$ with $u = \dfrac{2x}{1+x^2}$, $u' = \dfrac{2(1-x^2)}{(1+x^2)^2}$.
2. $1 - u^2 = \dfrac{(1+x^2)^2 - 4x^2}{(1+x^2)^2} = \dfrac{(1-x^2)^2}{(1+x^2)^2}$, so $\sqrt{1-u^2} = \dfrac{|1-x^2|}{1+x^2}$.
3. Hence $h' = \dfrac{2(1-x^2)}{(1+x^2)^2} \cdot \dfrac{1+x^2}{|1-x^2|} = \dfrac{2(1-x^2)}{(1+x^2)\,|1-x^2|} = \dfrac{2\ \mathrm{sgn}(1-x^2)}{1+x^2}$.
4. The $\mathrm{sgn}(1-x^2)$ is $+1$ for $|x| < 1$ and -1 for $|x| > 1$, reproducing $h'(\frac{1}{\sqrt{3}}) = \boxed{\dfrac{3}{2}}$, $h'(\sqrt{3}) = \boxed{-\dfrac{1}{2}}$.

INSIGHT

The hidden $|1-x^2|$ from $\sqrt{(1-x^2)^2}$ is what flips the sign at $|x| = 1$. Compare with Problem 'The Phantom Jump': for $\tan^{-1}\frac{2x}{1-x^2}$ the branch corrections were constants, so f' stayed $\frac{2}{1+x^2}$ throughout; here, for $\sin^{-1}\frac{2x}{1+x^2}$, the correction is the *non-constant* $\pi - 2\theta$, so the derivative genuinely changes sign. Always check whether the branch fix is a constant (slope unchanged) or a reflection (slope negated).

№74 A Cubic Whisper

●●●●○ L4

EVALUATE

$$\lim_{x\to 0} \frac{\tan^{-1} x - \sin^{-1} x}{x^3}.$$

limit · taylor-series · tan-inverse · sin-inverse

ANSWER

$$-\frac{1}{2}.$$

THE TRAP

A strong student rushes to L'Hopital or only keeps the first-order terms $\tan^{-1} x \approx x$, $\sin^{-1} x \approx x$, gets $\frac{x-x}{x^3} = 0$, and answers 0. The first-order parts cancel exactly; the limit is decided entirely by the x^3 coefficients, which have OPPOSITE signs: $\tan^{-1} x = x - \frac{x^3}{3} + \cdots$ but $\sin^{-1} x = x + \frac{x^3}{6} + \cdots$. Stopping at first order destroys the whole problem.

Maclaurin series

1. $\tan^{-1}x = x - \dfrac{x^3}{3} + O(x^5)$ and $\sin^{-1}x = x + \dfrac{x^3}{6} + O(x^5)$.
2. Subtract: $\tan^{-1}x - \sin^{-1}x = \Big(-\dfrac{1}{3} - \dfrac{1}{6}\Big)x^3 + O(x^5) = -\dfrac{1}{2}x^3 + O(x^5)$.
3. Divide by x^3 and let $x \to 0$: the limit is $\boxed{-\dfrac{1}{2}}$.

Repeated L'Hopital

1. The form is $\frac{0}{0}$. Differentiate top and bottom: $\dfrac{\frac{1}{1+x^2} - \frac{1}{\sqrt{1-x^2}}}{3x^2}$, still $\frac{0}{0}$.
2. Use binomial expansions $\frac{1}{1+x^2} = 1 - x^2 + \cdots$ and $\frac{1}{\sqrt{1-x^2}} = 1 + \frac{x^2}{2} + \cdots$; the numerator is $-x^2 - \frac{x^2}{2} + \cdots = -\frac{3}{2}x^2 + \cdots$.
3. Therefore the ratio $\to \dfrac{-\frac{3}{2}x^2}{3x^2} = -\dfrac{1}{2}$, so the limit is $\boxed{-\dfrac{1}{2}}$.

INSIGHT

The two famous cubic limits are $\lim_{x\to 0}\frac{x-\tan^{-1}x}{x^3} = +\frac{1}{3}$ and $\lim_{x\to 0}\frac{x-\sin^{-1}x}{x^3} = -\frac{1}{6}$. Their difference is what you see here. Memorise the cubic coefficients $\tan^{-1}: -\frac{1}{3}$, $\sin^{-1}: +\frac{1}{6}$, $\tan: +\frac{1}{3}$, $\sin: -\frac{1}{6}$ – the inverse functions have the opposite-sign cubic term of their forward counterparts, a fact that resolves an entire family of x^3 limits at sight.

№75 The Tilted Tangent

L4

DIFFERENTIATE

$$\text{For } x \in \left(0, \tfrac{3\pi}{4}\right) \text{ let } F(x) = \tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right). \text{ Find } F'(x), \text{ and find } F'(x) \text{ for } x \in \left(\tfrac{3\pi}{4}, \tfrac{7\pi}{4}\right).$$

tan-inverse · reduction · derivative-at-point · branch

ANSWER

$$F'(x) = -1 \text{ on } \left(0, \tfrac{3\pi}{4}\right) \text{ and also } F'(x) = -1 \text{ on } \left(\tfrac{3\pi}{4}, \tfrac{7\pi}{4}\right); \text{ but } F \text{ jumps by } \pi \text{ at } x = \tfrac{3\pi}{4}.$$

THE TRAP

Dividing numerator and denominator by $\cos x$ gives $\frac{1-\tan x}{1+\tan x} = \tan(\frac{\pi}{4} - x)$, so one writes $F(x) = \frac{\pi}{4} - x$ for ALL x and reports $F' = -1$ everywhere with no caveat. The reduction $\tan^{-1}(\tan(\frac{\pi}{4} - x)) = \frac{\pi}{4} - x$ is valid only while $\frac{\pi}{4} - x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, i.e. $x \in (-\frac{\pi}{4}, \frac{3\pi}{4})$. Beyond $x = \frac{3\pi}{4}$, $F(x) = \frac{5\pi}{4} - x$ — same slope -1, but the function leaps by $+\pi$ at $\frac{3\pi}{4}$. Claiming F is a single smooth line on the whole interval is the error.

Principal-value reduction

1. Divide top and bottom by $\cos x$ (valid where $\cos x \neq 0$): $F(x) = \tan^{-1}\big(\tan(\frac{\pi}{4} - x)\big)$.
2. On $(0, \frac{3\pi}{4})$, the angle $\frac{\pi}{4} - x \in (-\frac{\pi}{2}, \frac{\pi}{4}) \subset (-\frac{\pi}{2}, \frac{\pi}{2})$, so $F(x) = \frac{\pi}{4} - x$ and $F'(x) = \boxed{-1}$.
3. On $(\frac{3\pi}{4}, \frac{7\pi}{4})$, $\frac{\pi}{4} - x \in (-\frac{3\pi}{2}, -\frac{\pi}{2})$; add π to land in the principal range, giving $F(x) = \frac{5\pi}{4} - x$, so again $F'(x) = \boxed{-1}$.
4. The slope is -1 on each interval, but $F(\frac{3\pi}{4}^-) = -\frac{\pi}{2}$ while $F(\frac{3\pi}{4}^+) = +\frac{\pi}{2}$: a jump of $+\pi$, where $\cos x + \sin x = 0$ makes the argument blow up.

Chain rule on the quotient

1. Let $u = \dfrac{\cos x - \sin x}{\cos x + \sin x}$. By the quotient rule, $u' = \dfrac{-(\cos x + \sin x)^2 - (\cos x - \sin x)^2}{(\cos x + \sin x)^2} = \dfrac{-2}{(\cos x + \sin x)^2}$ (using $a^2 + b^2$ collapses to 2).

2. $1+u^2 = \dfrac{(\cos x+\sin x)^2+(\cos x-\sin x)^2}{(\cos x+\sin x)^2} = \dfrac{2}{(\cos x+\sin x)^2}$.

3. Thus $F' = \dfrac{u'}{1+u^2} = \dfrac{-2/(\cos x+\sin x)^2}{2/(\cos x+\sin x)^2} = -1$ wherever $\cos x+\sin x \neq 0$.

4. So $F'(x) = \boxed{-1}$ on both intervals; the only failure is at $x = \frac{3\pi}{4}$ where $\cos x+\sin x = 0$ and F has its π-jump.

INSIGHT

Identical lesson to 'The Phantom Jump' in trigonometric clothing: the chain-rule derivative is a constant -1 everywhere the expression is defined, and the branch corrections ($+\pi$) are constants that vanish on differentiation. The denominator vanishing ($\cos x+\sin x = 0 \Rightarrow x = \frac{3\pi}{4}$) is precisely where the $\tan^{-1}$ argument runs to $\pm\infty$ and F jumps. Locating zeros of the denominator finds every jump.

№76 Half the Angle, Half the Slope

L4

DIFFERENTIATE

For $x \in (0,\pi)$ let $P(x) = \tan^{-1}\sqrt{\dfrac{1-\cos x}{1+\cos x}}$. Find $P'(x)$, and find $Q'(x)$ where $Q(x) = \tan^{-1}\sqrt{\dfrac{1-\cos x}{1+\cos x}}$ on $(\pi, 2\pi)$.

half-angle · tan-inverse · domain · derivative-at-point

ANSWER

$$P'(x) = \frac{1}{2} \text{ on } (0,\pi); \quad Q'(x) = -\frac{1}{2} \text{ on } (\pi, 2\pi).$$

THE TRAP

Using $\frac{1-\cos x}{1+\cos x} = \tan^2\frac{x}{2}$, the student writes $\sqrt{\tan^2\frac{x}{2}} = \tan\frac{x}{2}$ and concludes $P(x) = \tan^{-1}(\tan\frac{x}{2}) = \frac{x}{2}$ for ALL x, hence $P' = \frac{1}{2}$ everywhere — including on $(\pi, 2\pi)$. The error: $\sqrt{\tan^2\frac{x}{2}} = \left|\tan\frac{x}{2}\right|$, and on $(\pi,2\pi)$ we have $\frac{x}{2} \in (\frac{\pi}{2}, \pi)$ where $\tan\frac{x}{2} < 0$, so $\left|\tan\frac{x}{2}\right| = -\tan\frac{x}{2} = \tan(\pi - \frac{x}{2})$, giving $Q(x) = \pi - \frac{x}{2}$ and $Q' = -\frac{1}{2}$.

Half-angle with modulus

1. $\dfrac{1-\cos x}{1+\cos x} = \dfrac{2\sin^2(x/2)}{2\cos^2(x/2)} = \tan^2\frac{x}{2}$, so $P(x) = \tan^{-1}\left|\tan\frac{x}{2}\right|$.

2. On $(0,\pi)$, $\frac{x}{2} \in (0, \frac{\pi}{2})$ so $\tan\frac{x}{2} > 0$: $P(x) = \tan^{-1}(\tan\frac{x}{2}) = \frac{x}{2} \Rightarrow P'(x) = \boxed{\frac{1}{2}}$.

3. On $(\pi, 2\pi)$, $\frac{x}{2} \in (\frac{\pi}{2}, \pi)$ so $\tan\frac{x}{2} < 0$: $Q(x) = \tan^{-1}(-\tan\frac{x}{2}) = \tan^{-1}\big(\tan(\pi - \frac{x}{2})\big) = \pi - \frac{x}{2}$.

4. Hence $Q'(x) = \boxed{-\frac{1}{2}}$ — the same expression has opposite slope on the two half-periods.

Chain rule

1. Set $u = \sqrt{\dfrac{1-\cos x}{1+\cos x}} = \left|\tan\frac{x}{2}\right|$ (valid on the whole domain, since $1+\cos x > 0$ there).

2. Differentiating $|f|$ gives $u' = \operatorname{sgn}\big(\tan\frac{x}{2}\big)\cdot\frac{1}{2}\sec^2\frac{x}{2}$, and $1+u^2 = 1+\tan^2\frac{x}{2} = \sec^2\frac{x}{2}$.

3. So $P' = \dfrac{u'}{1+u^2} = \dfrac{\frac{1}{2}\sec^2\frac{x}{2}\,\operatorname{sgn}(\tan\frac{x}{2})}{\sec^2\frac{x}{2}} = \dfrac{1}{2}\operatorname{sgn}\big(\tan\frac{x}{2}\big)$.

4. On $(0,\pi)$ the sign is $+$, giving $\boxed{\frac{1}{2}}$; on $(\pi, 2\pi)$ it is $-$, giving $\boxed{-\frac{1}{2}}$.

INSIGHT

Whenever a square root sits inside an inverse trig function, $\sqrt{(\cdot)^2} = |\cdot|$ is the hinge. Here the half-angle $\frac{x}{2}$ crosses $\frac{\pi}{2}$ at $x = \pi$, flipping the sign of $\tan\frac{x}{2}$ and hence the slope. The function P itself is the continuous tent $\min(\frac{x}{2}, \pi - \frac{x}{2})$-type wave; its derivative is the square wave $\frac{1}{2}\operatorname{sgn}(\tan\frac{x}{2})$.

№77 The Triple-Angle Mirage

●●●●● L5

DIFFERENTIATE

Let $S(x) = \sin^{-1}(3x - 4x^3),\ x \in [-1, 1]$. Find $S'(x)$ on each subinterval determined by $x = \pm\frac{1}{2}$, and give $S'(\frac{1}{4})$ and $S'(\frac{3}{4})$.

triple-angle · sin-inverse · domain · branch-sign · piecewise

ANSWER

$$S'(x) = \frac{3}{\sqrt{1-x^2}} \text{ on } \left(-\tfrac{1}{2}, \tfrac{1}{2}\right), \quad S'(x) = -\frac{3}{\sqrt{1-x^2}} \text{ on } \left(\tfrac{1}{2}, 1\right) \cup \left(-1, -\tfrac{1}{2}\right); \quad S'(\tfrac{1}{4}) = \frac{3}{\sqrt{15/4}} = \frac{12}{\sqrt{15}}, \quad S'(\tfrac{3}{4}) = -\frac{12}{\sqrt{7}}.$$

THE TRAP

Recognising $3x - 4x^3 = \sin 3\theta$ when $x = \sin\theta$, the student writes $S(x) = 3\sin^{-1} x$ on all of $[-1, 1]$ and reports $S'(x) = \frac{3}{\sqrt{1-x^2}}$ everywhere — so $S'(\frac{3}{4}) = +\frac{3}{\sqrt{7/4}} = +\frac{12}{\sqrt{7}}$. The identity $\sin^{-1}(\sin 3\theta) = 3\theta$ holds ONLY when $3\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, i.e. $\theta \in [-\frac{\pi}{6}, \frac{\pi}{6}]$, i.e. $|x| \le \frac{1}{2}$. For $\frac{1}{2} < x \le 1$, $3\theta \in (\frac{\pi}{2}, \frac{3\pi}{2}]$ and $\sin^{-1}(\sin 3\theta) = \pi - 3\theta$, flipping the sign of the derivative to $-\frac{3}{\sqrt{1-x^2}}$. The correct value is $S'(\frac{3}{4}) = -\frac{12}{\sqrt{7}}$.

Sine substitution with range bookkeeping

1. Let $x = \sin\theta$, $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$; then $3x - 4x^3 = \sin 3\theta$ and $S = \sin^{-1}(\sin 3\theta)$.
2. $|x| \le \frac{1}{2} \iff |\theta| \le \frac{\pi}{6} \iff 3\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, so $S = 3\theta = 3\sin^{-1} x \Rightarrow S' = \dfrac{3}{\sqrt{1-x^2}}$.
3. $\frac{1}{2} < x \le 1 \iff \frac{\pi}{6} < \theta \le \frac{\pi}{2} \iff 3\theta \in (\frac{\pi}{2}, \frac{3\pi}{2}]$, so $S = \pi - 3\theta = \pi - 3\sin^{-1} x \Rightarrow S' = -\dfrac{3}{\sqrt{1-x^2}}$ (and symmetrically for $x < -\frac{1}{2}$).
4. Evaluate: $S'(\frac{1}{4}) = \dfrac{3}{\sqrt{1-1/16}} = \dfrac{3}{\sqrt{15/4}} = \boxed{\dfrac{12}{\sqrt{15}}}$ and $S'(\frac{3}{4}) = -\dfrac{3}{\sqrt{1-9/16}} = -\dfrac{3}{\sqrt{7/4}} = \boxed{-\dfrac{12}{\sqrt{7}}}$.

Direct chain rule with modulus

1. $S' = \dfrac{(3-12x^2)}{\sqrt{1-(3x-4x^3)^2}}$. Factor: $3 - 12x^2 = 3(1-4x^2)$ and $1 - (3x-4x^3)^2 = (1-x^2)(1-4x^2)^2$ (a standard factorisation).
2. So $\sqrt{1-(3x-4x^3)^2} = \sqrt{1-x^2}\,|1-4x^2|$, giving $S' = \dfrac{3(1-4x^2)}{\sqrt{1-x^2}\,|1-4x^2|} = \dfrac{3\ \mathrm{sgn}(1-4x^2)}{\sqrt{1-x^2}}$.
3. $1 - 4x^2 > 0 \iff |x| < \frac{1}{2}$, so $S' = +\dfrac{3}{\sqrt{1-x^2}}$ there and $-\dfrac{3}{\sqrt{1-x^2}}$ for $\frac{1}{2} < |x| < 1$.
4. Hence $S'(\frac{1}{4}) = \boxed{\dfrac{12}{\sqrt{15}}}$ and $S'(\frac{3}{4}) = \boxed{-\dfrac{12}{\sqrt{7}}}$.

INSIGHT

The factor $|1-4x^2| = \sqrt{(1-4x^2)^2}$ is exactly the breakpoint detector: the triple-angle identity $\sin^{-1}(3x-4x^3) = 3\sin^{-1} x$ is true only on $|x| \le \frac{1}{2}$, and the sign of the derivative flips at $x = \pm\frac{1}{2}$. The analogous $\cos^{-1}(4x^3 - 3x)$ partitions at $x = \pm\frac{1}{2}$ too, and $\tan^{-1}\frac{3x-x^3}{1-3x^2}$ at $x = \pm\frac{1}{\sqrt{3}}$. Never apply a multiple-angle inverse identity without first checking the principal-range window for the inner angle.

№78 Integrating the Sawtooth

●●●●● L5

EVALUATE

$$\int_0^{2\pi} \sin^{-1}(\sin x)\,dx \quad \text{and} \quad \int_0^{\pi} \sin^{-1}(\sin x)\,dx.$$

definite-integral · sawtooth · sin-inverse · piecewise · symmetry

ANSWER

$$\int_0^{2\pi} \sin^{-1}(\sin x)\,dx = 0, \qquad \int_0^{\pi} \sin^{-1}(\sin x)\,dx = \frac{\pi^2}{4}.$$

THE TRAP

The fatal shortcut is $\sin^{-1}(\sin x) = x$, giving $\int_0^{2\pi} x\,dx = 2\pi^2$ and $\int_0^{\pi} x\,dx = \frac{\pi^2}{2}$. But $\sin^{-1}(\sin x) = x$ only on $[-\frac{\pi}{2}, \frac{\pi}{2}]$. The integrand is the triangle wave: it rises to $\frac{\pi}{2}$ at $x = \frac{\pi}{2}$, falls to $-\frac{\pi}{2}$ at $x = \frac{3\pi}{2}$, with a downward kink at $\frac{\pi}{2}$ — NOT the straight line $y = x$. Over a full period its positive and negative halves cancel to 0, and on $[0, \pi]$ it is x then $\pi - x$, not x throughout.

Piecewise on the triangle wave

1. On $[0, \frac{\pi}{2}]$, $\sin^{-1}(\sin x) = x$; on $[\frac{\pi}{2}, \pi]$, $\sin^{-1}(\sin x) = \pi - x$ (since $\sin x = \sin(\pi - x)$ and $\pi - x \in [0, \frac{\pi}{2}]$).
2. $\displaystyle\int_0^{\pi} = \int_0^{\pi/2} x\,dx + \int_{\pi/2}^{\pi} (\pi - x)\,dx = \frac{\pi^2}{8} + \frac{\pi^2}{8} = \boxed{\frac{\pi^2}{4}}.$
3. On $[\pi, \frac{3\pi}{2}]$, $\sin^{-1}(\sin x) = \pi - x$ (continuing the descent into negatives); on $[\frac{3\pi}{2}, 2\pi]$, $\sin^{-1}(\sin x) = x - 2\pi$. These two halves give $-\dfrac{\pi^2}{4}$.
4. Total over $[0, 2\pi]$: $\dfrac{\pi^2}{4} + \left(-\dfrac{\pi^2}{4}\right) = \boxed{0}.$

Symmetry / odd-function argument

1. Shift: over a full period the triangle wave $y = \sin^{-1}(\sin x)$ is odd about $x = \pi$ (it satisfies $y(2\pi - x) = -y(x)$ after centering), so $\int_0^{2\pi} = \boxed{0}$ immediately.
2. For $[0, \pi]$: the wave is symmetric about $x = \frac{\pi}{2}$ where it peaks at $\frac{\pi}{2}$. The region under it is an isosceles triangle of base π and height $\frac{\pi}{2}$.
3. Triangle area $= \frac{1}{2} \cdot \pi \cdot \frac{\pi}{2} = \dfrac{\pi^2}{4}$.
4. Hence $\int_0^{\pi} \sin^{-1}(\sin x)\,dx = \boxed{\dfrac{\pi^2}{4}}$, with the full-period integral $\boxed{0}$ by cancellation.

INSIGHT

Periodic inverse-trig integrands are graph problems, not antiderivative problems. $\sin^{-1}(\sin x)$ is a triangle wave of amplitude $\frac{\pi}{2}$; its integral over any whole number of periods is 0, and over a half-period (a single triangle) it is the triangle's area. The matching fact $\int_0^{\pi} \cos^{-1}(\cos x)\,dx = \frac{\pi^2}{2}$ (a full triangle of height π) is worth filing alongside.

№79 Two Sawtooths Collide

●●●●● L4

EVALUATE

Let $W(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$. Evaluate $W'(\frac{\pi}{3})$ and $W'(2)$.

sawtooth · sin-inverse · cos-inverse · derivative-at-point · piecewise

ANSWER

$$W'(\tfrac{\pi}{3}) = 2, \qquad W'(2) = 0.$$

THE TRAP

The tempting answer is to treat both pieces as x, so $W(x) = x + x = 2x$ and $W' = 2$ everywhere, giving $W'(2) = 2$. But on $[\frac{\pi}{2}, \pi]$, $\sin^{-1}(\sin x) = \pi - x$ (slope -1) while $\cos^{-1}(\cos x) = x$ (slope $+1$); the slopes CANCEL, so $W(x) = \pi$ is constant there and $W'(2) = 0$, not 2. Since $2\,\text{rad} \in (\frac{\pi}{2}, \pi)$, the constant-$\pi$ regime applies.

Add the two square waves

1. $\dfrac{d}{dx}\sin^{-1}(\sin x) = \mathrm{sgn}(\cos x)$ and $\dfrac{d}{dx}\cos^{-1}(\cos x) = \mathrm{sgn}(\sin x)$, since $\cos^{-1}(\cos x)$ rises (slope $+1$) exactly where $\sin x > 0$ and falls where $\sin x < 0$.
2. So $W'(x) = \mathrm{sgn}(\cos x) + \mathrm{sgn}(\sin x)$.
3. At $x = \frac{\pi}{3}$: $\cos\frac{\pi}{3} > 0$ and $\sin\frac{\pi}{3} > 0$, so $W' = (+1) + (+1) = \boxed{2}$.
4. At $x = 2\,\mathrm{rad} \in (\frac{\pi}{2}, \pi)$: $\cos 2 < 0$ and $\sin 2 > 0$, so $W' = (-1) + (+1) = \boxed{0}$ — the two square waves cancel.

Explicit piecewise on [0,π]

1. For $x \in [0, \frac{\pi}{2}]$: $\sin^{-1}(\sin x) = x$ and $\cos^{-1}(\cos x) = x$, so $W(x) = 2x$.
2. For $x \in [\frac{\pi}{2}, \pi]$: $\sin^{-1}(\sin x) = \pi - x$ and $\cos^{-1}(\cos x) = x$, so $W(x) = \pi$ (a flat shelf).
3. $\frac{\pi}{3} \approx 1.047 \in (0, \frac{\pi}{2})$ lies on the ramp $2x$, giving $W'(\frac{\pi}{3}) = \boxed{2}$.
4. $2\,\mathrm{rad} \approx 2.0 \in (\frac{\pi}{2}, \pi)$ lies on the flat shelf, giving $W'(2) = \boxed{0}$.

INSIGHT

$\sin^{-1}(\sin x)$ and $\cos^{-1}(\cos x)$ are both triangle waves but phase-shifted by $\frac{\pi}{2}$, so their slopes (± 1 square waves) sometimes reinforce ($\to \pm 2$) and sometimes cancel ($\to 0$). The composite W is a staircase of ramps and flats. The lesson: always reduce each inverse-trig term to its principal-value linear piece on the relevant interval before differentiating – never assume both reduce to x.

№80 The Even Derivative

●●●●● L4

DIFFERENTIATE

Let $R(x) = \sec^{-1} x,\ |x| \geq 1$. Find $R'(2)$ and $R'(-2),$ and state the sign relationship between them.

sec-inverse absolute-value derivative-at-point branch

ANSWER

$$R'(2) = \frac{1}{2\sqrt{3}}, \qquad R'(-2) = \frac{1}{2\sqrt{3}}; \quad \text{both are equal and POSITIVE } (R' \text{ is even}).$$

THE TRAP

Students who memorise $\dfrac{d}{dx}\sec^{-1} x = \dfrac{1}{x\sqrt{x^2-1}}$ (without the absolute value) compute $R'(-2) = \dfrac{1}{(-2)\sqrt{3}} = -\dfrac{1}{2\sqrt{3}}$, a NEGATIVE value, and conclude the two are opposite in sign. But with the principal range $\sec^{-1} : [0, \pi] \setminus \{\frac{\pi}{2}\}$, $\sec^{-1} x$ is INCREASING on both $(-\infty, -1]$ and $[1, \infty)$, so its derivative must be positive everywhere. The correct formula is $\dfrac{1}{|x|\sqrt{x^2-1}}$, giving $R'(-2) = +\dfrac{1}{2\sqrt{3}}$.

Correct formula with |x|

1. With the principal branch $\sec^{-1} : [0, \pi] \setminus \{\frac{\pi}{2}\}$, one derives $\dfrac{d}{dx}\sec^{-1} x = \dfrac{1}{|x|\sqrt{x^2-1}}$ — positive for all $|x| > 1$.
2. $R'(2) = \dfrac{1}{2\sqrt{4-1}} = \dfrac{1}{2\sqrt{3}}$.
3. $R'(-2) = \dfrac{1}{|-2|\sqrt{4-1}} = \dfrac{1}{2\sqrt{3}}$.
4. Both equal $\boxed{\dfrac{1}{2\sqrt{3}}}$; they are equal and positive, so R' is an even function.

Via cos inverse

1. For the principal branch, $\sec^{-1} x = \cos^{-1}\left(\frac{1}{x}\right)$ (valid for all $|x| \geq 1$).
2. Differentiate: $R'(x) = -\dfrac{1}{\sqrt{1-1/x^2}} \cdot \left(-\dfrac{1}{x^2}\right) = \dfrac{1}{x^2\sqrt{1-1/x^2}} = \dfrac{1}{x^2 \cdot \frac{\sqrt{x^2-1}}{|x|}} = \dfrac{1}{|x|\sqrt{x^2-1}}$.
3. At $x = 2$: $\dfrac{1}{2\sqrt{3}}$; at $x = -2$: $\dfrac{1}{2\sqrt{3}}$.
4. Both $\boxed{\dfrac{1}{2\sqrt{3}}}$ — manifestly even, confirming $\sec^{-1}$ increases on each ray.

INSIGHT

The absolute value in $\frac{d}{dx}\sec^{-1} x = \frac{1}{|x|\sqrt{x^2-1}}$ and $\frac{d}{dx}\csc^{-1} x = \frac{-1}{|x|\sqrt{x^2-1}}$ is not optional – it is forced by the principal range, which makes $\sec^{-1}$ monotone-increasing on both arms (so its slope cannot change sign). Writing $\sec^{-1} x = \cos^{-1}(1/x)$ is the safest way to recover the correct formula every time; the $|x|$ emerges automatically from $\sqrt{1-1/x^2} = \sqrt{x^2-1}/|x|$.

CHAPTER IX

Machin & Olympiad Identities

Machin's π, Gaussian-integer arctangents, the competition canon.

№81 Hutton's Hidden Twin

L4

PROVE THAT

$$2\tan^{-1}\tfrac{1}{3}+\tan^{-1}\tfrac{1}{7}=\frac{\pi}{4}$$

machin · gaussian-integer · arctan-sum · principal-value

ANSWER

$$\frac{\pi}{4}$$

THE TRAP

A student doubles the difference formula carelessly and writes $2\tan^{-1}\frac{1}{3}=\tan^{-1}\frac{2/3}{1-1/9}=\tan^{-1}\frac{3}{4}$, which is correct, but then adds $\tan^{-1}\frac{3}{4}+\tan^{-1}\frac{1}{7}$ using $\tan^{-1}a+\tan^{-1}b=\tan^{-1}\frac{a+b}{1-ab}$ and reports $\tan^{-1}\frac{3/4+1/7}{1-3/28}=\tan^{-1}1=\frac{\pi}{4}$ **without checking** that $ab=\frac{3}{28}<1$. Here the bound happens to hold, so the answer is right — but the same student will write $\frac{\pi}{4}$ for sums where $ab>1$ and the true value is $\frac{\pi}{4}+\pi$. The proof must state the inequality $ab<1$ that legitimises the unshifted formula; otherwise the chain is luck, not logic.

Tangent difference chain

1. First, $2\tan^{-1}\frac{1}{3}=\tan^{-1}\dfrac{2\cdot\frac{1}{3}}{1-\frac{1}{9}}=\tan^{-1}\dfrac{2/3}{8/9}=\tan^{-1}\frac{3}{4}$. Since $\frac{1}{3}<1$, $2\tan^{-1}\frac{1}{3}<2\cdot\frac{\pi}{4}=\frac{\pi}{2}$, so this single $\tan^{-1}$ value is legitimate (no $+\pi$ shift).
2. Now add: with $a=\frac{3}{4}$, $b=\frac{1}{7}$ we have $ab=\frac{3}{28}<1$, so $\tan^{-1}\frac{3}{4}+\tan^{-1}\frac{1}{7}=\tan^{-1}\dfrac{\frac{3}{4}+\frac{1}{7}}{1-\frac{3}{28}}=\tan^{-1}\dfrac{25/28}{25/28}=\tan^{-1}1$.
3. Because $ab<1$ throughout, the sum stays in $(0,\frac{\pi}{2})$, so it equals exactly $\boxed{\dfrac{\pi}{4}}$.

Gaussian-integer argument

1. The angle $\tan^{-1}\frac{1}{3}$ is $\arg(3+i)$ and $\tan^{-1}\frac{1}{7}=\arg(7+i)$, so the left side is $\arg\left((3+i)^2(7+i)\right)$.
2. Compute $(3+i)^2=8+6i$, then $(8+6i)(7+i)=56+8i+42i-6=50+50i$.
3. Since $50+50i=50(1+i)$ lies on the ray $\arg=\frac{\pi}{4}$ and every factor has positive real part (each argument is in $(0,\frac{\pi}{2})$, total $<\frac{3\pi}{2}$, with a first-quadrant product), the principal argument is exactly $\boxed{\dfrac{\pi}{4}}$.

INSIGHT

This is Hutton's relative of Machin. The Gaussian-integer method automates the branch bookkeeping: the *argument of a product* is the sum of arguments **mod** 2π, and you fix the representative by noting the product $50(1+i)$ sits in the first quadrant. Any identity $\sum c_k \tan^{-1}\frac{1}{n_k} = \frac{m\pi}{4}$ corresponds to $\prod(n_k+i)^{c_k} = r(1+i)^m$ up to a positive real factor.

№82 The Sum That Refuses to Vanish

●●●●● L4

PROVE THAT

$$\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3 = \pi$$

arctan-sum | branch | gaussian-integer | quadrant

ANSWER

$$\pi$$

THE TRAP

The seductive wrong answer is 0. A student computes $\tan$ of the sum and gets 0 (indeed $\tan\pi = 0$), then concludes the sum is 0 — or applies the two-term formula twice and, finding $\tan^{-1}2 + \tan^{-1}3 = \tan^{-1}\frac{5}{-5} = \tan^{-1}(-1) = -\frac{\pi}{4}$, reports $\frac{\pi}{4} - \frac{\pi}{4} = 0$. Both ignore that each of $\tan^{-1}1, \tan^{-1}2, \tan^{-1}3$ is **positive**, so the sum lies strictly between 0 and $\frac{3\pi}{2}$; a vanishing tangent in that window forces the value to be π, not 0.

Quadrant-bounded tangent

1. Each term is positive and at most $\tan^{-1}3 < \frac{\pi}{2}$, so $S = \tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3 \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ (already $\tan^{-1}2 + \tan^{-1}3 > \frac{\pi}{2}$).
2. Compute the tangent: $\tan(\tan^{-1}2 + \tan^{-1}3) = \frac{2+3}{1-6} = -1$, and adding $\tan^{-1}1$ gives $\tan S = \dfrac{1+(-1)}{1-(1)(-1)} = 0$.
3. The only multiple of π inside $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ with $\tan = 0$ is π, so $S = \boxed{\pi}$.

Gaussian-integer argument

1. $S = \arg(1+i) + \arg(1+2i) + \arg(1+3i) = \arg\big((1+i)(1+2i)(1+3i)\big)$.
2. $(1+i)(1+2i) = -1+3i$, then $(-1+3i)(1+3i) = -1-3i+3i-9 = -10$.
3. The product -10 is a negative real, so its principal argument is π. Since each factor is in the first quadrant the running sum never exceeds $\frac{3\pi}{2}$, fixing the representative at exactly $\boxed{\pi}$.

Two complementary pairs

1. Use $\tan^{-1}2 + \tan^{-1}3$: since $2\cdot 3 = 6 > 1$ and both terms positive, the formula needs the $+\pi$ shift: $\tan^{-1}2 + \tan^{-1}3 = \pi + \tan^{-1}\frac{2+3}{1-6} = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$.
2. Add the last term: $\frac{3\pi}{4} + \tan^{-1}1 = \frac{3\pi}{4} + \frac{\pi}{4} = \boxed{\pi}$.

INSIGHT

This is the cleanest illustration of the $+\pi$ branch correction: $\tan^{-1}a + \tan^{-1}b = \tan^{-1}\frac{a+b}{1-ab} + \pi$ when $a, b > 0$ and $ab > 1$. The integer triple $(1,2,3)$ works because $1+2+3 = 1\cdot 2\cdot 3$, the algebraic signature of three positive arctangents summing to π (a triangle's angles).

№83 Telescope to a Quarter Turn

●●●●● L4

EVALUATE

$$\sum_{n=1}^{\infty} \tan^{-1}\left(\frac{1}{n^2+n+1}\right)$$

telescoping | series | arctan-difference | limit

ANSWER

$$\frac{\pi}{4}$$

THE TRAP

A strong student spots the telescoping but mis-states the limit as $\frac{\pi}{2}$, forgetting the lower endpoint. The partial sum is $\tan^{-1}(N+1) - \tan^{-1} 1$, and they evaluate $\lim \tan^{-1}(N+1) = \frac{\pi}{2}$ but drop the $-\tan^{-1} 1 = -\frac{\pi}{4}$. The subtracted boundary term is exactly what turns $\frac{\pi}{2}$ into $\frac{\pi}{4}$.

Difference-formula telescoping

1. Write $n^2+n+1 = 1+n(n+1)$, so $\tan^{-1}\dfrac{1}{n^2+n+1} = \tan^{-1}\dfrac{(n+1)-n}{1+(n+1)n} = \tan^{-1}(n+1) - \tan^{-1} n$ — valid because $(n+1)n > 0$ keeps the difference in $(0, \frac{\pi}{2})$.
2. The partial sum collapses: $\sum_{n=1}^{N} \left[\tan^{-1}(n+1) - \tan^{-1} n\right] = \tan^{-1}(N+1) - \tan^{-1} 1$.
3. Let $N \to \infty$: $\tan^{-1}(N+1) \to \frac{\pi}{2}$, so the sum is $\frac{\pi}{2} - \frac{\pi}{4} = \boxed{\dfrac{\pi}{4}}$.

Antidifference via arctan ladder

1. Define $a_n = \tan^{-1} n$. Then $a_{n+1} - a_n = \tan^{-1}(n+1) - \tan^{-1} n$, and computing its tangent gives $\frac{1}{1+n(n+1)} = \frac{1}{n^2+n+1}$, matching the summand exactly (both sides in the first quadrant).
2. Hence the series is the telescoping image of the increasing bounded sequence $a_n \uparrow \frac{\pi}{2}$, giving total $a_\infty - a_1 = \frac{\pi}{2} - \frac{\pi}{4} = \boxed{\dfrac{\pi}{4}}$.

INSIGHT

The decomposition $\tan^{-1} \frac{1}{n^2+n+1} = \tan^{-1}(n+1) - \tan^{-1} n$ is the arctan analogue of partial fractions. The same trick handles $\sum \tan^{-1} \frac{1}{2n^2} = \frac{\pi}{4}$ (using $\tan^{-1} \frac{1}{2n-1} - \tan^{-1} \frac{1}{2n+1}$) – the art is matching the denominator to a $1+ab$ pattern.

№84 Hutton's Two-Term Machine

●●●●● L4

PROVE THAT

$$2\tan^{-1}\tfrac{1}{2} - \tan^{-1}\tfrac{1}{7} = \frac{\pi}{4}$$

machin | gaussian-integer | conjugate | arctan-difference

ANSWER

$$\frac{\pi}{4}$$

THE TRAP

A student writes $2\tan^{-1}\frac{1}{2} = \tan^{-1}\frac{1}{1-1/4} = \tan^{-1}\frac{4}{3}$ and then, treating $\tan^{-1}\frac{4}{3}$ as if it were below $\frac{\pi}{4}$, fumbles the subtraction. In fact $\tan^{-1}\frac{4}{3} > \frac{\pi}{4}$ (since $\frac{4}{3} > 1$), so the difference $\tan^{-1}\frac{4}{3} - \tan^{-1}\frac{1}{7}$ is genuinely positive and lands at $\frac{\pi}{4}$; a sign slip from mis-ordering the terms yields the tempting but wrong $-\frac{\pi}{4}$ or $\frac{3\pi}{4}$.

Double then difference

1. $2\tan^{-1}\frac{1}{2} = \tan^{-1}\dfrac{2\cdot\frac{1}{2}}{1-\frac{1}{4}} = \tan^{-1}\dfrac{1}{3/4} = \tan^{-1}\frac{4}{3}$, and since $\frac{1}{2} < 1$ this stays below $\frac{\pi}{2}$ (no shift).
2. Subtract: $\tan^{-1}\frac{4}{3} - \tan^{-1}\frac{1}{7} = \tan^{-1}\dfrac{\frac{4}{3}-\frac{1}{7}}{1+\frac{4}{21}} = \tan^{-1}\dfrac{25/21}{25/21} = \tan^{-1} 1$ (denominator $1+ab > 0$, so no shift).

3. Both intermediate angles are in $(0, \frac{\pi}{2})$ with positive difference, so the result is $\boxed{\frac{\pi}{4}}$.

Gaussian-integer argument

1. Subtraction of arguments means division by the conjugate: the left side is $\arg\dfrac{(2+i)^2}{7+i} = \arg\left((2+i)^2(7-i)\right)$ (multiplying num. and denom. by $7-i$ keeps the argument).
2. $(2+i)^2 = 3+4i$; then $(3+4i)(7-i) = 21 - 3i + 28i + 4 = 25 + 25i$.
3. $25 + 25i = 25(1+i)$ has principal argument $\frac{\pi}{4}$, and the combination $2\arg(2+i) - \arg(7+i)$ lies in $(0, \frac{\pi}{2})$, so the value is exactly $\boxed{\frac{\pi}{4}}$.

INSIGHT
Subtraction of arctangents is conjugation in the Gaussian integers: $-\tan^{-1}\frac{1}{n} \leftrightarrow$ a factor $n - i$. Here $(2+i)^2(7-i) = 25(1+i)$ encodes Hutton's formula, a cousin of Machin obtained by replacing $5 + 0i$ with $(2+i)^2 = 3+4i$ (the same Pythagorean primitive).

№85 Where Cosine Meets Cosecant

●●●●● L5

SOLVE FOR X

$$2\tan^{-1}(\cos x) = \tan^{-1}(2\csc x), \quad x \in (0, \pi)$$

equation · domain · principal-value · trig-substitution

ANSWER

$$x = \frac{\pi}{4}$$

THE TRAP
Taking tangents blindly gives $\dfrac{2\cos x}{1-\cos^2 x} = 2\csc x$, i.e. $\dfrac{2\cos x}{\sin^2 x} = \dfrac{2}{\sin x}$, leading to $\cos x = \sin x$ and the **pair** $x = \frac{\pi}{4}, \frac{5\pi}{4}$. The value $\frac{5\pi}{4}$ is outside $(0, \pi)$ — and even within reach it would fail because there $\cos x < 0$ makes the left side negative while $\csc x < 0$ flips the right side too, but the doubling formula $2\tan^{-1} u = \tan^{-1}\frac{2u}{1-u^2}$ silently shifts by π when $|u| >$ the branch threshold. The honest solution checks the range of each side, killing the spurious root.

Reduce on the valid domain

1. On $(0, \pi)$, $\sin x > 0$ so $\csc x > 0$ and $2\csc x \geq 2 > 0$; thus the right side $\tan^{-1}(2\csc x) \in (0, \frac{\pi}{2})$. For equality the left side must also be positive, forcing $\cos x > 0$, i.e. $x \in \left(0, \frac{\pi}{2}\right)$ where $|\cos x| < 1$ keeps the doubling unshifted.
2. Take tangents: $\dfrac{2\cos x}{1-\cos^2 x} = 2\csc x \Rightarrow \dfrac{2\cos x}{\sin^2 x} = \dfrac{2}{\sin x} \Rightarrow \cos x = \sin x$.
3. On $\left(0, \frac{\pi}{2}\right)$ the only solution of $\tan x = 1$ is $x = \frac{\pi}{4}$; checking, $\cos x = \frac{1}{\sqrt{2}}$ and $2\csc x = 2\sqrt{2}$, so both sides equal $\tan^{-1}(2\sqrt{2})$. Hence $x = \boxed{\frac{\pi}{4}}$.

Geometric / right-triangle reading

1. Set $\theta = \tan^{-1}(\cos x)$, so $\tan\theta = \cos x$. The equation says $2\theta = \tan^{-1}(2\csc x)$, and $2\theta \in (0, \pi)$ matches the codomain only if $2\theta < \frac{\pi}{2}$, i.e. $\cos x < 1$ with $\theta < \frac{\pi}{4}$, which holds for $x \in (0, \frac{\pi}{2})$.
2. Then $\tan 2\theta = \dfrac{2\cos x}{1-\cos^2 x} = \dfrac{2\cos x}{\sin^2 x}$, and equating to $2\csc x = \dfrac{2}{\sin x}$ gives $\dfrac{\cos x}{\sin x} = 1$.
3. So $x = \frac{\pi}{4}$, where both sides collapse to $\tan^{-1}(2\sqrt{2})$; the candidate $\frac{5\pi}{4} \notin (0, \pi)$ and is rejected, leaving $x = \boxed{\frac{\pi}{4}}$.

INSIGHT

The lesson: $2\tan^{-1}u = \tan^{-1}\frac{2u}{1-u^2}$ only for $|u| < 1$; for $|u| > 1$ a $\pm\pi$ correction enters. By first pinning each side's *sign and range*, you confine x to the interval where the naive algebra is legitimate, automatically discarding extraneous roots.

№86 Euler's Three Unit Fractions

●●●●● L4

PROVE THAT

$$\tan^{-1}\tfrac{1}{2} + \tan^{-1}\tfrac{1}{5} + \tan^{-1}\tfrac{1}{8} = \frac{\pi}{4}$$

arctan-sum gaussian-integer euler principal-value

ANSWER

$$\frac{\pi}{4}$$

THE TRAP

After combining the first two terms, $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{5} = \tan^{-1}\frac{7}{9}$, a student adds the third with the formula and gets $\tan^{-1}\frac{7/9+1/8}{1-7/72} = \tan^{-1}1$, but some will worry the product $\frac{7}{72}$· exceeds 1 and erroneously insert a $+\pi$, reporting $\frac{5\pi}{4}$. Here all partial products are < 1 and all terms positive and small, so the running sum never leaves $\left(0, \frac{\pi}{2}\right)$ — no shift is warranted.

Pairwise accumulation

1. $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{5}$: product $\frac{1}{10} < 1$, so $= \tan^{-1}\dfrac{\frac{1}{2}+\frac{1}{5}}{1-\frac{1}{10}} = \tan^{-1}\dfrac{7/10}{9/10} = \tan^{-1}\frac{7}{9}$.
2. Add the third: product $\frac{7}{9}\cdot\frac{1}{8} = \frac{7}{72} < 1$, so $\tan^{-1}\frac{7}{9} + \tan^{-1}\frac{1}{8} = \tan^{-1}\dfrac{\frac{7}{9}+\frac{1}{8}}{1-\frac{7}{72}} = \tan^{-1}\dfrac{65/72}{65/72} = \tan^{-1}1$.
3. All three terms are positive and the cumulative angle stays below $\frac{\pi}{2}$, so the sum is exactly $\boxed{\dfrac{\pi}{4}}$.

Gaussian-integer argument

1. The sum equals $\arg\big((2+i)(5+i)(8+i)\big)$.
2. $(2+i)(5+i) = 9+7i$; then $(9+7i)(8+i) = 72+9i+56i-7 = 65+65i$.
3. $65(1+i)$ lies on the $\frac{\pi}{4}$ ray, and since each factor is in the first quadrant the total argument is below $\frac{3\pi}{2}$ and equals exactly $\boxed{\dfrac{\pi}{4}}$.

INSIGHT

This is one of Euler's 1737 decompositions of $\frac{\pi}{4}$. Note $(2+i)(5+i)(8+i) = 65(1+i)$; the norm $65 = 5\cdot 13$ factors as Gaussian primes $(2\pm i)(2\mp i)(3\pm 2i)(3\mp 2i)$ – the existence of such clean arctan triples is exactly the existence of integers whose Gaussian factorisation rebalances to $(1+i)$.

№87 The Obtuse Sum

●●●●● L3

EVALUATE

$$\tan^{-1}2 + \tan^{-1}3$$

arctan-sum branch quadrant tangent-addition

ANSWER

$$\frac{3\pi}{4}$$

THE TRAP

The reflex answer is $-\dfrac{\pi}{4}$. Applying $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \frac{a+b}{1-ab}$ gives $\tan^{-1} \frac{5}{1-6} = \tan^{-1}(-1) = -\frac{\pi}{4}$. But $a = 2, b = 3$ are positive with $ab = 6 > 1$, so the formula's domain condition fails: each arctangent exceeds $\frac{\pi}{4}$, the true sum exceeds $\frac{\pi}{2}$ and cannot be negative. The correct value is $\pi + (-\frac{\pi}{4}) = \frac{3\pi}{4}$.

Shifted addition formula

1. Both terms are positive and $ab = 6 > 1$, so the addition formula carries a $+\pi$ correction: $\tan^{-1} 2 + \tan^{-1} 3 = \pi + \tan^{-1} \dfrac{2+3}{1-6} = \pi + \tan^{-1}(-1)$.
2. $\tan^{-1}(-1) = -\frac{\pi}{4}$, giving $\pi - \frac{\pi}{4} = \boxed{\dfrac{3\pi}{4}}$.

Bound then identify

1. Since $2, 3 > 1$, each arctangent is in $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$, so the sum $S \in \left(\frac{\pi}{2}, \pi\right)$ — second quadrant.
2. Compute $\tan S = \dfrac{2+3}{1-6} = -1$. The unique angle in $\left(\frac{\pi}{2}, \pi\right)$ with tangent -1 is $\frac{3\pi}{4}$, so $S = \boxed{\dfrac{3\pi}{4}}$.

Gaussian-integer argument

1. $S = \arg\big((2+i)(3+i)\big) = \arg(6 + 2i + 3i - 1) = \arg(5 + 5i)$.
2. Wait — recompute carefully: $(2+i)(3+i) = 6 + 2i + 3i + i^2 = 5 + 5i$? That is $\arg = \frac{\pi}{4}$, but each factor's argument $\tan^{-1} \frac{1}{2}, \tan^{-1} \frac{1}{3}$ — those are the reciprocals. For $\tan^{-1} 2, \tan^{-1} 3$ use $\arg(1+2i) + \arg(1+3i) = \arg\big((1+2i)(1+3i)\big) = \arg(1 + 5i - 6) = \arg(-5 + 5i)$.
3. $-5 + 5i$ is in the second quadrant on the ray $\arg = \frac{3\pi}{4}$, so $S = \boxed{\dfrac{3\pi}{4}}$.

INSIGHT

Whenever $a, b > 0$ with $ab > 1$, $\tan^{-1} a + \tan^{-1} b = \pi + \tan^{-1} \frac{a+b}{1-ab}$. In Gaussian-integer terms, $\tan^{-1} n = \arg(1 + ni)$ (not $\arg(n + i)$): the product $(1+2i)(1+3i) = -5 + 5i$ lands in quadrant II, instantly revealing the $\frac{3\pi}{4}$ answer that the bare addition formula hides.

№88 A Right Angle in Three Pieces

●●●●○ L4

PROVE THAT

$$\tan^{-1} 1 + \tan^{-1} \tfrac{1}{2} + \tan^{-1} \tfrac{1}{3} = \frac{\pi}{2}$$

arctan-sum · gaussian-integer · pure-imaginary · euler

ANSWER

$$\frac{\pi}{2}$$

THE TRAP

Students who know $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \frac{\pi}{4}$ correctly add $\tan^{-1} 1 = \frac{\pi}{4}$ to reach $\frac{\pi}{2}$ — but those who attack all three at once via repeated addition risk computing $\tan$ of the total, getting $\tan\left(\frac{\pi}{2}\right)$ undefined, and either declaring 'no closed form' or forcing a finite $\tan^{-1}$ value. The point is that the sum sits exactly on the boundary $\frac{\pi}{2}$ where $\tan$ blows up — the Gaussian product being **purely imaginary** is the clean certificate.

Known pair plus a quarter

1. First, $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3}$: product $\frac{1}{6} < 1$, so $= \tan^{-1} \dfrac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{6}} = \tan^{-1} \dfrac{5/6}{5/6} = \tan^{-1} 1 = \frac{\pi}{4}$.

2. Add $\tan^{-1} 1 = \frac{\pi}{4}$: total $= \frac{\pi}{4} + \frac{\pi}{4} = \boxed{\frac{\pi}{2}}$.

Gaussian-integer argument

1. The sum is $\arg\big((1+i)(2+i)(3+i)\big)$.
2. $(2+i)(3+i) = 5+5i$, then $(1+i)(5+5i) = 5+5i+5i-5 = 10i$.
3. $10i$ is a positive pure imaginary, so its principal argument is $\frac{\pi}{2}$; all factors first-quadrant keep the running sum below $\frac{3\pi}{2}$, fixing it at $\boxed{\frac{\pi}{2}}$.

Complementary-angle pairing

1. Note $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \tan^{-1} 1 = \frac{\pi}{4}$ means $\tan^{-1} \frac{1}{3} = \frac{\pi}{4} - \tan^{-1} \frac{1}{2}$.
2. Substitute: the total becomes $\tan^{-1} 1 + \tan^{-1} \frac{1}{2} + \left(\frac{\pi}{4} - \tan^{-1} \frac{1}{2}\right) = \frac{\pi}{4} + \frac{\pi}{4} = \boxed{\frac{\pi}{2}}$.

INSIGHT
A purely imaginary Gaussian product is the signature of an arctangent sum equal to $\frac{\pi}{2}$ (mod π). Equivalently, cot of the sum is 0; with $\cot^{-1}$ in $(0, \pi)$, this is also $\cot^{-1} 1 + \cot^{-1} 2 + \cot^{-1} 3$ rearranged – the same identity wearing a cotangent mask.

№89 Stormer's Hat-Trick

●●●●● L5

PROVE THAT

$$6 \tan^{-1} \frac{1}{8} + 2 \tan^{-1} \frac{1}{57} + \tan^{-1} \frac{1}{239} = \frac{\pi}{4}$$

machin · gaussian-integer · high-multiplicity · 239

ANSWER

$$\frac{\pi}{4}$$

THE TRAP
Chasing this by hand with the addition formula is a branch minefield: after a few combinations the intermediate $\tan^{-1}$ arguments exceed 1, and a student who forgets to track when $ab > 1$ inserts the wrong number of $\pm\pi$ corrections, arriving at $\frac{5\pi}{4}$ or $-\frac{3\pi}{4}$. The safe route is the Gaussian product, where the single check 'first-quadrant total $< \frac{\pi}{2}$' replaces six error-prone shifts.

Gaussian-integer argument

1. The left side is $\arg\big((8+i)^6(57+i)^2(239+i)\big)$; each factor is in the first quadrant and the small arguments sum to well under $\frac{\pi}{2}$ (numerically ≈ 0.785), so no branch shift can occur.
2. Computing the product (e.g. by repeated multiplication of Gaussian integers) gives $(8+i)^6(57+i)^2(239+i) = 150{,}837{,}781{,}250\,(1+i)$, a positive real multiple of $1+i$.
3. Lying exactly on the $\frac{\pi}{4}$ ray, its principal argument is $\boxed{\frac{\pi}{4}}$.

Reduction to Machin via known links

1. Use the building blocks $2 \tan^{-1} \frac{1}{8} = \tan^{-1} \frac{16}{63}$ and the chain that expresses $\tan^{-1} \frac{1}{5}$ and $\tan^{-1} \frac{1}{239}$ in terms of $\frac{1}{8}, \frac{1}{57}$; verifying each combination keeps products below 1 confirms no $+\pi$ shifts are needed.
2. Assembling the coefficients $(6, 2, 1)$ telescopes the expression onto Machin's $4 \tan^{-1} \frac{1}{5} - \tan^{-1} \frac{1}{239} = \frac{\pi}{4}$, so the value is $\boxed{\frac{\pi}{4}}$.

INSIGHT
This is Stormer's 1896 three-term Machin-like formula. Such formulas are catalogued by factoring the Gaussian primes dividing $5, 13, 1597, \ldots$; the coefficient vector $(6, 2, 1)$ is a solution of an integer linear system that forces $\prod(n_k + i)^{c_k}$ to be a real multiple of $1 + i$. High multiplicities (the exponent 6) are why $\frac{1}{239}$ appears only once.

№90 The Skipping Telescope

●●●●● L5

EVALUATE

$$\sum_{n=1}^{\infty} \tan^{-1}\left(\frac{2}{n^2}\right)$$

telescoping · series · arctan-difference · two-step

ANSWER

$$\frac{3\pi}{4}$$

THE TRAP
The natural guess after spotting telescoping is $\frac{\pi}{2}$ — by analogy with the single-step ladder. But the decomposition skips by **two**: $\tan^{-1}\frac{2}{n^2} = \tan^{-1}(n+1) - \tan^{-1}(n-1)$, so two boundary terms survive at the top ($\tan^{-1}(N+1) + \tan^{-1} N \to \pi$) and one at the bottom ($\tan^{-1} 1 + \tan^{-1} 0 = \frac{\pi}{4}$). Dropping the extra surviving term gives the wrong $\frac{\pi}{2}$; keeping all four yields $\pi - \frac{\pi}{4} = \frac{3\pi}{4}$.

Two-step telescoping

1. Write $\frac{2}{n^2} = \dfrac{(n+1)-(n-1)}{1+(n+1)(n-1)}$ since $1 + (n^2 - 1) = n^2$; hence $\tan^{-1}\frac{2}{n^2} = \tan^{-1}(n+1) - \tan^{-1}(n-1)$ for $n \geq 1$ (the case $n = 1$ reads $\tan^{-1} 2 = \tan^{-1} 2 - \tan^{-1} 0$, valid).
2. Partial sum: $\sum_{n=1}^{N} \left[\tan^{-1}(n+1) - \tan^{-1}(n-1)\right] = \left[\tan^{-1}(N+1) + \tan^{-1} N\right] - \left[\tan^{-1} 1 + \tan^{-1} 0\right]$ (even/odd indices telescope separately).
3. As $N \to \infty$: $\tan^{-1}(N+1) + \tan^{-1} N \to \frac{\pi}{2} + \frac{\pi}{2} = \pi$, while the lower terms give $\frac{\pi}{4} + 0$. The sum is $\pi - \frac{\pi}{4} = \boxed{\dfrac{3\pi}{4}}$.

Split into two single-step ladders

1. Group the surviving boundary terms by parity: the $\tan^{-1}(n+1)$ pieces telescope against $\tan^{-1}(n-1)$ leaving the two largest indices, and the bottom leaves the two smallest, as in any 'gap-2' telescope.
2. Concretely $S_N = \tan^{-1}(N+1) + \tan^{-1} N - \tan^{-1} 1 - \tan^{-1} 0$; taking $N \to \infty$ gives $\left(\frac{\pi}{2} + \frac{\pi}{2}\right) - \frac{\pi}{4} = \boxed{\dfrac{3\pi}{4}}$.

INSIGHT
Gap-k telescopes leave k terms at each end. The denominator pattern $1 + (n+1)(n-1) = n^2$ is the giveaway that the difference formula uses arguments two apart. Generalising, $\sum_{n=1}^{\infty} \tan^{-1}\frac{2}{n^2} = \frac{3\pi}{4}$ while the single-step $\sum \tan^{-1}\frac{1}{n^2+n+1} = \frac{\pi}{4}$ – same machinery, different boundary count.

CHAPTER X

The Grand Hybrids

Branch correction, telescoping, equations and calculus, fused.

№91 The Gap-Two Telescope That Overshoots

●●●●● L5

EVALUATE

$$\cos^{-1}\left(\cos\left(2\sum_{k=1}^{\infty}\tan^{-1}\frac{2}{k^2}\right)\right)$$

telescoping arctan cos-inverse-cos branch series

ANSWER

$$\frac{\pi}{2}$$

THE TRAP

The seductive wrong answer is $\pi/2$ reached by the wrong route, or worse $3\pi/2$. A strong student writes $\tan^{-1}\frac{2}{k^2} = \tan^{-1}(k+1) - \tan^{-1}(k-1)$ and assumes an ordinary telescope, getting a single survivor and the bogus inner sum $\pi/2$, then doubles to π and answers π. The error: this is a **gap-two** telescope, so **two** boundary terms survive at each end. The true inner sum is $\frac{3\pi}{4}$, the doubled argument $\frac{3\pi}{2}$ lies **outside** $[0,\pi]$, and $\cos^{-1}(\cos\frac{3\pi}{2})$ must be reduced by reflection — it equals $\pi/2$ only because $\cos\frac{3\pi}{2} = 0$, not because $\frac{3\pi}{2} \in [0,\pi]$.

Gap-two telescope, then periodic reduction

1. Since $\dfrac{2}{k^2} = \dfrac{(k+1)-(k-1)}{1+(k+1)(k-1)}$ and $(k+1)(k-1) = k^2 - 1 \geq 0$ keeps the product non-negative, the difference formula is valid with **no** $\pm\pi$: $\tan^{-1}\dfrac{2}{k^2} = \tan^{-1}(k+1) - \tan^{-1}(k-1)$ for all $k \geq 1$.
2. This is a gap-2 telescope.
$$\sum_{k=1}^{n}\left[\tan^{-1}(k+1) - \tan^{-1}(k-1)\right] = \tan^{-1}(n+1) + \tan^{-1}(n) - \tan^{-1}(1) - \tan^{-1}(0).$$
Two survivors at the top, two at the bottom.
3. Let $n \to \infty$: $\tan^{-1}(n+1) + \tan^{-1}(n) \to \frac{\pi}{2} + \frac{\pi}{2} = \pi$, and the bottom is $\frac{\pi}{4} + 0$. So $S = \pi - \frac{\pi}{4} = \dfrac{3\pi}{4}$.
4. Then $2S = \dfrac{3\pi}{2} \notin [0,\pi]$. Reduce: $\cos\frac{3\pi}{2} = 0$, so $\cos^{-1}(0) = \boxed{\dfrac{\pi}{2}}$.

Direct value of the inner sum via $\cos$

1. Compute $S = \sum_{k\geq 1}\tan^{-1}\frac{2}{k^2} = \frac{3\pi}{4}$ as above. Rather than reduce $2S$ by reflection, evaluate $\cos(2S)$ directly.
2. $2S = \frac{3\pi}{2}$, and $\cos\frac{3\pi}{2} = 0$. Since $\cos^{-1}$ asks for the angle in $[0,\pi]$ whose cosine is 0, that angle is $\frac{\pi}{2}$.
3. Therefore the value is $\boxed{\dfrac{\pi}{2}}$, independent of how far outside $[0,\pi]$ the argument $2S$ sits.

INSIGHT

Two instruments fuse here: a **telescoping arctan series** and the **$\cos^{-1}\circ\cos$ periodic reduction**. The lesson is twofold – gap-d telescopes leave d survivors at each end (here $d = 2$, so the limit is π, not $\pi/2$), and an inner angle of $\frac{3\pi}{2}$ must always be folded back into $[0, \pi]$ before $\cos^{-1}$ is applied.

№92 Half-Angle Trap on the Möbius Argument

●●●●● L4

SOLVE FOR X

$$\tan^{-1}\left(\frac{1-x}{1+x}\right) = \frac{1}{2}\tan^{-1} x$$

arctan | mobius-argument | branch | root-count | identity-collapse

ANSWER

$$x = \frac{1}{\sqrt{3}}$$

THE TRAP

The fatal move is $\tan^{-1}\frac{1-x}{1+x} = \frac{\pi}{4} - \tan^{-1} x$ applied **for all** x. This collapse holds only when $x > -1$; for $x < -1$ the correct value is $\frac{\pi}{4} - \tan^{-1} x - \pi$. A student who ignores the branch keeps both algebraic roots and may report a spurious second solution from the $x < -1$ region (or even $x = -1$, where the left side is undefined). Honoring the branch shows the $x < -1$ case forces $\tan^{-1} x = -\frac{\pi}{2}$, impossible on the open range $(-\frac{\pi}{2}, \frac{\pi}{2})$, leaving exactly one root.

Branch-aware identity collapse

1. For $x > -1$ the identity $\tan^{-1}\dfrac{1-x}{1+x} = \dfrac{\pi}{4} - \tan^{-1} x$ holds (the argument is $\tan(\frac{\pi}{4} - \tan^{-1} x)$ and $\frac{\pi}{4} - \tan^{-1} x \in (-\frac{\pi}{4}, \frac{\pi}{4})$).
2. Substitute: $\dfrac{\pi}{4} - \tan^{-1} x = \dfrac{1}{2}\tan^{-1} x \Rightarrow \dfrac{\pi}{4} = \dfrac{3}{2}\tan^{-1} x \Rightarrow \tan^{-1} x = \dfrac{\pi}{6}$, so $x = \tan\dfrac{\pi}{6} = \dfrac{1}{\sqrt{3}} > -1$. Valid.
3. For $x < -1$: $\tan^{-1}\frac{1-x}{1+x} = \frac{\pi}{4} - \tan^{-1} x - \pi$, giving $-\frac{3\pi}{4} = \frac{3}{2}\tan^{-1} x \Rightarrow \tan^{-1} x = -\frac{\pi}{2}$, unattainable. And $x = -1$ is excluded (left side undefined).
4. Hence the unique solution is $\boxed{x = \dfrac{1}{\sqrt{3}}}$.

Tangent of both sides plus a sign audit

1. Let $\theta = \frac{1}{2}\tan^{-1} x \in (-\frac{\pi}{4}, \frac{\pi}{4})$, so $\tan 2\theta = x$. Take tangent of the equation: $\dfrac{1-x}{1+x} = \tan\theta$.
2. Write $\dfrac{1-x}{1+x} = \tan(\frac{\pi}{4} - \tan^{-1} x) = \tan(\frac{\pi}{4} - 2\theta)$. So $\tan\theta = \tan(\frac{\pi}{4} - 2\theta)$, i.e. $\theta = \frac{\pi}{4} - 2\theta + k\pi$.
3. $3\theta = \frac{\pi}{4} + k\pi$. With $\theta \in (-\frac{\pi}{4}, \frac{\pi}{4})$ only $k = 0$ survives: $\theta = \frac{\pi}{12}$, so $x = \tan\frac{\pi}{6} = \dfrac{1}{\sqrt{3}}$. The candidate from $k = -1$ leaves the range and is rejected.
4. $\boxed{x = \dfrac{1}{\sqrt{3}}}$ is the only root.

INSIGHT

This couples the **Möbius-argument identity** $\tan^{-1}\frac{1-x}{1+x} = \frac{\pi}{4} - \tan^{-1} x$ (valid only on $x > -1$) with a **range-based root count**. Squaring or blindly taking tangents manufactures phantom roots; the principal-value range $(-\frac{\pi}{2}, \frac{\pi}{2})$ of $\tan^{-1}$ is what kills them.

№93 Differentiating Across the Fold

●●●●● L5

DIFFERENTIATE

$$\text{Let } f(x)=\sin^{-1}\!\left(2x\sqrt{1-x^2}\right).\ \text{Find } f'\!\left(\tfrac{\sqrt3}{2}\right).$$

derivative · double-angle · branch · sin-inverse · point-evaluation

ANSWER

$$-4$$

THE TRAP

The tempting answer is $+4$. A student recalls $\sin^{-1}(2x\sqrt{1-x^2})=2\sin^{-1}x$, differentiates to $\dfrac{2}{\sqrt{1-x^2}}$, and at $x=\frac{\sqrt3}{2}$ gets $\dfrac{2}{1/2}=4$. But the clean identity $f(x)=2\sin^{-1}x$ holds **only** for $|x|\le\frac{1}{\sqrt2}$. Since $\frac{\sqrt3}{2}>\frac{1}{\sqrt2}$, the correct branch is $f(x)=\pi-2\sin^{-1}x$, whose derivative is $-\dfrac{2}{\sqrt{1-x^2}}$, giving -4. The sign flip is the whole problem.

Branch-split by the threshold $1/\sqrt2$

1. Put $x=\sin\theta$, $\theta=\sin^{-1}x\in[-\frac{\pi}{2},\frac{\pi}{2}]$. Then $2x\sqrt{1-x^2}=2\sin\theta\cos\theta=\sin2\theta$, so $f=\sin^{-1}(\sin2\theta)$.
2. $\sin^{-1}(\sin2\theta)=2\theta$ only if $2\theta\in[-\frac{\pi}{2},\frac{\pi}{2}]$, i.e. $|x|\le\frac{1}{\sqrt2}$. For $\frac{1}{\sqrt2}<x\le1$ we have $2\theta\in(\frac{\pi}{2},\pi]$, so $\sin^{-1}(\sin2\theta)=\pi-2\theta=\pi-2\sin^{-1}x$.
3. At $x=\frac{\sqrt3}{2}$ we are above the threshold, so $f(x)=\pi-2\sin^{-1}x$ and $f'(x)=-\dfrac{2}{\sqrt{1-x^2}}$.
4. Evaluate: $\sqrt{1-\frac34}=\frac12$, so $f'(\frac{\sqrt3}{2})=-\dfrac{2}{1/2}=\boxed{-4}$.

Honest chain rule with the radical

1. Differentiate directly: $f'(x)=\dfrac{1}{\sqrt{1-(2x\sqrt{1-x^2})^2}}\cdot\dfrac{d}{dx}(2x\sqrt{1-x^2})$.
2. Inner derivative: $\dfrac{d}{dx}(2x\sqrt{1-x^2})=\dfrac{2(1-2x^2)}{\sqrt{1-x^2}}$. Also $1-4x^2(1-x^2)=(1-2x^2)^2$, so $\sqrt{\cdot}=|1-2x^2|$.
3. Thus $f'(x)=\dfrac{2(1-2x^2)}{|1-2x^2|\sqrt{1-x^2}}=\dfrac{2\,\mathrm{sgn}(1-2x^2)}{\sqrt{1-x^2}}$. At $x=\frac{\sqrt3}{2}$, $1-2x^2=1-\frac32=-\frac12<0$, so the sign is -1.
4. Hence $f'(\frac{\sqrt3}{2})=-\dfrac{2}{\sqrt{1-3/4}}=-\dfrac{2}{1/2}=\boxed{-4}$. The $|1-2x^2|$ is exactly the algebraic shadow of the branch.

INSIGHT
The **double-angle collapse** $f=\sin^{-1}(\sin2\theta)$ and a **point-evaluated derivative** combine. The square root $\sqrt{1-4x^2(1-x^2)}=|1-2x^2|$ silently carries the branch sign; forgetting the absolute value is the same sin as forgetting that $\sin^{-1}(2x\sqrt{1-x^2})=2\sin^{-1}x$ fails past $x=1/\sqrt2$.

№94 The Pi-Chain Plus a Sixth

●●●●● L5

EVALUATE

$$\cos^{-1}\!\left(\cos\left(\tan^{-1}1+\tan^{-1}2+\tan^{-1}3+\cos^{-1}(-\tfrac12)\right)\right)$$

arctan-chain · gaussian-integers · cos-inverse-cos · branch · reduction

ANSWER

$$\frac{\pi}{3}$$

THE TRAP

A student who writes $\tan^{-1}1+\tan^{-1}2+\tan^{-1}3=\tan^{-1}\frac{1+2+3-1\cdot2\cdot3}{1-(1\cdot2+2\cdot3+3\cdot1)}=\tan^{-1}0=0$ gets the inner angle $=\frac{2\pi}{3}$ and answers $\frac{2\pi}{3}$. The three-arctan formula gives a *tangent* of 0, but the actual sum is π, not 0 — each term is positive and their total exceeds $\frac{\pi}{2}$. So the inner angle is $\pi+\frac{2\pi}{3}=\frac{5\pi}{3}\notin[0,\pi]$, and $\cos^{-1}(\cos\frac{5\pi}{3})=\frac{\pi}{3}$ after reflecting through 2π.

Gaussian-integer product for the chain

1. The argument of $(1+i)(1+2i)(1+3i)$ equals $\tan^{-1}1+\tan^{-1}2+\tan^{-1}3$ (sum of principal arguments, each in $(0,\frac{\pi}{2})$). Multiply: $(1+i)(1+2i)=-1+3i$, then $(-1+3i)(1+3i)=-10$.
2. -10 is a negative real, so its argument is π. Since each summand is positive and they total more than $\frac{\pi}{2}$, the sum is exactly π (not 0).
3. Add $\cos^{-1}(-\frac{1}{2})=\frac{2\pi}{3}$: the inner angle is $\pi+\frac{2\pi}{3}=\dfrac{5\pi}{3}$.
4. $\dfrac{5\pi}{3}\notin[0,\pi]$; reflect: $\cos^{-1}(\cos\frac{5\pi}{3})=2\pi-\frac{5\pi}{3}=\boxed{\dfrac{\pi}{3}}$.

Pairwise arctan sum with branch bookkeeping

1. $\tan^{-1}2+\tan^{-1}3$: product $2\cdot3=6>1$, so add π: $\tan^{-1}2+\tan^{-1}3=\pi+\tan^{-1}\frac{2+3}{1-6}=\pi+\tan^{-1}(-1)=\pi-\frac{\pi}{4}=\frac{3\pi}{4}$.
2. Add $\tan^{-1}1=\frac{\pi}{4}$: the chain is $\frac{3\pi}{4}+\frac{\pi}{4}=\pi$.
3. Inner angle $=\pi+\cos^{-1}(-\frac{1}{2})=\pi+\frac{2\pi}{3}=\dfrac{5\pi}{3}$.
4. $\cos^{-1}(\cos\frac{5\pi}{3})=\boxed{\dfrac{\pi}{3}}$.

INSIGHT

Three instruments meet: the classic **$\tan^{-1}1+\tan^{-1}2+\tan^{-1}3=\pi$ chain (cleanest via Gaussian integers), a** special $\cos^{-1}$ value**, and the** $\cos^{-1}\circ\cos$ reflection** of an angle in $(\pi,2\pi)$. The branch fact – that the arctan-sum formula returns a tangent, not the angle – is what separates π from 0.

№95 Two Sines Equal a Cosine

L4

SOLVE FOR X

$$\sin^{-1}x+\sin^{-1}(1-x)=\cos^{-1}x$$

sin-inverse · cos-inverse · identity-collapse · root-count · domain

ANSWER

$$x\in\{\,0,\ \tfrac{1}{2}\,\}$$

THE TRAP

The trap is to take sine of both sides, expand, square, and report only $x=\frac{1}{2}$ while discarding $x=0$ as 'trivial', or to report extra roots created by squaring. Using $\cos^{-1}x=\frac{\pi}{2}-\sin^{-1}x$ to rewrite the right side as $\frac{\pi}{2}-\sin^{-1}x$ avoids squaring entirely and shows **both** $x=0$ and $x=\frac{1}{2}$ are genuine, while no other algebraic root survives the domain $x\in[0,1]$ (needed so $1-x\in[-1,1]$ and the residual angle stays in range).

Collapse via the complementary identity

1. Rewrite the right side: $\cos^{-1}x=\frac{\pi}{2}-\sin^{-1}x$, so the equation becomes $\sin^{-1}(1-x)=\frac{\pi}{2}-2\sin^{-1}x$.
2. Take sine: $1-x=\sin\left(\frac{\pi}{2}-2\sin^{-1}x\right)=\cos(2\sin^{-1}x)=1-2x^2$.
3. So $-x=-2x^2\Rightarrow x(2x-1)=0\Rightarrow x=0$ or $x=\frac{1}{2}$.
4. Both lie in the required domain $[0,1]$, and there $2\sin^{-1}x\in[0,\pi]$ so the residual angle $\frac{\pi}{2}-2\sin^{-1}x$ stays in $[-\frac{\pi}{2},\frac{\pi}{2}]$, the range where taking sine is reversible — hence no phantom root is introduced. Check: $x=0$ gives $0+\frac{\pi}{2}=\frac{\pi}{2}$

✓; $x = \frac{1}{2}$ gives $\frac{\pi}{6} + \frac{\pi}{6} = \frac{\pi}{3} = \cos^{-1}\frac{1}{2}$ ✓. $\boxed{x \in \{0, \frac{1}{2}\}}$.

Monotonicity / graphical count

1. Let $L(x) = \sin^{-1} x + \sin^{-1}(1-x)$ and $R(x) = \cos^{-1} x$ on the common domain $x \in [0,1]$, and study $F(x) = L(x) - R(x)$.
2. Endpoints: $L(0) = \frac{\pi}{2} = R(0)$ so $F(0) = 0$ (a root at $x = 0$); $L(1) = \frac{\pi}{2} > R(1) = 0$ so $F(1) > 0$. Note L is symmetric about $x = \frac{1}{2}$ (since $x \leftrightarrow 1-x$ fixes it) with a maximum there, while R strictly decreases.
3. On $(0,1]$ one finds $F(x) = \frac{\pi}{2} - 2\sin^{-1} x + \sin^{-1}(1-x)$ vanishes again only at $x = \frac{1}{2}$: F is negative on $(0, \frac{1}{2})$ (here $L < R$), touches 0 at $x = \frac{1}{2}$, and is positive on $(\frac{1}{2}, 1]$ — equivalently $1 - x - (1 - 2x^2) = x(2x-1)$ has the single interior sign change at $x = \frac{1}{2}$ (verified: $\frac{\pi}{3} = \cos^{-1}\frac{1}{2}$).
4. Hence F has zeros only at the endpoint $x = 0$ and the interior point $x = \frac{1}{2}$: exactly two solutions, $\boxed{x \in \{0, \frac{1}{2}\}}$.

INSIGHT

Here the **$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ collapse** turns a two-arc equation into the rational equation $1 - x = 1 - 2x^2$, and the **domain $[0,1]$ plus the range of the residual angle** certify the root count. The substitution sidesteps squaring, which is precisely where careless solvers breed false roots.

№96 A One-Sided Slope at the Origin

●●●●● L5

EVALUATE

$$\lim_{x \to 0^-} \frac{\cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right)}{x}$$

limit · cos-inverse · double-angle · branch · one-sided

ANSWER

$$-2$$

THE TRAP

The reflex is $\cos^{-1}\frac{1-x^2}{1+x^2} = 2\tan^{-1} x$, hence limit $= \lim \frac{2\tan^{-1} x}{x} = 2$. But $\cos^{-1}$ outputs values in $[0, \pi]$, so its result is **never negative**; the clean identity $\cos^{-1}\frac{1-x^2}{1+x^2} = 2\tan^{-1} x$ holds only for $x \geq 0$. For $x < 0$ the correct value is $-2\tan^{-1} x$ (which is positive). Dividing a positive numerator by $x \to 0^-$ yields -2, not $+2$. The two-sided limit does not exist; the left limit is -2.

Sign-correct the branch, then standard limit

1. Set $x = \tan\theta$, $\theta = \tan^{-1} x$. Then $\dfrac{1-x^2}{1+x^2} = \cos 2\theta$, so the numerator is $\cos^{-1}(\cos 2\theta)$.
2. Because $\cos^{-1}$ lands in $[0,\pi]$: for $x > 0$, $2\theta \in (0,\pi)$ gives $\cos^{-1}(\cos 2\theta) = 2\theta = 2\tan^{-1} x$; for $x < 0$, $2\theta \in (-\pi, 0)$ gives $\cos^{-1}(\cos 2\theta) = -2\theta = -2\tan^{-1} x$.
3. Approaching 0^-: numerator $= -2\tan^{-1} x$, so the ratio is $\dfrac{-2\tan^{-1} x}{x}$.
4. $\lim_{x\to 0} \dfrac{\tan^{-1} x}{x} = 1$, hence the left limit is $-2 \cdot 1 = \boxed{-2}$.

L'Hôpital with the honest derivative

1. The expression is $\frac{0}{0}$ as $x \to 0$. Differentiate numerator and denominator. Numerator $g(x) = \cos^{-1}\frac{1-x^2}{1+x^2}$ has $g'(x) = \dfrac{-1}{\sqrt{1 - \left(\frac{1-x^2}{1+x^2}\right)^2}} \cdot \dfrac{-4x}{(1+x^2)^2}$.
2. Since $1 - \left(\frac{1-x^2}{1+x^2}\right)^2 = \frac{4x^2}{(1+x^2)^2}$, its square root is $\frac{2|x|}{1+x^2}$. Thus $g'(x) = \dfrac{4x}{(1+x^2)^2} \cdot \dfrac{1+x^2}{2|x|} = \dfrac{2\,\mathrm{sgn}(x)}{1+x^2}$.

3. By L'Hôpital the ratio tends to $g'(x)/1 = \dfrac{2\,\mathrm{sgn}(x)}{1+x^2}$. As $x \to 0^-$, $\mathrm{sgn}(x) = -1$ and $1+x^2 \to 1$.

4. The left limit is $\boxed{-2}$; the $|x|$ in the square root is the branch made visible.

INSIGHT

A **branch-corrected composite** ($\cos^{-1}\frac{1-x^2}{1+x^2} = 2|\tan^{-1}x|$-style) meets a **one-sided limit.** The $\sqrt{\cdot} = \frac{2|x|}{1+x^2}$ encodes the same sign rule the substitution reveals: $\cos^{-1}$ cannot return a negative angle, so the slope flips across $x=0$ and the two-sided limit is undefined.

№97 Tripling the Three-Term Chain

●●●●● L5

EVALUATE

$$\cos^{-1}\Big(\cos\left(3\tan^{-1}1 + 3\tan^{-1}2 + 3\tan^{-1}3\right)\Big)$$

arctan-chain | cos-inverse-cos | branch | reduction | gaussian-integers

ANSWER

$$\pi$$

THE TRAP

Believing $\tan^{-1}1+\tan^{-1}2+\tan^{-1}3 = 0$ (because the arctan-sum tangent collapses to 0) gives inner angle 0 and answer 0. Even a student who knows the chain equals π may then write 3π and answer 3π — but $3\pi \notin [0,\pi]$, so it must be reduced. $\cos 3\pi = -1$, and $\cos^{-1}(-1) = \pi$. The pitfall is twofold: the chain is π (not 0), and the tripled angle 3π must be folded back into $[0,\pi]$.

Evaluate the chain, then reduce 3π

1. The chain $\tan^{-1}1+\tan^{-1}2+\tan^{-1}3 = \pi$: e.g. $\arg\big((1+i)(1+2i)(1+3i)\big) = \arg(-10) = \pi$, and the sum of three positive principal arctans exceeding $\frac{\pi}{2}$ forces π rather than 0.
2. Triple it: $3\tan^{-1}1+3\tan^{-1}2+3\tan^{-1}3 = 3\pi$.
3. $3\pi \notin [0,\pi]$, so evaluate $\cos 3\pi = -1$.
4. $\cos^{-1}(-1) = \boxed{\pi}$.

Reduce inside cosine using periodicity

1. $\cos$ has period 2π: $\cos(3\pi) = \cos(3\pi-2\pi) = \cos(\pi) = -1$.
2. Independently confirm the chain equals π via the pairwise rule: $\tan^{-1}2+\tan^{-1}3 = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$ (product > 1, so $+\pi$), plus $\tan^{-1}1 = \frac{\pi}{4}$ gives π; tripled is 3π.
3. Thus the inner cosine is -1 and $\cos^{-1}(-1)$ is the angle in $[0,\pi]$ with cosine -1.
4. That angle is $\boxed{\pi}$.

INSIGHT

Fuses the **arctan three-chain $=\pi$ **with the** $\cos^{-1}\circ\cos$ reduction** of a large multiple of π. Two independent branch hazards stack: getting the chain right (π, via Gaussian integers or the $+\pi$ pairwise rule), and folding 3π back into $[0,\pi]$.

№98 Sum of Squares Meets the Range Filter

●●●●● L5

SOLVE FOR X

$$(\sin^{-1}x)^2 + (\cos^{-1}x)^2 = \frac{5\pi^2}{8}$$

sin-inverse · cos-inverse · quadratic · range-filter · root-count

ANSWER

$$x = -\frac{1}{\sqrt{2}}$$

THE TRAP

Setting $s = \sin^{-1} x$ and reducing to $16s^2 - 8\pi s - 3\pi^2 = 0$ gives $s = \frac{3\pi}{4}$ or $s = -\frac{\pi}{4}$. The seductive error is to accept $s = \frac{3\pi}{4}$ and write $x = \sin\frac{3\pi}{4} = \frac{1}{\sqrt{2}}$ as a second solution. But $\sin^{-1} x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, and $\frac{3\pi}{4}$ lies outside that range — it is not a valid value of $\sin^{-1} x$. Only $s = -\frac{\pi}{4}$ survives, giving the single root $x = -\frac{1}{\sqrt{2}}$.

Reduce to a quadratic in $\sin^{-1}x$

1. Let $s = \sin^{-1} x$, $c = \cos^{-1} x$, with $s + c = \frac{\pi}{2}$. Then $s^2 + c^2 = (s+c)^2 - 2sc = \frac{\pi^2}{4} - 2s(\frac{\pi}{2} - s) = 2s^2 - \pi s + \frac{\pi^2}{4}$.
2. Set equal to $\frac{5\pi^2}{8}$: $2s^2 - \pi s - \frac{3\pi^2}{8} = 0$, i.e. $16s^2 - 8\pi s - 3\pi^2 = 0$.
3. Solve: $s = \dfrac{8\pi \pm \sqrt{64\pi^2 + 192\pi^2}}{32} = \dfrac{8\pi \pm 16\pi}{32}$, so $s = \frac{3\pi}{4}$ or $s = -\frac{\pi}{4}$.
4. Range filter: $s = \sin^{-1} x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ rules out $\frac{3\pi}{4}$. So $s = -\frac{\pi}{4}$, $x = \sin(-\frac{\pi}{4}) = \boxed{-\dfrac{1}{\sqrt{2}}}$.

Substitute the complement directly

1. With $c = \frac{\pi}{2} - s$, the equation is $s^2 + (\frac{\pi}{2} - s)^2 = \frac{5\pi^2}{8}$, expanding to $2s^2 - \pi s + \frac{\pi^2}{4} = \frac{5\pi^2}{8}$.
2. So $2s^2 - \pi s = \frac{3\pi^2}{8}$. Complete the square: $2(s - \frac{\pi}{4})^2 = \frac{3\pi^2}{8} + \frac{\pi^2}{8} = \frac{\pi^2}{2}$, giving $(s - \frac{\pi}{4})^2 = \frac{\pi^2}{4}$.
3. Thus $s - \frac{\pi}{4} = \pm\frac{\pi}{2}$, i.e. $s = \frac{3\pi}{4}$ or $s = -\frac{\pi}{4}$.
4. Only $s = -\frac{\pi}{4} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ is admissible: $x = -\dfrac{1}{\sqrt{2}}$. Unique root $\boxed{-\dfrac{1}{\sqrt{2}}}$.

INSIGHT

The **$s + c = \frac{\pi}{2}$ collapse **linearizes a sum of squares into a quadratic, and the** principal range of $\sin^{-1}$** acts as the decisive filter. A quadratic offers two roots; inverse-trig range admits only one. This is the cleanest place to see that 'algebraically valid' and 'range-admissible' are different tests.

№99 The Rotated Tangent, Read Far Right

L4

EVALUATE

$$\text{Let } h(x) = \tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right). \text{ Find } h(3).$$

tan-inverse-tan · phase-shift · branch · point-evaluation · piecewise

ANSWER

$$\frac{5\pi}{4} - 3$$

THE TRAP

Dividing numerator and denominator by $\cos x$ gives $\dfrac{\cos x - \sin x}{\cos x + \sin x} = \tan(\frac{\pi}{4} - x)$, so the tempting answer is $h(3) = \frac{\pi}{4} - 3$. But $\tan^{-1}(\tan\alpha) = \alpha$ only when $\alpha \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Here $\alpha = \frac{\pi}{4} - 3 \approx -2.21$, well below $-\frac{\pi}{2}$, so one must add π: $h(3) = \frac{\pi}{4} - 3 + \pi = \frac{5\pi}{4} - 3$. The naive $\frac{\pi}{4} - 3$ is the wrong branch.

Reduce the argument, then fix the branch

1. Divide top and bottom by $\cos x$: $\dfrac{\cos x - \sin x}{\cos x + \sin x} = \dfrac{1 - \tan x}{1 + \tan x} = \tan\left(\frac{\pi}{4} - x\right)$. So $h(x) = \tan^{-1}\left(\tan\left(\frac{\pi}{4} - x\right)\right)$.
2. $h(x) = \frac{\pi}{4} - x$ only while $\frac{\pi}{4} - x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, i.e. $x \in \left(-\frac{\pi}{4}, \frac{3\pi}{4}\right)$. Since $3 > \frac{3\pi}{4} \approx 2.356$, we are past this window.
3. Add the period: $\frac{\pi}{4} - 3 + \pi = \frac{5\pi}{4} - 3 \approx 0.927 \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, which is the principal value.
4. Therefore $h(3) = \boxed{\dfrac{5\pi}{4} - 3}$.

Compute the inner number and place it

1. At $x = 3$: $\cos 3 \approx -0.98999$, $\sin 3 \approx 0.14112$. Then $\dfrac{\cos 3 - \sin 3}{\cos 3 + \sin 3} \approx \dfrac{-1.1311}{-0.8489} \approx 1.3324$, a positive number.
2. So $h(3) = \tan^{-1}(1.3324) \approx 0.927$, a positive principal value in $\left(0, \frac{\pi}{2}\right)$.
3. Match to a clean form: $\frac{5\pi}{4} - 3 \approx 3.927 - 3 = 0.927$ ✓, whereas the naive $\frac{\pi}{4} - 3 \approx -2.21$ is negative — impossible, since the actual ratio is positive.
4. Hence $h(3) = \boxed{\dfrac{5\pi}{4} - 3}$.

INSIGHT
A **phase-shift collapse** $\tan^{-1}(\tan(\frac{\pi}{4} - x))$ evaluated at a **specific point outside the principal window.** The function h is piecewise-linear with slope -1 and jumps of π at $x = \frac{3\pi}{4} + k\pi$; reading it at $x = 3$ lands one jump to the right of the origin's branch, so the constant is $\frac{5\pi}{4}$, not $\frac{\pi}{4}$.

№100 Three Undoings at Five Radians

●●●●● L3

EVALUATE

$$\sin^{-1}(\sin 5) + \cos^{-1}(\cos 5) + \tan^{-1}(\tan 5)$$

sin-inverse-sin · cos-inverse-cos · tan-inverse-tan · quadrant · reduction

ANSWER

$$5 - 2\pi$$

THE TRAP
The instinctive answer is $5 + 5 + 5 = 15$ (or just '5'), assuming each inverse undoes its function. But 5 radians lies in $\left(\frac{3\pi}{2}, 2\pi\right)$ — the fourth quadrant past $\frac{3\pi}{2}$ — so none of the three returns 5. Each must be folded into its own principal range: $\sin^{-1}(\sin 5) = 5 - 2\pi$, $\cos^{-1}(\cos 5) = 2\pi - 5$, $\tan^{-1}(\tan 5) = 5 - 2\pi$. They do **not** all agree, and the $\cos^{-1}$ term has the opposite sign of the other two.

Reduce each into its principal range

1. Locate 5: since $\frac{3\pi}{2} \approx 4.712 < 5 < 2\pi \approx 6.283$, the angle 5 is in the fourth quadrant (a touch below 2π).
2. $\sin^{-1}(\sin 5)$: subtract 2π to reach $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. $5 - 2\pi \approx -1.283 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, and $\sin(5 - 2\pi) = \sin 5$. So this term $= 5 - 2\pi$.
3. $\cos^{-1}(\cos 5)$: need a value in $[0, \pi]$. $\cos 5 = \cos(2\pi - 5)$ and $2\pi - 5 \approx 1.283 \in [0, \pi]$. So this term $= 2\pi - 5$.
4. $\tan^{-1}(\tan 5)$: need a value in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. $5 - 2\pi \approx -1.283$ qualifies and $\tan(5 - 2\pi) = \tan 5$. So this term $= 5 - 2\pi$.
5. Add: $(5 - 2\pi) + (2\pi - 5) + (5 - 2\pi) = \boxed{5 - 2\pi}$.

Pair the cancelling terms first

1. The $\sin^{-1}$ and $\cos^{-1}$ reductions are negatives of each other here: $(5 - 2\pi) + (2\pi - 5) = 0$.
2. So the whole expression equals the lone $\tan^{-1}(\tan 5)$ term.
3. Since $5 \in \left(\frac{3\pi}{2}, 2\pi\right)$, $\tan^{-1}(\tan 5) = 5 - 2\pi$ (the unique representative in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$).

4. Hence the sum is $\boxed{5 - 2\pi}$.

INSIGHT

A compact capstone on the **three reduction maps** $\sin^{-1}\circ \sin$, $\cos^{-1}\circ \cos$, $\tan^{-1}\circ \tan$ at one out-of-range angle. The quadrant of 5 dictates three *different* corrections; the $\sin^{-1}$ and $\cos^{-1}$ pieces cancel, leaving the tangent reduction. Whenever $\theta \in (\frac{3\pi}{2}, 2\pi)$, this sum collapses to $\theta - 2\pi$.